

入院時の指示・薬・検査・当直・入院疾患の管理

内科大全病棟編 文献

下書き 2026-10-08 00:56

2026年9月

文献

本文の右肩の番号 (1)2) …) は、章ごとの文献番号である。章の番号と題は本文と同じにしてある。

第1部 入院時の指示と共通管理

1-1 入院時指示

- 1) Allen C, Glasziou P, Del Mar C. Bed rest: a potentially harmful treatment needing more careful evaluation. *Lancet* 1999;354:1229-33.
- 2) Covinsky KE, Palmer RM, Fortinsky RH, et al. Loss of independence in activities of daily living in older adults hospitalized with medical illnesses: increased vulnerability with age. *J Am Geriatr Soc* 2003;51:451-8.
- 3) 保健師助産師看護師法（昭和23年法律第203号）第5条、第37条.
- 4) 医師法（昭和23年法律第201号）第24条.
- 5) 日本医療機能評価機構. 医療安全情報 No.27 口頭指示による薬剤量間違い. 2009.
- 6) 厚生労働省. 「医薬品の安全使用のための業務手順書」作成マニュアル（平成30年改訂版）. 2018.
- 7) 日本医療機能評価機構. 医療安全情報 No.18 処方表記の解釈の違いによる薬剤量間違い. 2008.
- 8) Royal College of Physicians. National Early Warning Score (NEWS) 2: Standardising the assessment of acute-illness severity in the NHS. London: RCP; 2017.
- 9) Semler MW, Stover DG, Copland AP, et al. Flash mob research: a single-day, multicenter, resident-directed study of respiratory rate. *Chest* 2013;143:1740-4.
- 10) Lovett PB, Buchwald JM, Stürmann K, Bijur P. The vexatious vital: neither clinical measurements by nurses nor an electronic monitor provides accurate measurements of respiratory rate in triage. *Ann Emerg Med* 2005;45:68-76.
- 11) Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl* 2012;2:1-138.
- 12) 日本脳卒中学会 脳卒中ガイドライン委員会編. 脳卒中治療ガイドライン2021〔改訂2025〕. 協和企画；2025.
- 13) 日本腎臓学会 腎生検ガイドブック改訂委員会編. 腎生検ガイドブック2020. 東京医学社；2020.
- 14) Whittier WL, Korbet SM. Timing of complications in percutaneous renal biopsy. *J Am Soc Nephrol* 2004;15:142-7.
- 15) Neuberger J, Patel J, Caldwell H, et al. Guidelines on the use of liver biopsy in clinical practice from the British Society of Gastroenterology, the Royal College of Radiologists and the Royal College of Pathology. *Gut* 2020;69:1382-403.
- 16) Mohammady M, Heidari K, Akbari Sari A, et al. Early ambulation after diagnostic transfemoral catheterisation: a systematic review and meta-analysis. *Int J Nurs Stud* 2014;51:39-50.
- 17) AVERT Trial Collaboration group. Efficacy and safety of very early mobilisation within 24 h of stroke onset (AVERT): a randomised controlled trial. *Lancet* 2015;386:46-55.
- 18) Arevalo-Rodriguez I, Ciapponi A, Roqué i Figuls M, et al. Posture and fluids for preventing post-dural puncture headache. *Cochrane Database Syst Rev* 2016;3:CD009199.
- 19) Liu Z, Tao X, Chen Y, Fan Z, Li Y. Bed rest versus early ambulation with standard anticoagulation in the management of deep vein thrombosis: a meta-analysis. *PLoS One* 2015;10:e0121388.

- 20) 日本循環器学会, 日本肺高血圧・肺循環学会. 2025年改訂版 肺血栓塞栓症・深部静脈血栓症および肺高血圧症に関するガイドライン. 2025.
- 21) Martínez-Velilla N, Casas-Herrero A, Zambom-Ferraresi F, et al. Effect of exercise intervention on functional decline in very elderly patients during acute hospitalization: a randomized clinical trial. *JAMA Intern Med* 2019;179:28-36.
- 22) Brown CJ, Foley KT, Lowman JD Jr, et al. Comparison of posthospitalization function and community mobility in hospital mobility program and usual care patients: a randomized clinical trial. *JAMA Intern Med* 2016;176:921-7.
- 23) Schweickert WD, Pohlman MC, Pohlman AS, et al. Early physical and occupational therapy in mechanically ventilated, critically ill patients: a randomised controlled trial. *Lancet* 2009;373:1874-82.
- 24) TEAM Study Investigators and the ANZICS Clinical Trials Group. Early active mobilization during mechanical ventilation in the ICU. *N Engl J Med* 2022;387:1747-58.
- 25) Maeda K, Koga T, Akagi J. Tentative nil per os leads to poor outcomes in older adults with aspiration pneumonia. *Clin Nutr* 2016;35:1147-52.
- 26) 小山珠美, 若林秀隆, 前田圭介, ほか. 誤嚥性肺炎患者に対するチーム医療による早期経口摂取が在院日数と退院時経口摂取に及ぼす影響. *日摂食嚥下リハ会誌* 2020;24:14-25.
- 27) Finucane TE, Christmas C, Travis K. Tube feeding in patients with advanced dementia: a review of the evidence. *JAMA* 1999;282:1365-70.
- 28) Davies N, Barrado-Martín Y, Vickerstaff V, et al. Enteral tube feeding for people with severe dementia. *Cochrane Database Syst Rev* 2021;8:CD013503.
- 29) Gomes CA Jr, Andriolo RB, Bennett C, et al. Percutaneous endoscopic gastrostomy versus nasogastric tube feeding for adults with swallowing disturbances. *Cochrane Database Syst Rev* 2015;(5):CD008096.
- 30) JAID/JSC感染症治療ガイド・ガイドライン作成委員会. JAID/JSC感染症治療ガイドライン2015—腸管感染症一. *日本化学療法学会雑誌* 2016;64:31-65.
- 31) Shane AL, Mody RK, Crump JA, et al. 2017 Infectious Diseases Society of America clinical practice guidelines for the diagnosis and management of infectious diarrhea. *Clin Infect Dis* 2017;65:e45-e80.
- 32) 急性膵炎診療ガイドライン2021改訂出版委員会編. 急性膵炎診療ガイドライン2021 第5版. 金原出版; 2021.
- 33) Al-Omran M, Albalawi ZH, Tashkandi MF, Al-Ansary LA. Enteral versus parenteral nutrition for acute pancreatitis. *Cochrane Database Syst Rev* 2010;(1):CD002837.
- 34) 藤城光弘, 井口幹崇, 小野敏嗣, ほか. 非静脈瘤性上部消化管出血における内視鏡診療ガイドライン (第2版). *日本消化器内視鏡学会雑誌* 2024;66:1515-38.
- 35) 日本消化管学会編. 大腸憩室症 (憩室出血・憩室炎) ガイドライン. *日本消化管学会雑誌* 2017;1(Suppl):1-52.
- 36) 日本摂食嚥下リハビリテーション学会嚥下調整食委員会. 日本摂食嚥下リハビリテーション学会嚥下調整食分類2021. *日摂食嚥下リハ会誌* 2021;25:135-49.
- 37) 厚生労働省. 診療報酬点数表 (令和8年度) H004 摂食機能療法.
- 38) 日本麻酔科学会. 術前絶飲食ガイドライン. 2012.
- 39) ACR Committee on Drugs and Contrast Media. ACR Manual on Contrast Media. American College of Radiology; 2024.
- 40) Neeman Z, et al. Is fasting still necessary prior to contrast-enhanced computed tomography? A randomized clinical study. *Eur Radiol* 2021;31:1451-9.

- 41) Austin MA, Wills KE, Blizzard L, Walters EH, Wood-Baker R. Effect of high flow oxygen on mortality in chronic obstructive pulmonary disease patients in prehospital setting: randomised controlled trial. *BMJ* 2010;341:c5462.
- 42) O'Driscoll BR, Howard LS, Earis J, et al. BTS guideline for oxygen use in adults in healthcare and emergency settings. *Thorax* 2017;72(Suppl 1):iii1-ii90.
- 43) Sandau KE, Funk M, Auerbach A, et al. Update to practice standards for electrocardiographic monitoring in hospital settings: a scientific statement from the American Heart Association. *Circulation* 2017;136:e273-e344.
- 44) Najafi N, Auerbach A. Use and outcomes of telemetry monitoring on a medicine service. *Arch Intern Med* 2012;172:1349-50.
- 45) Drew BJ, Harris P, Zègre-Hemsey JK, et al. Insights into the problem of alarm fatigue with physiologic monitor devices: a comprehensive observational study of consecutive intensive care unit patients. *PLoS One* 2014;9:e110274.
- 46) 日本医療機能評価機構. 医療事故情報収集等事業 第38回報告書. 2014.
- 47) Bulger J, Nickel W, Messler J, et al. Choosing wisely in adult hospital medicine: five opportunities for improved healthcare value. *J Hosp Med* 2013;8:486-92.
- 48) 日本静脈経腸栄養学会編. 静脈経腸栄養ガイドライン 第3版. 照林社; 2013.
- 49) Rickard CM, Webster J, Wallis MC, et al. Routine versus clinically indicated replacement of peripheral intravenous catheters: a randomised controlled equivalence trial. *Lancet* 2012;380:1066-74.
- 50) O'Grady NP, Alexander M, Burns LA, et al. Guidelines for the prevention of intravascular catheter-related infections. *Clin Infect Dis* 2011;52:e162-93.
- 51) Gould CV, Umscheid CA, Agarwal RK, et al. Guideline for prevention of catheter-associated urinary tract infections 2009. *Infect Control Hosp Epidemiol* 2010;31:319-26.
- 52) Adams D, Bucior H, Day G, Rimmer JA. HOUDINI: make that urinary catheter disappear – nurse-led protocol. *J Infect Prev* 2012;13:44-6.
- 53) Fisher E, Subramonian K, Omar MI. The role of alpha blockers prior to removal of urethral catheter for acute urinary retention in men. *Cochrane Database Syst Rev* 2014;(6):CD006744.
- 54) Fitzpatrick JM, Desgrandchamps F, Adjali K, et al. Management of acute urinary retention: a worldwide survey of 6074 men with benign prostatic hyperplasia. *BJU Int* 2012;109:88-95.
- 55) Chrouser K, Fowler KE, Mann JD, et al. Urinary retention evaluation and catheterization algorithm for adult inpatients. *JAMA Netw Open* 2024;7:e2422281.
- 56) Verathon. BladderScan BVI 9400 取扱説明書 (日本語版) .
- 57) Palese A, Buchini S, Deroma L, Barbone F. The effectiveness of the ultrasound bladder scanner in reducing urinary tract infections: a meta-analysis. *J Clin Nurs* 2010;19:2970-9.
- 58) 日本泌尿器科学会編. 男性下部尿路症状・前立腺肥大症診療ガイドライン. リッチヒルメディカル; 2017.
- 59) 厚生労働省. 診療報酬点数表 (令和8年度) A251 排尿自立支援加算.
- 60) Yoder JC, Yuen TC, Churpek MM, Arora VM, Edelson DP. A prospective study of nighttime vital sign monitoring frequency and risk of clinical deterioration. *JAMA Intern Med* 2013;173:1554-5.
- 61) Thavendiranathan P, Bagai A, Ebidia A, Detsky AS, Choudhry NK. Do blood tests cause anemia in hospitalized patients? The effect of diagnostic phlebotomy on hemoglobin and hematocrit levels. *J Gen Intern Med* 2005;20:520-4.
- 62) Salisbury AC, Reid KJ, Alexander KP, et al. Diagnostic blood loss from phlebotomy and hospital-acquired anemia during acute myocardial infarction. *Arch Intern Med* 2011;171:1646-53.
- 63) Corson AH, Fan VS, White T, et al. A multifaceted hospitalist quality improvement intervention: decreased frequency of common labs. *J Hosp Med* 2015;10:390-5.

1-2 頓用指示

- 1) Manterola C, Vial M, Moraga J, Astudillo P. Analgesia in patients with acute abdominal pain. Cochrane Database Syst Rev 2011;(1):CD005660.
- 2) 各薬剤の電子添文（KEGG MEDICUSで2026年9～10月に確認）。
- 3) Young P, Saxena M, Bellomo R, et al. Acetaminophen for fever in critically ill patients with suspected infection. N Engl J Med 2015;373:2215-24.
- 4) Coxib and traditional NSAID Trialists' (CNT) Collaboration. Vascular and upper gastrointestinal effects of non-steroidal anti-inflammatory drugs: meta-analyses of individual participant data from randomised trials. Lancet 2013;382:769-79.
- 5) Hansen TM, Matzen P, Madsen P. Endoscopic evaluation of the effect of indomethacin capsules and suppositories on the gastric mucosa in rheumatic patients. J Rheumatol 1984;11:484-7.
- 6) Strom BL, Berlin JA, Kinman JL, et al. Parenteral ketorolac and risk of gastrointestinal and operative site bleeding: a postmarketing surveillance study. JAMA 1996;275:376-82.
- 7) 日本医療機能評価機構. 医療事故情報収集等事業 第55回報告書（分析テーマ：小児へ投与する薬剤に関連した事例）. 2018.
- 8) Cantais A, Schnell D, Vincent F, et al. Acetaminophen-induced changes in systemic blood pressure in critically ill patients: results of a multicenter cohort study. Crit Care Med 2016;44:2192-8.
- 9) 厚生労働省. 重篤副作用疾患別対応マニュアル 薬剤性せん妄. 2022.
- 10) 厚生労働科学研究・障害者対策総合研究事業「睡眠薬の適正使用及び減量・中止のための診療ガイドラインに関する研究班」, 日本睡眠学会・睡眠薬使用ガイドライン作成ワーキンググループ編. 睡眠薬の適正な使用と休薬のための診療ガイドライン. 2013.
- 11) Girard TD, Exline MC, Carson SS, et al. Haloperidol and ziprasidone for treatment of delirium in critical illness. N Engl J Med 2018;379:2506-16.
- 12) Andersen-Ranberg NC, Poulsen LM, Perner A, et al. Haloperidol for the treatment of delirium in ICU patients. N Engl J Med 2022;387:2425-35.
- 13) Agar MR, Lawlor PG, Quinn S, et al. Efficacy of oral risperidone, haloperidol, or placebo for symptoms of delirium among patients in palliative care: a randomized clinical trial. JAMA Intern Med 2017;177:34-42.
- 14) 厚生労働省保険局医療課. 医薬品の適応外使用に係る保険診療上の取扱いについて（保医発0928第1号）. 2011.
- 15) 厚生労働省医薬局. 医薬品・医療機器等安全性情報 No.420（ドンペリドンの使用上の注意の改訂）. 2025.
- 16) 日本消化管学会編. 便通異常症診療ガイドライン2023—慢性便秘症. 南江堂；2023.（英語版：Ihara E, et al. Digestion 2025;106:62-89）
- 17) 医薬品医療機器総合機構. 酸化マグネシウム製剤 適正使用に関するお願い—高マグネシウム血症—. 2020.
- 18) Lewis SJ, Heaton KW. Stool form scale as a useful guide to intestinal transit time. Scand J Gastroenterol 1997;32:920-4.
- 19) 日本化学療法学会, 日本感染症学会 CDI診療ガイドライン作成委員会編. Clostridioides difficile感染症診療ガイドライン2022.
- 20) Siegel DL, Edelstein PH, Nachamkin I. Inappropriate testing for diarrheal diseases in the hospital. JAMA 1990;263:979-82.
- 21) Guerrant RL, Van Gilder T, Steiner TS, et al. Practice guidelines for the management of infectious diarrhea. Clin Infect Dis 2001;32:331-51.

- 22) Bauer TM, Lalvani A, Fehrenbach J, et al. Derivation and validation of guidelines for stool cultures for enteropathogenic bacteria other than *Clostridium difficile* in hospitalized adults. JAMA 2001;285:313-9.
- 23) 日本高血圧学会. 高血圧管理・治療ガイドライン2025. 2025. (英語版: Hypertens Res 2026;49:9-235)
- 24) Rastogi R, Sheehan MM, Hu B, et al. Treatment and outcomes of inpatient hypertension among adults with noncardiac admissions. JAMA Intern Med 2021;181:345-52.
- 25) Umpierrez GE, Smiley D, Zisman A, et al. Randomized study of basal-bolus insulin therapy in the inpatient management of patients with type 2 diabetes (RABBIT 2 trial). Diabetes Care 2007;30:2181-6.
- 26) 日本糖尿病学会. 糖尿病診療ガイドライン2024. 第20章 糖尿病における急性代謝失調・シックデイ (感染症を含む). 2024.
- 27) 日本糖尿病学会編・著. 糖尿病治療ガイド2018-2019. 文光堂; 2018.
- 28) Holstein A, Plaschke A, Hammer C, Egberts EH. Characteristics and time course of severe glimepiride- versus glibenclamide-induced hypoglycaemia. Eur J Clin Pharmacol 2003;59:91-7.
- 29) Namba M, Iwakura T, Nishimura R, et al. The current status of treatment-related severe hypoglycemia in Japanese patients with diabetes mellitus: a report from the committee on a survey of severe hypoglycemia in the Japan Diabetes Society. J Diabetes Investig 2018;9:642-56.
- 30) Galvin R, Bråthen G, Ivashynka A, et al. EFNS guidelines for diagnosis, therapy and prevention of Wernicke encephalopathy. Eur J Neurol 2010;17:1408-18.
- 31) 日本神経学会監修. 重症筋無力症／ランバート・イートン筋無力症候群診療ガイドライン作成委員会編. 重症筋無力症／ランバート・イートン筋無力症候群診療ガイドライン2022. 南江堂; 2022.
- 32) Walter EJ, Hanna-Jumma S, Carraretto M, Forni L. The pathophysiological basis and consequences of fever. Crit Care 2016;20:200.
- 33) 日本救急医学会. 熱中症診療ガイドライン2024. 2024.
- 34) Heyneman CA, Lawless-Liday C, Wall GC. Oral versus topical NSAIDs in rheumatic diseases: a comparison. Drugs 2000;60:555-74.
- 35) Mu R, Bao CD, Chen ZW, et al. Efficacy and safety of loxoprofen hydrogel patch versus loxoprofen tablet in patients with knee osteoarthritis: a randomized controlled non-inferiority trial. Clin Rheumatol 2016;35:165-73.
- 36) 日本医療機能評価機構. 医療安全情報 No.98 カリウム製剤の投与方法間違い. 2015.

1-3 持参薬の扱い

- 1) 厚生労働省. 高齢者の医薬品適正使用の指針 (各論編 (療養環境別)). 2019.
- 2) 各薬剤の電子添文・インタビューフォーム (KEGG MEDICUSで2026年9月に確認).
- 3) Baldeweg SE, et al. Society for Endocrinology clinical guidance: inpatient management of cranial diabetes insipidus. Endocr Connect 2018;7:G8-11.
- 4) 日本神経学会監修. パーキンソン病診療ガイドライン2018. 医学書院; 2018.
- 5) Beckers M, Hommel D, Lennaerts H, et al. Discontinuation or acute unplanned cessation of oral dopaminergic medications in persons with Parkinson's disease: a practice review. J Parkinsons Dis 2025;15:1062-77.
- 6) Kashiwara Y, et al. Effects of magnesium oxide on pharmacokinetics of L-dopa/carbidopa and assessment of pharmacodynamic changes by a model-based simulation. Eur J Clin Pharmacol 2019;75:351-61.
- 7) Broersen LH, Pereira AM, Jørgensen JO, Dekkers OM. Adrenal insufficiency in corticosteroids use: systematic review and meta-analysis. J Clin Endocrinol Metab 2015;100:2171-80.

- 8) Beuschlein F, Else T, Bancos I, et al. European Society of Endocrinology and Endocrine Society joint clinical guideline: diagnosis and therapy of glucocorticoid-induced adrenal insufficiency. *J Clin Endocrinol Metab* 2024;109:1657-83.
- 9) Serhal DI, Nasrallah MP, Arafah BM. Rapid rise in serum thyrotropin concentrations after thyroidectomy or withdrawal of suppressive thyroxine therapy in preparation for radioactive iodine administration to patients with differentiated thyroid cancer. *J Clin Endocrinol Metab* 2004;89:3285-9.
- 10) Ono Y, et al. Clinical characteristics and outcomes of myxedema coma: analysis of a national inpatient database in Japan. *J Epidemiol* 2017;27:117-22.
- 11) Jonklaas J, Bianco AC, Bauer AJ, et al. Guidelines for the treatment of hypothyroidism: prepared by the American Thyroid Association task force on thyroid hormone replacement. *Thyroid* 2014;24:1670-751.
- 12) 厚生労働省. 使用薬剤の薬価（薬価基準）に収載されている医薬品について（注射薬）. 2026年8月.
- 13) Meyer JM. Individual changes in clozapine levels after smoking cessation: results and a predictive model. *J Clin Psychopharmacol* 2001;21:569-74.
- 14) 厚生労働科学研究・障害者対策総合研究事業「睡眠薬の適正使用及び減量・中止のための診療ガイドラインに関する研究班」, 日本睡眠学会・睡眠薬使用ガイドライン作成ワーキンググループ編. 睡眠薬の適正な使用と休薬のための診療ガイドライン. 2013.
- 15) Brunner E, et al. Joint clinical practice guideline on benzodiazepine tapering: considerations when risks outweigh benefits. *J Gen Intern Med* 2025;40:2814-59.
- 16) 日本糖尿病学会 SGLT2阻害薬の適正使用に関する委員会. 糖尿病治療におけるSGLT2阻害薬の適正使用に関するRecommendation. 2022年7月改訂.
- 17) 日本糖尿病学会 ビグアナイド薬の適正使用に関する委員会. メトホルミンの適正使用に関するRecommendation. 2020年3月改訂.
- 18) 日本リウマチ学会MTX診療ガイドライン策定小委員会編. 関節リウマチ治療におけるメトトレキサート（MTX）診療ガイドライン2023年改訂版.
- 19) 日本リウマチ学会. 関節リウマチ（RA）に対するTNF阻害薬使用の手引き. 2024年7月改訂.
- 20) Umpierrez GE, Davis GM, ElSayed NA, et al. Hyperglycaemic crises in adults with diabetes: a consensus report. *Diabetologia* 2024;67:1455-79. (*Diabetes Care* 2024;47:1257-75と同時掲載)
- 21) Silveira SQ, et al. Relationship between perioperative semaglutide use and residual gastric content: a retrospective analysis of patients undergoing elective upper endoscopy. *J Clin Anesth* 2023;87:111091.
- 22) Klein SR, Hobai IA. Semaglutide, delayed gastric emptying, and intraoperative pulmonary aspiration: a case report. *Can J Anaesth* 2023;70:1394-6.
- 23) American Society of Anesthesiologists. American Society of Anesthesiologists consensus-based guidance on preoperative management of patients (adults and children) on glucagon-like peptide-1 (GLP-1) receptor agonists. 2023.
- 24) Kindel TL, et al. Multisociety clinical practice guidance for the safe use of glucagon-like peptide-1 receptor agonists in the perioperative period. *Surg Endosc* 2025;39:180-3.
- 25) Sodhi M, Rezaeianzadeh R, Kezouh A, Etminan M. Risk of gastrointestinal adverse events associated with glucagon-like peptide-1 receptor agonists for weight loss. *JAMA* 2023;330:1795-7.
- 26) 藤島一郎監修, 倉田なおみ編. 内服薬 経管投与ハンドブック 第5版. じほう; 2026.
- 27) Bauer LA. Interference of oral phenytoin absorption by continuous nasogastric feedings. *Neurology* 1982;32:570-2.
- 28) Dickerson RN, et al. Clinical relevancy of the levothyroxine-continuous enteral nutrition interaction. *Nutr Clin Pract* 2010;25:646-52.

- 29) Penrod LE, et al. Warfarin resistance and enteral feedings: 2 case reports and a supporting in vitro study. *Arch Phys Med Rehabil* 2001;82:1270-3.
- 30) 厚生労働省. 高齢者の医薬品適正使用の指針（総論編）. 2018.
- 31) Rochon PA, Gurwitz JH. Optimising drug treatment for elderly people: the prescribing cascade. *BMJ* 1997;315:1096-9.
- 32) Savage RD, Visentin JD, Bronskill SE, et al. Evaluation of a common prescribing cascade of calcium channel blockers and diuretics in older adults with hypertension. *JAMA Intern Med* 2020;180:643-51.
- 33) Gurwitz JH, Avorn J, Bohn RL, et al. Initiation of antihypertensive treatment during nonsteroidal anti-inflammatory drug therapy. *JAMA* 1994;272:781-6.
- 34) Avorn J, Gurwitz JH, Bohn RL, et al. Increased incidence of levodopa therapy following metoclopramide use. *JAMA* 1995;274:1780-2.
- 35) Gill SS, Mamdani M, Naglie G, et al. A prescribing cascade involving cholinesterase inhibitors and anticholinergic drugs. *Arch Intern Med* 2005;165:808-13.
- 36) Rochon PA, et al. Association of a calcium channel blocker and diuretic prescribing cascade with adverse events: a population-based cohort study. *J Am Geriatr Soc* 2024;72:467-78.
- 37) Page AT, Clifford RM, Potter K, et al. The feasibility and effect of deprescribing in older adults on mortality and health: a systematic review and meta-analysis. *Br J Clin Pharmacol* 2016;82:583-623.
- 38) Blum MR, Sallevelt BTGM, Spinewine A, et al. Optimizing Therapy to Prevent Avoidable Hospital Admissions in Multimorbid Older Adults (OPERAM): cluster randomised controlled trial. *BMJ* 2021;374:n1585.
- 39) Ceraso A, Lin JJ, Schneider-Thoma J, et al. Maintenance treatment with antipsychotic drugs for schizophrenia. *Cochrane Database Syst Rev* 2020;8:CD008016.
- 40) Handley SA, Every-Palmer S, Ismail A, Flanagan RJ. Clozapine-induced gastrointestinal hypomotility: presenting features and outcomes, UK pharmacovigilance reports, 1992-2017. *Br J Psychiatry* 2022;220:355-63.
- 41) Legrand M, Falcone J, Cholley B, et al. Continuation vs discontinuation of renin-angiotensin system inhibitors before major noncardiac surgery: the Stop-or-Not randomized clinical trial. *JAMA* 2024;332:970-8.
- 42) Ackland GL, Patel A, Abbott TEF, et al. Discontinuation vs. continuation of renin-angiotensin system inhibition before non-cardiac surgery: the SPACE trial. *Eur Heart J* 2024;45:1146-55.

1-4 静脈血栓塞栓症（VTE）の予防

- 1) Barbar S, Noventa F, Rossetto V, et al. A risk assessment model for the identification of hospitalized medical patients at risk for venous thromboembolism: the Padua Prediction Score. *J Thromb Haemost* 2010;8:2450-7.
- 2) 日本循環器学会, 日本肺高血圧・肺循環学会. 2025年改訂版 肺血栓塞栓症・深部静脈血栓症および肺高血圧症に関するガイドライン. 2025.
- 3) Kahn SR, Lim W, Dunn AS, et al. Prevention of VTE in nonsurgical patients: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest* 2012;141(2 Suppl):e195S-e226S.
- 4) Decousus H, Tapson VF, Bergmann JF, et al. Factors at admission associated with bleeding risk in medical patients: findings from the IMPROVE investigators. *Chest* 2011;139:69-79.
- 5) Phung OJ, Kahn SR, Cook DJ, et al. Dosing frequency of unfractionated heparin thromboprophylaxis: a meta-analysis. *Chest* 2011;140:374-81.
- 6) 各薬剤の電子添文（KEGG MEDICUSで2026年9～10月に確認）.

- 7) CLOTS (Clots in Legs Or sTockings after Stroke) Trials Collaboration. Effectiveness of intermittent pneumatic compression in reduction of risk of deep vein thrombosis in patients who have had a stroke (CLOTS 3): a multicentre randomised controlled trial. *Lancet* 2013;382:516-24.
- 8) CLOTS Trials Collaboration. Effectiveness of thigh-length graduated compression stockings to reduce the risk of deep vein thrombosis after stroke (CLOTS trial 1): a multicentre, randomised controlled trial. *Lancet* 2009;373:1958-65.
- 9) Arabi YM, Al-Hameed F, Burns KEA, et al. Adjunctive intermittent pneumatic compression for venous thromboprophylaxis. *N Engl J Med* 2019;380:1305-15.
- 10) 厚生労働省. 診療報酬点数表 (令和8年度) B001-6 肺血栓塞栓症予防管理料.
- 11) Schünemann HJ, Cushman M, Burnett AE, et al. American Society of Hematology 2018 guidelines for management of venous thromboembolism: prophylaxis for hospitalized and nonhospitalized medical patients. *Blood Adv* 2018;2:3198-225.
- 12) Cohen AT, Spiro TE, Büller HR, et al. Rivaroxaban for thromboprophylaxis in acutely ill medical patients. *N Engl J Med* 2013;368:513-23.
- 13) Cohen AT, Harrington RA, Goldhaber SZ, et al. Extended thromboprophylaxis with betrixaban in acutely ill medical patients. *N Engl J Med* 2016;375:534-44.
- 14) Spyropoulos AC, Ageno W, Albers GW, et al. Rivaroxaban for thromboprophylaxis after hospitalization for medical illness. *N Engl J Med* 2018;379:1118-27.
- 15) 日本血栓止血学会HIT診療ガイドライン作成委員会. ヘパリン起因性血小板減少症の診断・治療ガイドライン. *日本血栓止血学会誌* 2021;32:737-82.
- 16) Cuker A, Arepally GM, Chong BH, et al. American Society of Hematology 2018 guidelines for management of venous thromboembolism: heparin-induced thrombocytopenia. *Blood Adv* 2018;2:3360-92.
- 17) Lederle FA, Zylla D, MacDonald R, Wilt TJ. Venous thromboembolism prophylaxis in hospitalized medical patients and those with stroke: a background review for an American College of Physicians clinical practice guideline. *Ann Intern Med* 2011;155:602-15.
- 18) Kakkar AK, Cimminiello C, Goldhaber SZ, et al. Low-molecular-weight heparin and mortality in acutely ill medical patients. *N Engl J Med* 2011;365:2463-72.

1-5 ストレス潰瘍の予防

- 1) Cook DJ, Fuller HD, Guyatt GH, et al. Risk factors for gastrointestinal bleeding in critically ill patients. *N Engl J Med* 1994;330:377-81.
- 2) Herzig SJ, Howell MD, Ngo LH, Marcantonio ER. Acid-suppressive medication use and the risk for hospital-acquired pneumonia. *JAMA* 2009;301:2120-8.
- 3) Bulger J, Nickel W, Messler J, et al. Choosing wisely in adult hospital medicine: five opportunities for improved healthcare value. *J Hosp Med* 2013;8:486-92.
- 4) Herzig SJ, Vaughn BP, Howell MD, Ngo LH, Marcantonio ER. Acid-suppressive medication use and the risk for nosocomial gastrointestinal tract bleeding. *Arch Intern Med* 2011;171:991-7.
- 5) MacLaren R, Dionne JC, Granholm A, et al. Society of Critical Care Medicine and American Society of Health-System Pharmacists guideline for the prevention of stress-related gastrointestinal bleeding in critically ill adults. *Crit Care Med* 2024;52:e421-e430.
- 6) Granholm A, Zeng L, Dionne JC, et al. Predictors of gastrointestinal bleeding in adult ICU patients: a systematic review and meta-analysis. *Intensive Care Med* 2019;45:1347-59.
- 7) 日本集中治療医学会, 日本救急医学会. 日本版敗血症診療ガイドライン2024 (J-SSCG 2024) . 2024.
- 8) Ye Z, Reintam Blaser A, Lytvy L, et al. Gastrointestinal bleeding prophylaxis for critically ill patients: a clinical practice guideline. *BMJ* 2020;368:l6722.

10 文献

- 9) Krag M, Marker S, Perner A, et al. Pantoprazole in patients at risk for gastrointestinal bleeding in the ICU. *N Engl J Med* 2018;379:2199-208.
- 10) PEPTIC Investigators. Effect of stress ulcer prophylaxis with proton pump inhibitors vs histamine-2 receptor blockers on in-hospital mortality among ICU patients receiving invasive mechanical ventilation: the PEPTIC randomized clinical trial. *JAMA* 2020;323:616-26.
- 11) Cook D, Deane A, Lauzier F, et al. Stress ulcer prophylaxis during invasive mechanical ventilation. *N Engl J Med* 2024;391:9-20.
- 12) 各薬剤の電子添文（KEGG MEDICUSで2026年9～10月に確認）。
- 13) 2023 American Geriatrics Society Beers Criteria Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc* 2023;71:2052-81.
- 14) Farley KJ, Barned KL, Crozier TM. Inappropriate continuation of stress ulcer prophylaxis beyond the intensive care setting. *Crit Care Resusc* 2013;15:147-51.
- 15) Conn HO, Poynard T. Corticosteroids and peptic ulcer: meta-analysis of adverse events during steroid therapy. *J Intern Med* 1994;236:619-32.
- 16) 日本消化器病学会編. 消化性潰瘍診療ガイドライン2020（改訂第3版）. 南江堂；2020.
- 17) Piper JM, Ray WA, Daugherty JR, Griffin MR. Corticosteroid use and peptic ulcer disease: role of nonsteroidal anti-inflammatory drugs. *Ann Intern Med* 1991;114:735-40.
- 18) Narum S, Westergren T, Klemp M. Corticosteroids and risk of gastrointestinal bleeding: a systematic review and meta-analysis. *BMJ Open* 2014;4:e004587.

1-6 せん妄の予防

- 1) Gibb K, Seeley A, Quinn T, et al. The consistent burden in published estimates of delirium occurrence in medical inpatients over four decades: a systematic review and meta-analysis study. *Age Ageing* 2020;49:352-60.
- 2) Witlox J, Eurelings LS, de Jonghe JF, et al. Delirium in elderly patients and the risk of postdischarge mortality, institutionalization, and dementia: a meta-analysis. *JAMA* 2010;304:443-51.
- 3) 厚生労働省. 診療報酬点数表（令和8年度）A247-2 せん妄ハイリスク患者ケア加算，別紙様式7の3.
- 4) National Institute for Health and Care Excellence. Delirium: prevention, diagnosis and management in hospital and long-term care (CG103). Last reviewed 2023.
- 5) 日本集中治療医学会J-PADガイドライン作成委員会. 日本版・集中治療室における成人重症患者に対する痛み・不穏・せん妄管理のための臨床ガイドライン. *日集中医誌* 2014;21:539-79.
- 6) Inouye SK, Charpentier PA. Precipitating factors for delirium in hospitalized elderly persons: predictive model and interrelationship with baseline vulnerability. *JAMA* 1996;275:852-7.
- 7) Inouye SK, Bogardus ST Jr, Charpentier PA, et al. A multicomponent intervention to prevent delirium in hospitalized older patients. *N Engl J Med* 1999;340:669-76.
- 8) Hshieh TT, Yue J, Oh E, et al. Effectiveness of multicomponent nonpharmacological delirium interventions: a meta-analysis. *JAMA Intern Med* 2015;175:512-20.
- 9) Ogawa A, Okumura Y, Fujisawa D, et al. Quality of care in hospitalized cancer patients before and after implementation of a systematic prevention program for delirium: the DELTA exploratory trial. *Support Care Cancer* 2019;27:557-65.
- 10) van den Boogaard M, Slooter AJC, Brüggemann RJM, et al. Effect of haloperidol on survival among critically ill adults with a high risk of delirium: the REDUCE randomized clinical trial. *JAMA* 2018;319:680-90.
- 11) Oh ES, Needham DM, Nikoos R, et al. Antipsychotics for preventing delirium in hospitalized adults: a systematic review. *Ann Intern Med* 2019;171:474-84.

- 12) Hatta K, Kishi Y, Wada K, et al. Preventive effects of ramelteon on delirium: a randomized placebo-controlled trial. *JAMA Psychiatry* 2014;71:397-403.
- 13) Hatta K, Kishi Y, Wada K, et al. Suvorexant for reduction of delirium in older adults after hospitalization: a randomized clinical trial. *JAMA Netw Open* 2024;7:e2427691.
- 14) 日本サイコオンコロジー学会, 日本がんサポートティブケア学会編. がん患者におけるせん妄ガイドライン 2025年版. 金原出版; 2025.
- 15) 2023 American Geriatrics Society Beers Criteria Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc* 2023;71:2052-81.
- 16) Tiegies Z, MacLulich AMJ, Anand A, et al. Diagnostic accuracy of the 4AT for delirium detection in older adults: systematic review and meta-analysis. *Age Ageing* 2021;50:733-43.
- 17) Inouye SK, van Dyck CH, Alessi CA, et al. Clarifying confusion: the confusion assessment method. A new method for detection of delirium. *Ann Intern Med* 1990;113:941-8.
- 18) Ely EW, Inouye SK, Bernard GR, et al. Delirium in mechanically ventilated patients: validity and reliability of the confusion assessment method for the intensive care unit (CAM-ICU). *JAMA* 2001;286:2703-10.
- 19) Bergeron N, Dubois MJ, Dumont M, et al. Intensive Care Delirium Screening Checklist: evaluation of a new screening tool. *Intensive Care Med* 2001;27:859-64.
- 20) Najafi N, Robinson A, Pletcher MJ, Patel S. Effectiveness of an analytics-based intervention for reducing sleep interruption in hospitalized patients: a randomized clinical trial. *JAMA Intern Med* 2022;182:172-7.

1-7 転倒・転落の予防

- 1) 日本病院会. QIプロジェクト2023 結果報告. 2024.
- 2) 日本医療安全調査機構 医療事故調査・支援センター. 医療事故の再発防止に向けた提言 第9号 入院中に発生した転倒・転落による頭部外傷に係る死亡事例の分析. 2019.
- 3) 日本医師会. 転倒・転落防止マニュアル (アセスメントスコアシート) .
- 4) National Institute for Health and Care Excellence. Falls: assessment and prevention in older people and in people 50 and over at higher risk (NG249). 2025.
- 5) Montero-Odasso M, van der Velde N, Martin FC, et al. World guidelines for falls prevention and management for older adults: a global initiative. *Age Ageing* 2022;51:afac205.
- 6) Freeman R, Wieling W, Axelrod FB, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clin Auton Res* 2011;21:69-72.
- 7) Hill AM, McPhail SM, Waldron N, et al. Fall rates in hospital rehabilitation units after individualised patient and staff education programmes: a pragmatic, stepped-wedge, cluster-randomised controlled trial. *Lancet* 2015;385:2592-9.
- 8) McLennan C, Dyer SM, Kwok WS, et al. Interventions for preventing falls in older people in hospitals. *Cochrane Database Syst Rev* 2026;5:CD016065.
- 9) Shorr RI, Chandler AM, Mion LC, et al. Effects of an intervention to increase bed alarm use to prevent falls in hospitalized patients: a cluster randomized trial. *Ann Intern Med* 2012;157:692-9.
- 10) 日本医療機能評価機構. 医療事故情報収集等事業 第71回報告書 分析テーマ「離床センサーが電源の入れ忘れや使用方法の間違いにより作動しなかった事例」. 2022.
- 11) Köpke S, Mühlhauser I, Gerlach A, et al. Effect of a guideline-based multicomponent intervention on use of physical restraints in nursing homes: a randomized controlled trial. *JAMA* 2012;307:2177-84.

12 文献

- 12) Hitcho EB, Krauss MJ, Birge S, et al. Characteristics and circumstances of falls in a hospital setting: a prospective analysis. *J Gen Intern Med* 2004;19:732-9.
- 13) Seppala LJ, Petrovic M, Ryg J, et al. STOPPFall (Screening Tool of Older Persons Prescriptions in older adults with high fall risk): a Delphi study by the EuGMS Task and Finish Group on Fall-Risk-Increasing Drugs. *Age Ageing* 2021;50:1189-99.
- 14) 日本老年医学会, 日本医療研究開発機構研究費・高齢者の薬物治療の安全性に関する研究研究班編. 高齢者の安全な薬物療法ガイドライン2015. メジカルビュー社; 2015.
- 15) Treves N, Perlman A, Kolenberg Geron L, et al. Z-drugs and risk for falls and fractures in older adults: a systematic review and meta-analysis. *Age Ageing* 2018;47:201-8.

1-8 褥瘡の予防

- 1) 厚生労働省. 基本診療料の施設基準等及びその届出に関する手続きの取扱いについて（令和8年3月5日保医発0305第7号）. 2026.
- 2) Bergstrom N, Braden BJ, Laguzza A, Holman V. The Braden Scale for predicting pressure sore risk. *Nurs Res* 1987;36:205-10.
- 3) 創傷・褥瘡・熱傷ガイドライン策定委員会（褥瘡グループ）. 創傷・褥瘡・熱傷ガイドライン（2023）—2 褥瘡診療ガイドライン（第3版）. 日皮会誌 2023;133:2735-97.
- 4) Moore ZE, Patton D. Risk assessment tools for the prevention of pressure ulcers. *Cochrane Database Syst Rev* 2019;1:CD006471.
- 5) Nixon J, Smith IL, Brown S, et al. Pressure relieving support surfaces for pressure ulcer prevention (PRESSURE 2): clinical and health economic results of a randomised controlled trial. *EclinicalMedicine* 2019;14:42-52.
- 6) Latimer SL, Chaboyer WP, Probst S, et al. Repositioning for pressure injury prevention in adults. *Cochrane Database Syst Rev* 2026;6:CD009958.
- 7) Bergstrom N, Horn SD, Rapp MP, et al. Turning for Ulcer Reduction: a multisite randomized clinical trial in nursing homes. *J Am Geriatr Soc* 2013;61:1705-13.
- 8) Defloor T, De Bacquer D, Grypdonck MH. The effect of various combinations of turning and pressure reducing devices on the incidence of pressure ulcers. *Int J Nurs Stud* 2005;42:37-46.
- 9) Patton D, Moore ZE, Boland F, et al. Dressings and topical agents for preventing pressure ulcers. *Cochrane Database Syst Rev* 2024;12:CD009362.
- 10) Langer G, Wan CS, Fink A, et al. Nutritional interventions for preventing and treating pressure ulcers. *Cochrane Database Syst Rev* 2024;2:CD003216.
- 11) Cereda E, Klersy C, Serioli M, et al. A nutritional formula enriched with arginine, zinc, and antioxidants for the healing of pressure ulcers: a randomized trial. *Ann Intern Med* 2015;162:167-74.
- 12) 日本褥瘡学会編. ベストプラクティス 医療関連機器圧迫創傷の予防と管理. 日本褥瘡学会; 2016.
- 13) 日本褥瘡学会編. 改定DESIGN-R®2020コンセンサス・ドキュメント. 照林社; 2020.
- 14) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）.
- 15) Reddy M, Gill SS, Wu W, Kalkar SR, Rochon PA. Does this patient have an infection of a chronic wound? *JAMA* 2012;307:605-11.

1-9 嚥下評価と誤嚥の予防

- 1) 日本呼吸器学会. 成人肺炎診療ガイドライン2024. 2024.
- 2) Maeda K, Koga T, Akagi J. Tentative nil per os leads to poor outcomes in older adults with aspiration pneumonia. *Clin Nutr* 2016;35:1147-52.
- 3) Skoretz SA, Flowers HL, Martino R. The incidence of dysphagia following endotracheal intubation: a systematic review. *Chest* 2010;137:665-73.

- 4) 日本脳卒中学会 脳卒中ガイドライン委員会（改訂2025）．脳卒中治療ガイドライン2021〔改訂2025〕（改訂項目の公開版）．2025.
- 5) 日本摂食嚥下リハビリテーション学会医療検討委員会．摂食嚥下障害の評価2019．2019.
- 6) 日本耳鼻咽喉科頭頸部外科学会編．嚥下障害診療ガイドライン2024年版．金原出版；2024.
- 7) Brodsky MB, Suiter DM, González-Fernández M, et al. Screening accuracy for aspiration using bedside water swallow tests: a systematic review and meta-analysis. *Chest* 2016;150:148-63.
- 8) 日本摂食嚥下リハビリテーション学会嚥下調整食委員会．日本摂食嚥下リハビリテーション学会嚥下調整食分類2021．日摂食嚥下リハ会誌 2021;25:135-49.
- 9) Robbins J, Gensler G, Hind J, et al. Comparison of 2 interventions for liquid aspiration on pneumonia incidence: a randomized trial. *Ann Intern Med* 2008;148:509-18.
- 10) 日本集中治療医学会日本版重症患者の栄養療法ガイドライン検討委員会．日本版重症患者の栄養療法ガイドライン2024．日集中医誌 2025;32(Suppl 3).
- 11) Drakulovic MB, Torres A, Bauer TT, et al. Supine body position as a risk factor for nosocomial pneumonia in mechanically ventilated patients: a randomised trial. *Lancet* 1999;354:1851-8.
- 12) Yoneyama T, Yoshida M, Ohru T, et al. Oral care reduces pneumonia in older patients in nursing homes. *J Am Geriatr Soc* 2002;50:430-3.
- 13) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）．

1-10 院内感染対策

- 1) Siegel JD, Rhinehart E, Jackson M, Chiarello L; Healthcare Infection Control Practices Advisory Committee. 2007 Guideline for isolation precautions: preventing transmission of infectious agents in health care settings. *Am J Infect Control* 2007;35(10 Suppl 2):S65-164.（付録Aは2026年9月更新版で確認）
- 2) 厚生労働省医政局地域医療計画課長．医療機関における院内感染対策について（医政地発1219第1号）．2014.
- 3) 厚生労働省インフルエンザ等新興再興感染症研究事業．結核院内（施設内）感染対策の手引き 平成26年版．2014.
- 4) World Health Organization. WHO guidelines on hand hygiene in health care. Geneva: WHO; 2009.
- 5) 日本感染症学会，日本呼吸器学会，日本化学療法学会，日本臨床微生物学会，日本環境感染学会．5学会による新型コロナウイルス感染症 診療の指針2025．2025.
- 6) 国立感染症研究所 感染症疫学センター．医療機関での麻疹対応ガイドライン 第七版．2018.
- 7) Centers for Disease Control and Prevention. Infection prevention and control strategies for seasonal influenza in healthcare settings. 2025.
- 8) MacCannell T, Umscheid CA, Agarwal RK, et al. Guideline for the prevention and control of norovirus gastroenteritis outbreaks in healthcare settings. *Infect Control Hosp Epidemiol* 2011;32:939-69.
- 9) 日本環境感染学会 CDI対策ガイドライン策定委員会編．*Clostridioides difficile*感染対策ガイド 改訂版．環境感染誌 2026;41(Suppl III).
- 10) 日本皮膚科学会疥癬診療ガイドライン策定委員会．疥癬診療ガイドライン（第3版）．日皮会誌 2015;125:2023-48.
- 11) 日本呼吸器学会．成人肺炎診療ガイドライン2024．2024.
- 12) Zifodya JS, Kreniske JS, Schiller I, et al. Xpert Ultra versus Xpert MTB/RIF for pulmonary tuberculosis and rifampicin resistance in adults with presumptive pulmonary tuberculosis. *Cochrane Database Syst Rev* 2021;2:CD009593.
- 13) Harris AD, Pineles L, Belton B, et al. Universal glove and gown use and acquisition of antibiotic-resistant bacteria in the ICU: a randomized trial. *JAMA* 2013;310:1571-80.

14 文献

- 14) Maechler F, Schwab F, Hansen S, et al. Contact isolation versus standard precautions to decrease acquisition of extended-spectrum β -lactamase-producing Enterobacterales in non-critical care wards: a cluster-randomised crossover trial. *Lancet Infect Dis* 2020;20:575-84.
- 15) 日本感染症学会. インフルエンザ病院内感染対策の考え方について（高齢者施設を含めて）（日本感染症学会提言2012）. 2012.
- 16) タミフルカプセル75 電子添文（KEGG MEDICUSで2026年10月に確認）.
- 17) Centers for Disease Control and Prevention. Interim infection prevention and control recommendations for measles in healthcare settings. 2025.
- 18) 厚生労働省, 経済産業省. 新型コロナウイルス感染症により亡くなられた方及びその疑いがある方の処置、搬送、葬儀、火葬等に関するガイドライン（令和5年1月6日改正）. 2023.
- 19) Radonovich LJ Jr, Simberkoff MS, Bessesen MT, et al. N95 respirators vs medical masks for preventing influenza among health care personnel: a randomized clinical trial. *JAMA* 2019;322:824-33.
- 20) Loeb M, Dafoe N, Mahony J, et al. Surgical mask vs N95 respirator for preventing influenza among health care workers: a randomized trial. *JAMA* 2009;302:1865-71.
- 21) Loeb M, Bartholomew A, Hashmi M, et al. Medical masks versus N95 respirators for preventing COVID-19 among health care workers: a randomized trial. *Ann Intern Med* 2022;175:1629-38.

1-11 身体拘束

- 1) 厚生労働省老健局. 介護施設・事業所等で働く方々への身体拘束廃止・防止の手引き. 2025.
- 2) Inouye SK, Charpentier PA. Precipitating factors for delirium in hospitalized elderly persons: predictive model and interrelationship with baseline vulnerability. *JAMA* 1996;275:852-7.
- 3) 最高裁判所第三小法廷 平成22年1月26日判決（平成20年(受)第2029号）. 民集64巻1号219頁.
- 4) 厚生労働省. 令和6年度診療報酬改定の概要【重点分野Ⅱ（認知症、精神医療、難病患者に対する医療）】. 2024.
- 5) 厚生労働省. 令和8年度診療報酬改定の概要 6. 入院（共通事項）. 2026.
- 6) 日本看護協会臨床倫理ガイドライン検討委員会. 身体拘束予防ガイドライン. 2015.
- 7) Sonnevile R, Couffignal C, Sigaud F, et al. Restrictive vs liberal physical restraint strategies in critically ill patients: the R2D2-ICU randomized clinical trial. *JAMA* 2026;335:1232-42.
- 8) Köpke S, Mühlhauser I, Gerlach A, et al. Effect of a guideline-based multicomponent intervention on use of physical restraints in nursing homes: a randomized controlled trial. *JAMA* 2012;307:2177-84.
- 9) 厚生省. 精神保健及び精神障害者福祉に関する法律第三十六条第三項の規定に基づき厚生労働大臣が定める行動の制限（昭和63年厚生省告示第129号）.
- 10) 厚生省. 精神保健及び精神障害者福祉に関する法律第三十七条第一項の規定に基づき厚生労働大臣が定める基準（昭和63年厚生省告示第130号、最終改正 令和5年厚生労働省告示第117号）.

1-12 コードステータスと意思決定支援

- 1) 日本集中治療医学会. Do Not Attempt Resuscitation（DNAR）指示のあり方についての勧告. *日集中医誌* 2017;24:208-9.
- 2) 厚生労働省. 人生の最終段階における医療・ケアの決定プロセスに関するガイドライン（解説編を含む）. 2018.
- 3) 日本集中治療医学会, 日本救急医学会, 日本循環器学会, 日本緩和医療学会. 救急・集中治療における生命維持治療の終了／差し控えに関する4学会合同ガイドライン. 2026.
- 4) 厚生労働省. 令和6年度診療報酬改定の概要. 2024.

- 5) Ehlenbach WJ, Barnato AE, Curtis JR, et al. Epidemiologic study of in-hospital cardiopulmonary resuscitation in the elderly. *N Engl J Med* 2009;361:22-31.
- 6) Detering KM, Hancock AD, Reade MC, Silvester W. The impact of advance care planning on end of life care in elderly patients: randomised controlled trial. *BMJ* 2010;340:c1345.
- 7) 日本透析医学会. 透析の開始と継続に関する意思決定プロセスについての提言. *透析会誌* 2020;53:173-217.
- 8) Lautrette A, Darmon M, Megarbane B, et al. A communication strategy and brochure for relatives of patients dying in the ICU. *N Engl J Med* 2007;356:469-78.
- 9) 厚生労働行政推進調査事業「医療現場における成年後見制度への理解及び病院が身元保証人に求める役割等の実態把握に関する研究」班. 身寄りがいない人の入院及び医療に係る意思決定が困難な人への支援に関するガイドライン. 2019.
- 10) 日本臨床救急医学会. 人生の最終段階にある傷病者の意思に沿った救急現場での心肺蘇生等のあり方に関する提言. 2017.
- 11) 日本救急医学会, 日本臨床救急医学会, 日本老年医学会, ほか. 高齢者救急問題の現状とその対応策についての提言2024. *日救急医学会誌* 2025;36:63-81.
- 12) 日本臨床倫理学会. 日本版POLST (DNAR指示を含む) 作成指針. 2015.
- 13) Murphy DJ, Burrows D, Santilli S, et al. The influence of the probability of survival on patients' preferences regarding cardiopulmonary resuscitation. *N Engl J Med* 1994;330:545-9.
- 14) El-Jawahri A, Mitchell SL, Paasche-Orlow MK, et al. A randomized controlled trial of a CPR and intubation video decision support tool for hospitalized patients. *J Gen Intern Med* 2015;30:1071-80.
- 15) Diem SJ, Lantos JD, Tulskey JA. Cardiopulmonary resuscitation on television: miracles and misinformation. *N Engl J Med* 1996;334:1578-82.

1-13 報告・相談・記録・病状説明

- 1) Starmer AJ, Spector ND, Srivastava R, et al. Changes in medical errors after implementation of a handoff program. *N Engl J Med* 2014;371:1803-12.
- 2) Müller M, Jürgens J, Redaelli M, Klingberg K, Hautz WE, Stock S. Impact of the communication and patient hand-off tool SBAR on patient safety: a systematic review. *BMJ Open* 2018;8:e022202.
- 3) Marshall S, Harrison J, Flanagan B. The teaching of a structured tool improves the clarity and content of interprofessional clinical communication. *Qual Saf Health Care* 2009;18:137-40.
- 4) Kessler CS, Afshar Y, Sardar G, Yudkowsky R, Ankel F, Schwartz A. A prospective, randomized, controlled study demonstrating a novel, effective model of transfer of care between physicians: the 5 Cs of consultation. *Acad Emerg Med* 2012;19:968-74.
- 5) Horwitz LI, Moin T, Krumholz HM, Wang L, Bradley EH. Consequences of inadequate sign-out for patient care. *Arch Intern Med* 2008;168:1755-60.
- 6) 医師法（昭和23年法律第201号）第24条.
- 7) Weed LL. Medical records that guide and teach. *N Engl J Med* 1968;278:593-600.
- 8) Wang MD, Khanna R, Najafi N. Characterizing the source of text in electronic health record progress notes. *JAMA Intern Med* 2017;177:1212-3.
- 9) 厚生労働省医政局長. 診療情報の提供等に関する指針の策定について（平成15年9月12日 医政発第0912001号）.
- 10) 厚生労働省. 医師臨床研修指導ガイドライン 2023年度版. 2023.
- 11) 厚生労働省保険局医療課長. 基本診療料の施設基準等及びその届出に関する手続きの取扱いについて（令和8年度）. 診療録管理体制加算の施設基準.

- 12) Kripalani S, LeFevre F, Phillips CO, Williams MV, Basaviah P, Baker DW. Deficits in communication and information transfer between hospital-based and primary care physicians: implications for patient safety and continuity of care. *JAMA* 2007;297:831-41.
- 13) Baile WF, Buckman R, Lenzi R, Glober G, Beale EA, Kudelka AP. SPIKES—a six-step protocol for delivering bad news: application to the patient with cancer. *Oncologist* 2000;5:302-11.
- 14) Fujimori M, Shirai Y, Asai M, Kubota K, Katsumata N, Uchitomi Y. Effect of communication skills training program for oncologists based on patient preferences for communication when receiving bad news: a randomized controlled trial. *J Clin Oncol* 2014;32:2166-72.

第2部 治療の基本

2-1 輸液

- 1) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. 日集中医誌 2024;31:S1165-313.
- 2) Monnet X, Marik P, Teboul JL. Passive leg raising for predicting fluid responsiveness: a systematic review and meta-analysis. *Intensive Care Med* 2016;42:1935-47.
- 3) Semler MW, Self WH, Wanderer JP, et al. Balanced crystalloids versus saline in critically ill adults. *N Engl J Med* 2018;378:829-39.
- 4) Zampieri FG, Machado FR, Biondi RS, et al. Effect of intravenous fluid treatment with a balanced solution vs 0.9% saline solution on mortality in critically ill patients: the BaSICS randomized clinical trial. *JAMA* 2021;326:818-29.
- 5) Finfer S, Micallef S, Hammond N, et al. Balanced multielectrolyte solution versus saline in critically ill adults. *N Engl J Med* 2022;386:815-26.
- 6) Meyhoff TS, Hjortrup PB, Wetterslev J, et al. Restriction of intravenous fluid in ICU patients with septic shock. *N Engl J Med* 2022;386:2459-70.
- 7) 各薬剤の電子添文（ラクテック注、ヴィーンF輸液、ピカーボン輸液、ソリタ-T1号輸液、ソリタ-T3号輸液、ロセフィン静注用。2026年10月に確認）。
- 8) National Institute for Health and Care Excellence. Intravenous fluid therapy in adults in hospital (CG174). 2013 (updated 2017).
- 9) O'Malley CMN, Frumento RJ, Hardy MA, et al. A randomized, double-blind comparison of lactated Ringer's solution and 0.9% NaCl during renal transplantation. *Anesth Analg* 2005;100:1518-24.
- 10) McGee S, Abernethy WB 3rd, Simel DL. The rational clinical examination. Is this patient hypovolemic? *JAMA* 1999;281:1022-9.
- 11) Van Regenmortel N, Hendrickx S, Roelant E, et al. 154 compared to 54 mmol per liter of sodium in intravenous maintenance fluid therapy for adult patients undergoing major thoracic surgery (TOPMAST): a single-center randomized controlled double-blind trial. *Intensive Care Med* 2019;45:1422-32.
- 12) McNab S, Duke T, South M, et al. 140 mmol/L of sodium versus 77 mmol/L of sodium in maintenance intravenous fluid therapy for children in hospital (PIMS): a randomised controlled double-blind trial. *Lancet* 2015;385:1190-7.

2-2 栄養管理

- 1) Schuetz P, Fehr R, Baechli V, et al. Individualised nutritional support in medical inpatients at nutritional risk: a randomised clinical trial. *Lancet* 2019;393:2312-21.
- 2) 厚生労働省. 令和6年度診療報酬改定の概要. 2024.

- 3) Cederholm T, Jensen GL, Correia MITD, et al. GLIM criteria for the diagnosis of malnutrition—a consensus report from the global clinical nutrition community. *Clin Nutr* 2019;38:1-9.
- 4) Evans DC, Corkins MR, Malone A, et al. The use of visceral proteins as nutrition markers: an ASPEN position paper. *Nutr Clin Pract* 2021;36:22-8.
- 5) Gomes F, Schuetz P, Bounoure L, et al. ESPEN guidelines on nutritional support for polymorbid internal medicine patients. *Clin Nutr* 2018;37:336-53.
- 6) 日本静脈経腸栄養学会編. 静脈経腸栄養ガイドライン 第3版. 照林社; 2013.
- 7) Japanese Critical Care Nutrition Guideline Committee. The Japanese Critical Care Nutrition Guideline 2024. *J Intensive Care* 2025;13:18.
- 8) Casaer MP, Mesotten D, Hermans G, et al. Early versus late parenteral nutrition in critically ill adults. *N Engl J Med* 2011;365:506-17.
- 9) 各薬剤の電子添文（ビーフリード輸液、イントラリポス輸液、エルネオパNF輸液、メタボリン注射液・チアミン塩化物塩酸塩注、ドグマチール錠。2026年10月に確認）。
- 10) National Institute for Health and Care Excellence. Nutrition support for adults: oral nutrition support, enteral tube feeding and parenteral nutrition (CG32). 2006 (updated 2017).
- 11) da Silva JSV, Seres DS, Sabino K, et al. ASPEN consensus recommendations for refeeding syndrome. *Nutr Clin Pract* 2020;35:178-95.
- 12) 日本医療研究開発機構（AMED）障害者対策総合研究開発事業「摂食障害の治療支援ネットワークの指針と簡易治療プログラムの開発」神経性やせ症の簡易治療プログラム作成ワーキンググループ. 神経性やせ症（AN）初期診療の手引き 第1版. 2019.
- 13) Harper CG, Giles M, Finlay-Jones R. Clinical signs in the Wernicke-Korsakoff complex: a retrospective analysis of 131 cases diagnosed at necropsy. *J Neurol Neurosurg Psychiatry* 1986;49:341-5.
- 14) Volkert D, Beck AM, Cederholm T, et al. ESPEN practical guideline: clinical nutrition and hydration in geriatrics. *Clin Nutr* 2022;41:958-89.
- 15) Gay L, Sheth S, Horstman MJ. Things We Do for No Reason: prescribing appetite stimulants to hospitalized older adults with unintentional weight loss. *J Hosp Med* 2022;17:389-92.
- 16) Harvey SE, Parrott F, Harrison DA, et al. Trial of the route of early nutritional support in critically ill adults. *N Engl J Med* 2014;371:1673-84.
- 17) Reignier J, Boisramé-Helms J, Brisard L, et al. Enteral versus parenteral early nutrition in ventilated adults with shock: a randomised, controlled, multicentre, open-label, parallel-group study (NUTRIREA-2). *Lancet* 2018;391:133-43.

2-3 酸素療法と呼吸補助

- 1) Chu DK, Kim LH, Young PJ, et al. Mortality and morbidity in acutely ill adults treated with liberal versus conservative oxygen therapy (IOTA): a systematic review and meta-analysis. *Lancet* 2018;391:1693-705.
- 2) Hofmann R, James SK, Jernberg T, et al. Oxygen therapy in suspected acute myocardial infarction. *N Engl J Med* 2017;377:1240-9.
- 3) O'Driscoll BR, Howard LS, Earis J, et al. BTS guideline for oxygen use in adults in healthcare and emergency settings. *Thorax* 2017;72(Suppl 1):ii1-ii90.
- 4) 日本呼吸ケア・リハビリテーション学会 酸素療法マニュアル作成委員会, 日本呼吸器学会 肺生理専門委員会編. 酸素療法マニュアル. メディカルレビュー社; 2017.
- 5) Siemieniuk RAC, Chu DK, Kim LH, et al. Oxygen therapy for acutely ill medical patients: a clinical practice guideline. *BMJ* 2018;363:k4169.
- 6) 日本集中治療医学会, 日本呼吸器学会, 日本呼吸療法医学会. ARDS診療ガイドライン2021. 日集中医誌 2022;29:295-332.

- 7) Oczkowski S, Ergan B, Bos L, et al. ERS clinical practice guidelines: high-flow nasal cannula in acute respiratory failure. *Eur Respir J* 2022;59:2101574.
- 8) Frat JP, Quenot JP, Guitton C, et al. High-flow or standard oxygen in acute hypoxemic respiratory failure. *N Engl J Med* 2026;394:2095-106.
- 9) Frat JP, Thille AW, Mercat A, et al. High-flow oxygen through nasal cannula in acute hypoxemic respiratory failure. *N Engl J Med* 2015;372:2185-96.
- 10) Roca O, Caralt B, Messika J, et al. An index combining respiratory rate and oxygenation to predict outcome of nasal high-flow therapy. *Am J Respir Crit Care Med* 2019;199:1368-76.
- 11) Kang BJ, Koh Y, Lim CM, et al. Failure of high-flow nasal cannula therapy may delay intubation and increase mortality. *Intensive Care Med* 2015;41:623-32.
- 12) 日本呼吸器学会NPPVガイドライン作成委員会編. NPPV（非侵襲的陽圧換気療法）ガイドライン 改訂第2版. 南江堂；2015.
- 13) Davidson AC, Banham S, Elliott M, et al. BTS/ICS guideline for the ventilatory management of acute hypercapnic respiratory failure in adults. *Thorax* 2016;71(Suppl 2):ii1-35.

2-4 入院中の血糖管理

- 1) NICE-SUGAR Study Investigators. Intensive versus conventional glucose control in critically ill patients. *N Engl J Med* 2009;360:1283-97.
- 2) 日本糖尿病学会編. 糖尿病診療ガイドライン2024. 南江堂；2024.
- 3) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. 日集中医誌 2024;31:S1165-313.
- 4) American Diabetes Association Professional Practice Committee. 16. Diabetes care in the hospital: Standards of Care in Diabetes—2026. *Diabetes Care* 2026;49(Suppl 1):S339-55.
- 5) Umpierrez GE, Smiley D, Zisman A, et al. Randomized study of basal-bolus insulin therapy in the inpatient management of patients with type 2 diabetes (RABBIT 2 trial). *Diabetes Care* 2007;30:2181-6.
- 6) Umpierrez GE, Smiley D, Hermayer K, et al. Randomized study comparing a basal-bolus with a basal plus correction insulin regimen for the hospital management of medical and surgical patients with type 2 diabetes: basal plus trial. *Diabetes Care* 2013;36:2169-74.
- 7) Pasquel FJ, Gianchandani R, Rubin DJ, et al. Efficacy of sitagliptin for the hospital management of general medicine and surgery patients with type 2 diabetes (Sita-Hospital): a multicentre, prospective, open-label, non-inferiority randomised trial. *Lancet Diabetes Endocrinol* 2017;5:125-33.
- 8) Brooks D, Schulman-Rosenbaum R, Griff M, Lester J, Low Wang CC. Glucocorticoid-induced hyperglycemia including dexamethasone-associated hyperglycemia in COVID-19 infection: a systematic review. *Endocr Pract* 2022;28:1166-77.
- 9) Bellido V, Suarez L, Rodriguez MG, et al. Comparison of basal-bolus and premixed insulin regimens in hospitalized patients with type 2 diabetes. *Diabetes Care* 2015;38:2211-6.
- 10) 日本医療機能評価機構. 医療安全情報 No.131 インスリン単位の誤解（第2報）. 2017.

2-5 抗菌薬（総論と主要薬剤）

- 1) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. 日集中医誌 2024;31:S1165-313.
- 2) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認。キュビシン、塩酸バンコマイシン点滴静注用、フラジール内服錠、アネメトロ点滴静注液は2026年10月に再確認）。
- 3) 日本呼吸器学会成人肺炎診療ガイドライン2024作成委員会編. 成人肺炎診療ガイドライン2024. 日本呼吸器学会；2024（参考資料：代表的な抗菌薬名と用法・用量）。

- 4) 日本神経学会監修. 重症筋無力症／ランバート・イートン筋無力症候群診療ガイドライン2022. 南江堂；2022.
- 5) 日本化学療法学会, 日本TDM学会 抗菌薬TDMガイドライン作成委員会. 抗菌薬TDM臨床実践ガイドライン2022. 日化療会誌 2022;70:1-72.
- 6) 厚生労働省健康・生活衛生局感染症対策部感染症対策課編. 抗微生物薬適正使用の手引き 第四版 薬剤耐性菌感染症の抗菌薬適正使用編. 2026.
- 7) European Committee on Antimicrobial Susceptibility Testing (EUCAST). Expected resistant phenotypes. Version 1.2. 2023.
- 8) Daneman N, Rishu A, Pinto R, et al. Antibiotic treatment for 7 versus 14 days in patients with bloodstream infections. *N Engl J Med* 2025;392:1065-78.
- 9) Shenoy ES, Macy E, Rowe T, Blumenthal KG. Evaluation and management of penicillin allergy: a review. *JAMA* 2019;321:188-99.
- 10) Trubiano JA, Vogrin S, Chua KYL, et al. Development and validation of a penicillin allergy clinical decision rule. *JAMA Intern Med* 2020;180:745-52.
- 11) Copaesu AM, Vogrin S, James F, et al. Efficacy of a clinical decision rule to enable direct oral challenge in patients with low-risk penicillin allergy: the PALACE randomized clinical trial. *JAMA Intern Med* 2023;183:944-52.
- 12) Dark P, Hossain A, McAuley DF, et al. Biomarker-guided antibiotic duration for hospitalized patients with suspected sepsis: the ADAPT-Sepsis randomized clinical trial. *JAMA* 2025;333:682-93.
- 13) Li HK, Rombach I, Zambellas R, et al. Oral versus intravenous antibiotics for bone and joint infection. *N Engl J Med* 2019;380:425-36.
- 14) Iversen K, Ihlemann N, Gill SU, et al. Partial oral versus intravenous antibiotic treatment of endocarditis. *N Engl J Med* 2019;380:415-24.
- 15) Tamma PD, Conley AT, Cosgrove SE, et al. Association of 30-day mortality with oral step-down vs continued intravenous therapy in patients hospitalized with Enterobacteriaceae bacteremia. *JAMA Intern Med* 2019;179:316-23.

2-6 鎮痛薬

- 1) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認。ナロキソン、アセリオは2026年10月に再確認、タリージェ、ソセゴン、トラムセット、オキファストは2026年10月に確認）。
- 2) Tokoro A, Imai H, Fumita S, et al. Incidence of opioid-induced constipation in Japanese patients with cancer pain: a prospective observational cohort study. *Cancer Med* 2019;8:4883-91.
- 3) 日本緩和医療学会ガイドライン統括委員会編. がん疼痛の薬物療法に関するガイドライン2020年版. 金原出版；2020.
- 4) Lapi F, Azoulay L, Yin H, Nessim SJ, Suissa S. Concurrent use of diuretics, angiotensin converting enzyme inhibitors, and angiotensin receptor blockers with non-steroidal anti-inflammatory drugs and risk of acute kidney injury: nested case-control study. *BMJ* 2013;346:e8525.
- 5) Hamano J, Higashibata T, Kessoku T, et al. Naldemedine for opioid-induced constipation in patients with cancer: a multicenter, double-blind, randomized, placebo-controlled trial. *J Clin Oncol* 2024;42:4206-17.
- 6) Ozaki A, Kessoku T, Tanaka K, et al. Effectiveness of naldemedine compared with magnesium oxide in preventing opioid-induced constipation: a randomized controlled trial. *Cancers (Basel)* 2022;14:2112.

2-7 睡眠薬・抗精神病薬

- 1) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認。レスリン、ザイボックスの相互作用、アキネトン注、クービピックは2026年10月に確認）。
- 2) 厚生労働科学研究・障害者対策総合研究事業「睡眠薬の適正使用及び減量・中止のための診療ガイドラインに関する研究班」，日本睡眠学会・睡眠薬使用ガイドライン作成ワーキンググループ編，睡眠薬の適正な使用と休薬のための診療ガイドライン，2013。
- 3) 社会保険診療報酬支払基金．審査情報提供事例（薬剤）165～168．2011。
- 4) van Harten PN, Hoek HW, Kahn RS. Acute dystonia induced by drug treatment. *BMJ* 1999;319:623-6.
- 5) Boyer EW, Shannon M. The serotonin syndrome. *N Engl J Med* 2005;352:1112-20.
- 6) Dunkley EJ, Isbister GK, Sibbritt D, Dawson AH, Whyte IM. The Hunter Serotonin Toxicity Criteria: simple and accurate diagnostic decision rules for serotonin toxicity. *QJM* 2003;96:635-42.
- 7) Chiew AL, Isbister GK. Management of serotonin syndrome (toxicity). *Br J Clin Pharmacol* 2025;91:654-61.

2-8 ステロイド

- 1) Beuschlein F, Else T, Bancos I, et al. European Society of Endocrinology and Endocrine Society joint clinical guideline: diagnosis and therapy of glucocorticoid-induced adrenal insufficiency. *J Clin Endocrinol Metab* 2024;109:1657-83.
- 2) 日本肝臓学会肝炎診療ガイドライン作成委員会編．B型肝炎治療ガイドライン（第5版）．2026。
- 3) 日本結核病学会予防委員会・治療委員会．潜在性結核感染症治療指針．結核 2013;88:497-512。
- 4) 各薬剤の電子添文（KEGG MEDICUSで2026年9月、ダイフェン配合錠、ソル・メドロール静注用、ストロメクトール錠、プレドニン錠は10月に確認）。
- 5) Ikuno H, Ishikawa T, Norose K. Status of strongyloidiasis in Japan, 2000-2017. *Am J Trop Med Hyg* 2020;103:727-34.
- 6) Buonfrate D, Requena-Mendez A, Angheben A, et al. Severe strongyloidiasis: a systematic review of case reports. *BMC Infect Dis* 2013;13:78.
- 7) Fragoulis GE, Nikiphorou E, Dey M, et al. 2022 EULAR recommendations for screening and prophylaxis of chronic and opportunistic infections in adults with autoimmune inflammatory rheumatic diseases. *Ann Rheum Dis* 2023;82:742-53.
- 8) 日本骨代謝学会グルココルチコイド誘発性骨粗鬆症の管理と治療のガイドライン作成委員会編．グルココルチコイド誘発性骨粗鬆症の管理と治療のガイドライン2023．南山堂；2023。
- 9) Tanaka Y, Soen S, Hirata S, et al. The 2023 Guidelines for the management and treatment of glucocorticoid-induced osteoporosis. *J Bone Miner Metab* 2024;42:143-54.
- 10) 日本版敗血症診療ガイドライン2024特別委員会．日本版敗血症診療ガイドライン2024．日集中医誌 2024;31:S1165-313.

2-9 輸血と輸血副反応

- 1) 日本輸血・細胞治療学会ガイドライン委員会．輸血療法実践ガイド．日本輸血細胞治療学会誌 2026;72:521-70。
- 2) Villanueva C, Colomo A, Bosch A, et al. Transfusion strategies for acute upper gastrointestinal bleeding. *N Engl J Med* 2013;368:11-21.
- 3) Carson JL, Brooks MM, Hébert PC, et al. Restrictive or liberal transfusion strategy in myocardial infarction and anemia. *N Engl J Med* 2023;389:2446-56.

- 4) 園木孝志, ほか. 科学的根拠に基づいた赤血球製剤の使用ガイドライン (改訂第3版). 日本輸血細胞治療学会誌 2024;70:579-96.
- 5) 厚生労働省医薬・生活衛生局血液対策課. 輸血療法の実施に関する指針 (平成17年9月、令和2年3月一部改正).
- 6) 岡崎仁, ほか. 科学的根拠に基づいた輸血有害事象対応ガイドライン (改訂第2版). 日本輸血細胞治療学会誌 2025;71:512-22.
- 7) Wiersum-Osselton JC, Whitaker B, Grey S, et al. Revised international surveillance case definition of transfusion-associated circulatory overload: a classification agreement validation study. *Lancet Haematol* 2019;6:e350-8.
- 8) Hébert PC, Wells G, Blajchman MA, et al. A multicenter, randomized, controlled clinical trial of transfusion requirements in critical care. *N Engl J Med* 1999;340:409-17.
- 9) Carson JL, Stanworth SJ, Dennis JA, et al. Transfusion thresholds and other strategies for guiding red blood cell transfusion. *Cochrane Database Syst Rev* 2025;10:CD002042.

2-10 抗凝固薬・抗血小板薬

- 1) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認。リクシアナの用法及び用量、ブラザキサの効能・効果と適用上の注意、エリキュースとイグザレルトの相互作用は2026年10月に確認)。
- 2) Okumura K, Akao M, Yoshida T, et al. Low-dose edoxaban in very elderly patients with atrial fibrillation. *N Engl J Med* 2020;383:1735-45.
- 3) 日本循環器学会, 日本不整脈心電学会. 2020年改訂版 不整脈薬物治療ガイドライン. 2020.
- 4) 日本循環器学会, 日本不整脈心電学会. 2024年JCS/JHRSガイドライン フォーカスアップデート版 不整脈治療. 2024.
- 5) Raschke RA, Reilly BM, Guidry JR, et al. The weight-based heparin dosing nomogram compared with a "standard care" nomogram: a randomized controlled trial. *Ann Intern Med* 1993;119:874-81.
- 6) 日本循環器学会, 日本肺高血圧・肺循環学会. 2025年改訂版 肺血栓塞栓症・深部静脈血栓症および肺高血圧症に関するガイドライン. 2025.
- 7) Sugimoto M, Kamada T, Ito M, et al. Evidence-based clinical practice guidelines for peptic ulcer disease 2026. *J Gastroenterol* 2026;61:1361-81.
- 8) Tunkel DE, Anne S, Payne SC, et al. Clinical practice guideline: nosebleed (epistaxis). *Otolaryngol Head Neck Surg* 2020;162(1_suppl):S1-38.
- 9) Kamada T, Satoh K, Itoh T, et al. Evidence-based clinical practice guidelines for peptic ulcer disease 2020. *J Gastroenterol* 2021;56:303-22.
- 10) Douketis JD, Spyropoulos AC, Kaatz S, et al. Perioperative bridging anticoagulation in patients with atrial fibrillation. *N Engl J Med* 2015;373:823-33.
- 11) 加藤元嗣, ほか. 抗血栓薬服用者に対する消化器内視鏡診療ガイドライン 直接経口抗凝固薬 (DOAC) を含めた抗凝固薬に関する追補2017. *Gastroenterol Endosc* 2017;59:1547-58.
- 12) 日本循環器学会, ほか. 2022年改訂版 非心臓手術における合併心疾患の評価と管理に関するガイドライン. 2022.
- 13) 藤本一眞, ほか. 抗血栓薬服用者に対する消化器内視鏡診療ガイドライン. *Gastroenterol Endosc* 2012;54:2075-102.
- 14) 日本腎臓学会 腎生検ガイドブック改訂委員会編. 腎生検ガイドブック2020. 東京医学社; 2020.
- 15) Ruff CT, Giugliano RP, Braunwald E, et al. Comparison of the efficacy and safety of new oral anticoagulants with warfarin in patients with atrial fibrillation: a meta-analysis of randomised trials. *Lancet* 2014;383:955-62.

- 16) Joosten LPT, van Doorn S, van de Ven PM, et al. Safety of switching from a vitamin K antagonist to a non-vitamin K antagonist oral anticoagulant in frail older patients with atrial fibrillation: results of the FRAIL-AF randomized controlled trial. *Circulation* 2024;149:279-89.
- 17) Connolly SJ, Karthikeyan G, Ntsekhe M, et al. Rivaroxaban in rheumatic heart disease-associated atrial fibrillation. *N Engl J Med* 2022;387:978-88.

2-11 腎機能と薬剤、造影剤

- 1) 日本腎臓学会, 日本医学放射線学会, 日本循環器学会編. 腎障害患者におけるヨード造影剤使用に関するガイドライン2018. *日腎会誌* 2019;61:933-1081.
- 2) Matsuo S, Imai E, Horio M, et al. Revised equations for estimated GFR from serum creatinine in Japan. *Am J Kidney Dis* 2009;53:982-92.
- 3) Cockcroft DW, Gault MH. Prediction of creatinine clearance from serum creatinine. *Nephron* 1976;16:31-41.
- 4) 日本腎臓学会編. エビデンスに基づくCKD診療ガイドライン2023. 東京医学社; 2023.
- 5) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認。メトグルコ、コルヒチンは2026年10月に再確認、イーケブラは2026年10月に確認)。
- 6) NSFとガドリニウム造影剤使用に関する合同委員会 (日本医学放射線学会, 日本腎臓学会). 腎障害患者におけるガドリニウム造影剤使用に関するガイドライン (第3版: 2024年5月20日改訂). 2024.
- 7) McDonald JS, McDonald RJ, Comin J, et al. Frequency of acute kidney injury following intravenous contrast medium administration: a systematic review and meta-analysis. *Radiology* 2013;267:119-28.
- 8) Davenport MS, Khalatbari S, Cohan RH, et al. Contrast material-induced nephrotoxicity and intravenous low-osmolality iodinated contrast material: risk stratification by using estimated glomerular filtration rate. *Radiology* 2013;268:719-28.
- 9) Nijssen EC, Rennenberg RJ, Nelemans PJ, et al. Prophylactic hydration to protect renal function from intravascular iodinated contrast material in patients at high risk of contrast-induced nephropathy (AMACING): a prospective, randomised, phase 3, controlled, open-label, non-inferiority trial. *Lancet* 2017;389:1312-22.

2-12 高齢者の処方と減薬

- 1) 厚生労働省. 高齢者の医薬品適正使用の指針 (総論編). 2018.
- 2) Blum MR, Sallevelt BTGM, Spinewine A, et al. Optimizing therapy to prevent avoidable hospital admissions in multimorbid older adults (OPERAM): cluster randomised controlled trial. *BMJ* 2021;374:n1585.
- 3) 日本老年医学会編. 高齢者の安全な薬物療法ガイドライン2025. メジカルビュー社; 2025.
- 4) 日本老年医学会, 日本医療研究開発機構研究費・高齢者の薬物治療の安全性に関する研究研究班編. 高齢者の安全な薬物療法ガイドライン2015. メジカルビュー社; 2015.
- 5) Mizokami F, Mizuno T, Taguchi R, et al. Development of the Japanese Anticholinergic Risk Scale: English translation of the Japanese article. *Geriatr Gerontol Int* 2025;25:5-13.
- 6) Yamada Y, Nakashima H, Nagae M, et al. Anticholinergic burden on admission assessed by the Japanese Anticholinergic Risk Scale and incident in-hospital delirium in older adults: a multicenter prospective cohort study. *Arch Gerontol Geriatr* 2026;148:106272.
- 7) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認。テルネリン、ロゼレム、テオドール、ジプレキサは2026年10月に確認)。

2-13 退院の準備

- 1) Forster AJ, Murff HJ, Peterson JF, Gandhi TK, Bates DW. The incidence and severity of adverse events affecting patients after discharge from the hospital. *Ann Intern Med* 2003;138:161-7.
- 2) Roy CL, Poon EG, Karson AS, et al. Patient safety concerns arising from test results that return after hospital discharge. *Ann Intern Med* 2005;143:121-8.
- 3) 厚生労働省. 診療報酬点数表（令和8年度）A308-3 地域包括ケア病棟入院料.
- 4) 厚生労働省老健局老人保健課長. 要介護認定における「認定調査票記入の手引き」、「主治医意見書記入の手引き」及び「特定疾病にかかる診断基準」について（平成21年9月30日 老老発0930第2号、令和8年4月改正）.
- 5) 厚生労働省地方厚生局. 訪問看護の基本的な仕組み（医療保険と介護保険の訪問看護、特別訪問看護指示書）. 2024.
- 6) 厚生労働省健康局がん・疾病対策課長. 手術前等に行われる肝炎ウイルス検査の結果を踏まえた受診・受療・フォローアップの推進等の医療機関管理者への協力依頼（健が発0309第1号）. 2023.

第3部 検査の読み方と異常値への対応

3-1 血算の読み方と貧血

- 1) Srygley FD, Gerardo CJ, Tran T, Fisher DA. Does this patient have a severe upper gastrointestinal bleed? *JAMA* 2012;307:1072-9.
- 2) Ebell MH, Call M, Shinholser J, Gardner J. Does this patient have infectious mononucleosis? The rational clinical examination systematic review. *JAMA* 2016;315:1502-9.
- 3) 厚生労働省. 重篤副作用疾患別対応マニュアル 薬剤性過敏症症候群.
- 4) 日本鉄バイオサイエンス学会編. 鉄欠乏性貧血の診療指針. フジメディカル出版；2024.
- 5) 日本透析医学会. 2015年版 日本透析医学会 慢性腎臓病患者における腎性貧血治療のガイドライン. 透析会誌 2016;49:89-158.
- 6) Garratty G. Drug-induced immune hemolytic anemia. *Hematology Am Soc Hematol Educ Program* 2009;2009:73-9.
- 7) 各薬剤の電子添文（KEGG MEDICUSで確認。フェロミア錠、フェジン静注、フェインジェクト、モノヴァー、メチコパール注射液・錠、フォリアミン錠は2026年10月に再確認）.
- 8) Stoffel NU, Cercamondi CI, Brittenham G, et al. Iron absorption from oral iron supplements given on consecutive versus alternate days and as single morning doses versus twice-daily split dosing in iron-depleted women: two open-label, randomised controlled trials. *Lancet Haematol* 2017;4:e524-33.
- 9) Snook J, Bhala N, Beales ILP, et al. British Society of Gastroenterology guidelines for the management of iron deficiency anaemia in adults. *Gut* 2021;70:2030-51.
- 10) Wolf M, Rubin J, Achebe M, et al. Effects of iron isomaltoside vs ferric carboxymaltose on hypophosphatemia in iron-deficiency anemia: two randomized clinical trials. *JAMA* 2020;323:432-43.
- 11) Savage DG, Ogundipe A, Allen RH, Stabler SP, Lindenbaum J. Etiology and diagnostic evaluation of macrocytosis. *Am J Med Sci* 2000;319:343-52.
- 12) Wang H, Li L, Qin LL, Song Y, Vidal-Alaball J, Liu TH. Oral vitamin B12 versus intramuscular vitamin B12 for vitamin B12 deficiency. *Cochrane Database Syst Rev* 2018;3:CD004655.
- 13) Röllig C, Ehninger G. How I treat hyperleukocytosis in acute myeloid leukemia. *Blood* 2015;125:3246-52.

- 14) Zini G, d'Onofrio G, Erber WN, et al. 2021 update of the 2012 ICSH Recommendations for identification, diagnostic value, and quantitation of schistocytes: impact and revisions. *Int J Lab Hematol* 2021;43:1264-71.
- 15) Henter JI, Horne A, Aricó M, et al. HLH-2004: diagnostic and therapeutic guidelines for hemophagocytic lymphohistiocytosis. *Pediatr Blood Cancer* 2007;48:124-31.
- 16) Pasupathy E, Kandasamy R, Thomas K, Basheer A. Alternate day versus daily oral iron for treatment of iron deficiency anemia: a randomized controlled trial. *Sci Rep* 2023;13:1818.
- 17) von Siebenthal HK, Gessler S, Vallelian F, et al. Alternate day versus consecutive day oral iron supplementation in iron-depleted women: a randomized double-blind placebo-controlled study. *EClinicalMedicine* 2023;65:102286.

3-2 血小板減少

- 1) 日本血栓止血学会HIT診療ガイドライン作成委員会. ヘパリン起因性血小板減少症の診断・治療ガイドライン. *日本血栓止血学会誌* 2021;32:737-82.
- 2) Lo GK, Juhl D, Warkentin TE, et al. Evaluation of pretest clinical score (4 T's) for the diagnosis of heparin-induced thrombocytopenia in two clinical settings. *J Thromb Haemost* 2006;4:759-65.
- 3) Cuker A, Gimotty PA, Crowther MA, Warkentin TE. Predictive value of the 4Ts scoring system for heparin-induced thrombocytopenia: a systematic review and meta-analysis. *Blood* 2012;120:4160-7.
- 4) Cuker A, Arepally GM, Chong BH, et al. American Society of Hematology 2018 guidelines for management of venous thromboembolism: heparin-induced thrombocytopenia. *Blood Adv* 2018;2:3360-92.
- 5) アルガトロバン水和物注射液（スロンノンHI注）の電子添文（KEGG MEDICUSで2026年10月に確認）.
- 6) 日本輸血・細胞治療学会ガイドライン委員会. 輸血療法実践ガイド. *日本輸血細胞治療学会誌* 2026;72:521-70.
- 7) Warkentin TE, Elavathil LJ, Hayward CP, et al. The pathogenesis of venous limb gangrene associated with heparin-induced thrombocytopenia. *Ann Intern Med* 1997;127:804-12.
- 8) Bendapudi PK, Hurwitz S, Fry A, et al. Derivation and external validation of the PLASMIC score for rapid assessment of adults with thrombotic microangiopathies: a cohort study. *Lancet Haematol* 2017;4:e157-64.
- 9) Paydary K, Banwell E, Tong J, Chen Y, Cuker A. Diagnostic accuracy of the PLASMIC score in patients with suspected thrombotic thrombocytopenic purpura: a systematic review and meta-analysis. *Transfusion* 2020;60:2047-57.

3-3 凝固異常とDIC

- 1) 日本検査血液学会標準化委員会凝固検査標準化ワーキンググループ. 凝固検査検体取扱いに関するコンセンサス. *日本検査血液学会雑誌* 2016;17:149-57.
- 2) 各薬剤の電子添文（KEGG MEDICUSで確認。ケイツーN静注、ノイアート、リコモジュリンは2026年10月に再確認）.
- 3) 日本血栓止血学会DIC診断基準作成委員会. 日本血栓止血学会DIC診断基準2017年版. *日本血栓止血学会誌* 2017;28:369-92.
- 4) Taylor FB Jr, Toh CH, Hoots WK, et al. Towards definition, clinical and laboratory criteria, and a scoring system for disseminated intravascular coagulation. *Thromb Haemost* 2001;86:1327-30.
- 5) Gando S, Iba T, Eguchi Y, et al. A multicenter, prospective validation of disseminated intravascular coagulation diagnostic criteria for critically ill patients: comparing current criteria. *Crit Care Med* 2006;34:625-31.

- 6) 日本輸血・細胞治療学会ガイドライン委員会. 輸血療法実践ガイド. 日本輸血細胞治療学会誌 2026;72:521-70.
- 7) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. 日集中医誌 2024;31:S1165-313.
- 8) Yamakawa K, Okamoto K, Seki Y, et al. Clinical practice guidelines for management of disseminated intravascular coagulation in Japan 2024. Part 1: sepsis. Int J Hematol 2025;121:592-604.
- 9) Vincent JL, Francois B, Zabolotskikh I, et al. Effect of a recombinant human soluble thrombomodulin on mortality in patients with sepsis-associated coagulopathy: the SCARLET randomized clinical trial. JAMA 2019;321:1993-2002.
- 10) Warren BL, Eid A, Singer P, et al. Caring for the critically ill patient. High-dose antithrombin III in severe sepsis: a randomized controlled trial. JAMA 2001;286:1869-78.
- 11) 日本血栓止血学会 DIC診療ガイドライン作成委員会. 日本血栓止血学会 播種性血管内凝固 (DIC) 診療ガイドライン2024. 日本血栓止血学会誌 2025;36:68-156.
- 12) Abdel-Wahab OI, Healy B, Dzik WH. Effect of fresh-frozen plasma transfusion on prothrombin time and bleeding in patients with mild coagulation abnormalities. Transfusion 2006;46:1279-85.
- 13) Segal JB, Dzik WH; Transfusion Medicine/Hemostasis Clinical Trials Network. Paucity of studies to support that abnormal coagulation test results predict bleeding in the setting of invasive procedures: an evidence-based review. Transfusion 2005;45:1413-25.
- 14) Müller MC, Arbous MS, Spoelstra-de Man AM, et al. Transfusion of fresh-frozen plasma in critically ill patients with a coagulopathy before invasive procedures: a randomized clinical trial. Transfusion 2015;55:26-35.

3-4 クレアチニン上昇と急性腎障害

- 1) STARRT-AKI Investigators. Timing of initiation of renal-replacement therapy in acute kidney injury. N Engl J Med 2020;383:240-51.
- 2) Gaudry S, Hajage D, Martin-Lefevre L, et al. Comparison of two delayed strategies for renal replacement therapy initiation for severe acute kidney injury (AKIKI 2): a multicentre, open-label, randomised, controlled trial. Lancet 2021;397:1293-300.
- 3) AKI (急性腎障害) 診療ガイドライン作成委員会. AKI (急性腎障害) 診療ガイドライン2016. 日腎会誌 2017;59:419-533.
- 4) Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney Int Suppl 2012;2:1-138.
- 5) Carvounis CP, Nisar S, Guro-Razuman S. Significance of the fractional excretion of urea in the differential diagnosis of acute renal failure. Kidney Int 2002;62:2223-9.
- 6) Scolari F, Ravani P. Atheroembolic renal disease. Lancet 2010;375:1650-60.
- 7) Scolari F, Tardanico R, Zani R, et al. Cholesterol crystal embolism: a recognizable cause of renal disease. Am J Kidney Dis 2000;36:1089-109.
- 8) フロセミド注射液 (ラシックス注100 mg) の電子添文 (2026年10月に確認) .
- 9) Vaara ST, Ostermann M, Bitker L, et al. Restrictive fluid management versus usual care in acute kidney injury (REVERSE-AKI): a pilot randomized controlled feasibility trial. Intensive Care Med 2021;47:665-73.
- 10) Bouchard J, Soroko SB, Chertow GM, et al. Fluid accumulation, survival and recovery of kidney function in critically ill patients with acute kidney injury. Kidney Int 2009;76:422-7.

3-5 低ナトリウム血症

- 1) Spasovski G, Vanholder R, Allolio B, et al. Clinical practice guideline on diagnosis and treatment of hyponatraemia. *Eur J Endocrinol* 2014;170:G1-47.
- 2) Baek SH, Jo YH, Ahn S, et al. Risk of overcorrection in rapid intermittent bolus vs slow continuous infusion therapies of hypertonic saline for patients with symptomatic hyponatremia: the SALSA randomized clinical trial. *JAMA Intern Med* 2021;181:81-92.
- 3) Verbalis JG, Goldsmith SR, Greenberg A, et al. Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med* 2013;126(10 Suppl 1):S1-42.
- 4) トルバプタン（サムスカ錠）の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 5) Wijdicks EF, Vermeulen M, Hijdra A, van Gijn J. Hyponatremia and cerebral infarction in patients with ruptured intracranial aneurysms: is fluid restriction harmful? *Ann Neurol* 1985;17:137-40.
- 6) Katayama K, Tokuda Y. Mineralocorticoid responsive hyponatremia of the elderly: a systematic review. *Medicine (Baltimore)* 2017;96:e7154.
- 7) Hoorn EJ, Zietse R. Diagnosis and treatment of hyponatremia: compilation of the guidelines. *J Am Soc Nephrol* 2017;28:1340-9.
- 8) European Association for the Study of the Liver. EASL clinical practice guidelines for the management of patients with decompensated cirrhosis. *J Hepatol* 2018;69:406-60.
- 9) MacMillan TE, Shin S, Topf J, et al. Osmotic demyelination syndrome in patients hospitalized with hyponatremia. *NEJM Evid* 2023;2:EVIDoa2200215.
- 10) Seethapathy H, Zhao S, Ouyang T, et al. Severe hyponatremia correction, mortality, and central pontine myelinolysis. *NEJM Evid* 2023;2:EVIDoa2300107.
- 11) Sterns RH. Treatment of severe hyponatremia. *Clin J Am Soc Nephrol* 2018;13:641-9.
- 12) Lien YH, Shapiro JI, Chan L. Study of brain electrolytes and organic osmolytes during correction of chronic hyponatremia. Implications for the pathogenesis of central pontine myelinolysis. *J Clin Invest* 1991;88:303-9.

3-6 高ナトリウム血症

- 1) Adrogue HJ, Madias NE. Hypernatremia. *N Engl J Med* 2000;342:1493-9.
- 2) Feigin E, Feigin L, Ingber M, et al. Rate of correction and all-cause mortality in patients with severe hypernatremia. *JAMA Netw Open* 2023;6:e2335415.
- 3) Chauhan K, Pattharanitima P, Patel N, et al. Rate of correction of hypernatremia and health outcomes in critically ill patients. *Clin J Am Soc Nephrol* 2019;14:656-63.
- 4) Yun G, Baek SH, Kim S. Evaluation and management of hypernatremia in adults: clinical perspectives. *Korean J Intern Med* 2023;38:290-302.

3-7 低カリウム血症

- 1) 各薬剤の電子添文（KEGG MEDICUSで確認。KCL注、アスパラカリウム注、グルコンサンK、塩化カリウム徐放錠600 mg「St」は2026年10月に再確認）。
- 2) 医薬品医療機器総合機構。PMDA医療安全情報 No.19 カリウム（K）製剤の誤投与について。2010.
- 3) Kardalas E, Paschou SA, Anagnostis P, Muscogiuri G, Siasos G, Vryonidou A. Hypokalemia: a clinical update. *Endocr Connect* 2018;7:R135-46.
- 4) Lin SH, Lin YF, Chen DT, Chu P, Hsu CW, Halperin ML. Laboratory tests to determine the cause of hypokalemia and paralysis. *Arch Intern Med* 2004;164:1561-6.

- 5) Kruse JA, Carlson RW. Rapid correction of hypokalemia using concentrated intravenous potassium chloride infusions. Arch Intern Med 1990;150:613-7.
- 6) Hamill RJ, Robinson LM, Wexler HR, Moote C. Efficacy and safety of potassium infusion therapy in hypokalemic critically ill patients. Crit Care Med 1991;19:694-9.

3-8 高カリウム血症

- 1) UK Kidney Association. Clinical practice guideline: management of hyperkalaemia in adults. Updated July 2026.
- 2) 各薬剤の電子添文（KEGG MEDICUSで確認。カルチコール、ロケルマ、カリメート、大塚塩カル注2%は2026年10月に再確認）。
- 3) Moussavi K, Garcia J, Tellez-Corrales E, Fitter S. Reduced alternative insulin dosing in hyperkalemia: a meta-analysis of effects on hypoglycemia and potassium reduction. Pharmacotherapy 2021;41:598-607.
- 4) Piktel JS, Wan X, Kouk S, Laurita KR, Wilson LD. Beneficial effect of calcium treatment for hyperkalemia is not due to "membrane stabilization". Crit Care Med 2024;52:1499-508.
- 5) Batterink J, Cessford TA, Taylor RA. Pharmacological interventions for the acute management of hyperkalaemia in adults. Cochrane Database Syst Rev 2015;(10):CD010344.
- 6) Martin TJ, Kang Y, Robertson KM, Virji MA, Marquez JM. Ionization and hemodynamic effects of calcium chloride and calcium gluconate in the absence of hepatic function. Anesthesiology 1990;73:62-5.
- 7) Gruy-Kapral C, Emmett M, Santa Ana CA, Porter JL, Fordtran JS, Fine KD. Effect of single dose resin-cathartic therapy on serum potassium concentration in patients with end-stage renal disease. J Am Soc Nephrol 1998;9:1924-30.
- 8) Noel JA, Bota SE, Petrcich W, et al. Risk of hospitalization for serious adverse gastrointestinal events associated with sodium polystyrene sulfonate use in patients of advanced age. JAMA Intern Med 2019;179:1025-33.
- 9) Peacock WF, Rafique Z, Vishnevskiy K, et al. Emergency potassium normalization treatment including sodium zirconium cyclosilicate: a phase II, randomized, double-blind, placebo-controlled study (ENERGIZE). Acad Emerg Med 2020;27:475-86.

3-9 Ca・Mg・Pの異常

- 1) 医薬品医療機器総合機構. 酸化マグネシウム製剤 適正使用に関するお願い—高マグネシウム血症—. 2020.
- 2) Payne RB, Little AJ, Williams RB, Milner JR. Interpretation of serum calcium in patients with abnormal serum proteins. Br Med J 1973;4:643-6.
- 3) Desgagnés N, King JA, Kline GA, Seiden-Long I, Leung AA. Use of albumin-adjusted calcium measurements in clinical practice. JAMA Netw Open 2025;8:e2455251.
- 4) 各薬剤の電子添文（KEGG MEDICUSで確認。エルシトニン注、ゾメタ、カルチコール、硫酸Mg補正液、リン酸Na補正液、大塚塩カル注2%、ホスリボン配合顆粒は2026年10月に再確認）。
- 5) El-Hajj Fuleihan G, Clines GA, Hu MI, et al. Treatment of hypercalcemia of malignancy in adults: an Endocrine Society clinical practice guideline. J Clin Endocrinol Metab 2023;108:507-28.
- 6) National Institute for Health and Care Excellence. Nutrition support for adults: oral nutrition support, enteral tube feeding and parenteral nutrition (CG32). 2006 (updated 2017).
- 7) Whang R, Whang DD, Ryan MP. Refractory potassium repletion. A consequence of magnesium deficiency. Arch Intern Med 1992;152:40-5.

- 8) Huang CL, Kuo E. Mechanism of hypokalemia in magnesium deficiency. *J Am Soc Nephrol* 2007;18:2649-52.
- 9) Rude RK, Oldham SB, Singer FR. Functional hypoparathyroidism and parathyroid hormone end-organ resistance in human magnesium deficiency. *Clin Endocrinol (Oxf)* 1976;5:209-24.

3-10 血液ガスと酸塩基平衡

- 1) Byrne AL, Bennett M, Chatterji R, et al. Peripheral venous and arterial blood gas analysis in adults: are they comparable? A systematic review and meta-analysis. *Respirology* 2014;19:168-75.
- 2) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. *日集中医誌* 2024;31:S1165-313.
- 3) Iberti TJ, Leibowitz AB, Papadakos PJ, Fischer EP. Low sensitivity of the anion gap as a screen to detect hyperlactatemia in critically ill patients. *Crit Care Med* 1990;18:275-7.
- 4) Batlle DC, Hizon M, Cohen E, Gutterman C, Gupta R. The use of the urinary anion gap in the diagnosis of hyperchloremic metabolic acidosis. *N Engl J Med* 1988;318:594-9.
- 5) Palmer BF, Kelepouris E, Clegg DJ. Renal tubular acidosis and management strategies: a narrative review. *Adv Ther* 2021;38:949-68.
- 6) Jaber S, Paugam C, Futier E, et al. Sodium bicarbonate therapy for patients with severe metabolic acidemia in the intensive care unit (BICAR-ICU): a multicentre, open-label, randomised controlled, phase 3 trial. *Lancet* 2018;392:31-40.
- 7) Jung B, Jabaudon M, De Jong A, et al. Sodium bicarbonate for severe metabolic acidemia and acute kidney injury: the BICARICU-2 randomized clinical trial. *JAMA* 2025;334:2000-10.
- 8) メイロン静注7% 電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 9) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認。ラクテック注、ヴィーンF輸液、ピカーボン輸液) .
- 10) Zitek T, Skaggs ZD, Rahbar A, Patel J, Khan M. Does intravenous lactated Ringer's solution raise serum lactate? *J Emerg Med* 2018;55:313-8.

3-11 肝胆道系酵素の上昇

- 1) 持田智, 滝川康裕, 中山伸朗, ほか. 我が国における「急性肝不全」の概念, 診断基準の確立: 厚生労働省科学研究費補助金(難治性疾患克服研究事業)「難治性の肝・胆道疾患に関する調査研究」班, ワーキンググループ-1, 研究報告. *肝臓* 2011;52:393-8.
- 2) Kumagai K, Mawatari S, Moriuchi A, et al. Early-phase prothrombin time-international normalized ratio in acute liver injury indicates the timing of therapeutic intervention and predicts prognostic improvement. *Hepatol Res* 2023;53:160-71.
- 3) 田中篤. 薬物性肝障害スコアリングシステム—RECAM-J 2023—. *肝臓* 2024;65:482-90.
- 4) Fontana RJ, Liou I, Reuben A, et al. AASLD practice guidance on drug, herbal, and dietary supplement-induced liver injury. *Hepatology* 2023;77:1036-65.
- 5) Tapper EB, Sengupta N, Bonder A. The incidence and outcomes of ischemic hepatitis: a systematic review with meta-analysis. *Am J Med* 2015;128:1314-21.
- 6) Cassidy WM, Reynolds TB. Serum lactic dehydrogenase in the differential diagnosis of acute hepatocellular injury. *J Clin Gastroenterol* 1994;19:118-21.
- 7) 日本肝臓学会 肝炎診療ガイドライン作成委員会編. B型肝炎治療ガイドライン (第5版) . 2026.
- 8) U.S. Food and Drug Administration. Guidance for industry. Drug-induced liver injury: premarketing clinical evaluation. 2009.
- 9) Kwo PY, Cohen SM, Lim JK. ACG clinical guideline: evaluation of abnormal liver chemistries. *Am J Gastroenterol* 2017;112:18-35.

- 10) 日本消化器がん検診学会 超音波検診委員会, ほか. 腹部超音波検診判定マニュアル改訂版 (2021年). 日消がん検診誌 2022;60:123-81.
- 11) Gurusamy KS, Giljaca V, Takwoingi Y, et al. Ultrasound versus liver function tests for diagnosis of common bile duct stones. Cochrane Database Syst Rev 2015;(2):CD011548.
- 12) Kiriyama S, Takada T, Strasberg SM, et al. New diagnostic criteria and severity assessment of acute cholangitis in revised Tokyo Guidelines. J Hepatobiliary Pancreat Sci 2012;19:548-56.

3-12 CK上昇と横紋筋融解症

- 1) Bosch X, Poch E, Grau JM. Rhabdomyolysis and acute kidney injury. N Engl J Med 2009;361:62-72.
- 2) Kodadek L, Carmichael SP 2nd, Seshadri A, et al. Rhabdomyolysis: an American Association for the Surgery of Trauma Critical Care Committee Clinical Consensus Document. Trauma Surg Acute Care Open 2022;7:e000836.
- 3) McMahon GM, Zeng X, Waikar SS. A risk prediction score for kidney failure or mortality in rhabdomyolysis. JAMA Intern Med 2013;173:1821-8.
- 4) キュビシン 静注用350 mg 電子添文 (KEGG MEDICUSで2026年10月に確認) .

3-13 尿検査

- 1) 日本腎臓学会編. エビデンスに基づくCKD診療ガイドライン2023. 東京医学社; 2023.
- 2) 血尿診断ガイドライン改訂委員会編. 血尿診断ガイドライン2023. ライフサイエンス出版; 2023.
- 3) Muriithi AK, Nasr SH, Leung N. Utility of urine eosinophils in the diagnosis of acute interstitial nephritis. Clin J Am Soc Nephrol 2013;8:1857-62.
- 4) Nicolle LE, Gupta K, Bradley SF, et al. Clinical practice guideline for the management of asymptomatic bacteriuria: 2019 update by the Infectious Diseases Society of America. Clin Infect Dis 2019;68:e83-110.
- 5) Devillé WL, Yzermans JC, van Duijn NP, et al. The urine dipstick test useful to rule out infections. A meta-analysis of the accuracy. BMC Urol 2004;4:4.

3-14 血液培養

- 1) Lee A, Mirrett S, Reller LB, et al. Detection of bloodstream infections in adults: how many blood cultures are needed? J Clin Microbiol 2007;45:3546-8.
- 2) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. 日集中医誌 2024;31:S1165-313.
- 3) Cheng MP, Stenstrom R, Paquette K, et al. Blood culture results before and after antimicrobial administration in patients with severe manifestations of sepsis: a diagnostic study. Ann Intern Med 2019;171:547-54.
- 4) 日本臨床微生物学会編. 血液培養検査ガイド. 南江堂; 2013.
- 5) Mermel LA, Maki DG. Detection of bacteremia in adults: consequences of culturing an inadequate volume of blood. Ann Intern Med 1993;119:270-2.
- 6) Weinstein MP, Towns ML, Quartey SM, et al. The clinical significance of positive blood cultures in the 1990s: a prospective comprehensive evaluation of the microbiology, epidemiology, and outcome of bacteremia and fungemia in adults. Clin Infect Dis 1997;24:584-602.
- 7) Lambregts MMC, Bernards AT, van der Beek MT, Visser LG, de Boer MG. Time to positivity of blood cultures supports early re-evaluation of empiric broad-spectrum antimicrobial therapy. PLoS One 2019;14:e0208819.

- 8) Yahav D, Franceschini E, Koppel F, et al. Seven versus 14 days of antibiotic therapy for uncomplicated gram-negative bacteremia: a noninferiority randomized controlled trial. *Clin Infect Dis* 2019;69:1091-8.
- 9) Coburn B, Morris AM, Tomlinson G, Detsky AS. Does this adult patient with suspected bacteremia require blood cultures? *JAMA* 2012;308:502-11.
- 10) Tokuda Y, Miyasato H, Stein GH, Kishaba T. The degree of chills for risk of bacteremia in acute febrile illness. *Am J Med* 2005;118:1417.
- 11) Komatsu T, Takahashi E, Mishima K, et al. A simple algorithm for predicting bacteremia using food consumption and shaking chills: a prospective observational study. *J Hosp Med* 2017;12:510-5.
- 12) Riedel S, Bourbeau P, Swartz B, et al. Timing of specimen collection for blood cultures from febrile patients with bacteremia. *J Clin Microbiol* 2008;46:1381-5.
- 13) Tabriz MS, Riederer K, Baran J Jr, Khatib R. Repeating blood cultures during hospital stay: practice pattern at a teaching hospital and a proposal for guidelines. *Clin Microbiol Infect* 2004;10:624-7.
- 14) Grace CJ, Lieberman J, Pierce K, Littenberg B. Usefulness of blood culture for hospitalized patients who are receiving antibiotic therapy. *Clin Infect Dis* 2001;32:1651-5.

3-15 心電図

- 1) Bogun F, Anh D, Kalahasty G, et al. Misdiagnosis of atrial fibrillation and its clinical consequences. *Am J Med* 2004;117:636-42.
- 2) Drew BJ, Ackerman MJ, Funk M, et al. Prevention of torsade de pointes in hospital settings: a scientific statement from the American Heart Association and the American College of Cardiology Foundation. *Circulation* 2010;121:1047-60.
- 3) 日本循環器学会, ほか. 急性冠症候群ガイドライン (2018年改訂版). 2019.
- 4) Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction (2018). *Circulation* 2018;138:e618-51.
- 5) Gulati M, Levy PD, Mukherjee D, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain. *Circulation* 2021;144:e368-454.
- 6) de Winter RJ, Verouden NJ, Wellens HJ, Wilde AA. A new ECG sign of proximal LAD occlusion. *N Engl J Med* 2008;359:2071-3.
- 7) de Zwaan C, Bär FW, Wellens HJ. Characteristic electrocardiographic pattern indicating a critical stenosis high in left anterior descending coronary artery in patients admitted because of impending myocardial infarction. *Am Heart J* 1982;103:730-6.
- 8) Smith SW, Dodd KW, Henry TD, Dvorak DM, Pearce LA. Diagnosis of ST-elevation myocardial infarction in the presence of left bundle branch block with the ST-elevation to S-wave ratio in a modified Sgarbossa rule. *Ann Emerg Med* 2012;60:766-76.
- 9) Bischof JE, Worrall C, Thompson P, Marti D, Smith SW. ST depression in lead aVL differentiates inferior ST-elevation myocardial infarction from pericarditis. *Am J Emerg Med* 2016;34:149-54.
- 10) Marchick MR, Courtney DM, Kabrhel C, et al. 12-lead ECG findings of pulmonary hypertension occur more frequently in emergency department patients with pulmonary embolism than in patients without pulmonary embolism. *Ann Emerg Med* 2010;55:331-5.

3-16 胸部X線

- 1) Claessens YE, Debray MP, Tubach F, et al. Early chest computed tomography scan to assist diagnosis and guide treatment decision for suspected community-acquired pneumonia. *Am J Respir Crit Care Med* 2015;192:974-82.

- 2) Collins SP, Lindsell CJ, Storrow AB, Abraham WT; ADHERE Scientific Advisory Committee, Investigators and Study Group. Prevalence of negative chest radiography results in the emergency department patient with decompensated heart failure. *Ann Emerg Med* 2006;47:13-8.
- 3) Blackmore CC, Black WC, Dallas RV, Crow HC. Pleural fluid volume estimation: a chest radiograph prediction rule. *Acad Radiol* 1996;3:103-9.
- 4) Yousefifard M, Baikpour M, Ghelichkhani P, et al. Screening performance characteristic of ultrasonography and radiography in detection of pleural effusion; a meta-analysis. *Emerg (Tehran)* 2016;4:1-10.
- 5) Goodman LR, Conrardy PA, Laing F, Singer MM. Radiographic evaluation of endotracheal tube position. *AJR Am J Roentgenol* 1976;127:433-4.
- 6) 日本麻酔科学会安全委員会. 安全な中心静脈カテーテル挿入・管理のためのプラクティカルガイド 2017. 2017.
- 7) 日本医療機能評価機構. 医療安全情報No.121 経鼻栄養チューブの誤挿入. 2016.
- 8) 日本医療機能評価機構. 医療安全情報No.218 X線画像確認時の経鼻栄養チューブ誤挿入の見落とし. 2025.
- 9) Hejblum G, Chalumeau-Lemoine L, Ioos V, et al. Comparison of routine and on-demand prescription of chest radiographs in mechanically ventilated adults: a multicentre, cluster-randomised, two-period crossover study. *Lancet* 2009;374:1687-93.

3-17 ベッドサイドの超音波

- 1) Lichtenstein DA, Mezière GA. Relevance of lung ultrasound in the diagnosis of acute respiratory failure: the BLUE protocol. *Chest* 2008;134:117-25.
- 2) Maw AM, Hassanin A, Ho PM, et al. Diagnostic accuracy of point-of-care lung ultrasonography and chest radiography in adults with symptoms suggestive of acute decompensated heart failure: a systematic review and meta-analysis. *JAMA Netw Open* 2019;2:e190703.
- 3) Volpicelli G, Elbarbary M, Blaivas M, et al. International evidence-based recommendations for point-of-care lung ultrasound. *Intensive Care Med* 2012;38:577-91.
- 4) Chavez MA, Shams N, Ellington LE, et al. Lung ultrasound for the diagnosis of pneumonia in adults: a systematic review and meta-analysis. *Respir Res* 2014;15:50.
- 5) Mukherjee M, Rudski LG, Addetia K, et al. Guidelines for the echocardiographic assessment of the right heart in adults and special considerations in pulmonary hypertension: recommendations from the American Society of Echocardiography. *J Am Soc Echocardiogr* 2025;38:141-86.
- 6) Long E, Oakley E, Duke T, Babl FE; Paediatric Research in Emergency Departments International Collaborative (PREDICT). Does respiratory variation in inferior vena cava diameter predict fluid responsiveness: a systematic review and meta-analysis. *Shock* 2017;47:550-9.
- 7) Pomero F, Dentali F, Borretta V, et al. Accuracy of emergency physician-performed ultrasonography in the diagnosis of deep-vein thrombosis: a systematic review and meta-analysis. *Thromb Haemost* 2013;109:137-45.
- 8) Bernardi E, Camporese G, Büller HR, et al. Serial 2-point ultrasonography plus D-dimer vs whole-leg color-coded Doppler ultrasonography for diagnosing suspected symptomatic deep vein thrombosis: a randomized controlled trial. *JAMA* 2008;300:1653-9.

第4部 病棟当直

4-1 入院中の発熱

- 1) Kaul DR, Flanders SA, Beck JM, Saint S. Brief report: incidence, etiology, risk factors, and outcome of hospital-acquired fever: a systematic, evidence-based review. *J Gen Intern Med* 2006;21:1184-7.
- 2) Prescott HC, Antonelli M, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2026. *Crit Care Med* 2026;54:725-812.
- 3) High KP, Bradley SF, Gravenstein S, et al. Clinical practice guideline for the evaluation of fever and infection in older adult residents of long-term care facilities: 2008 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2009;48:149-71.
- 4) Cheng MP, Stenstrom R, Paquette K, et al. Blood culture results before and after antimicrobial administration in patients with severe manifestations of sepsis: a diagnostic study. *Ann Intern Med* 2019;171:547-54.
- 5) Tokuda Y, Miyasato H, Stein GH, Kishaba T. The degree of chills for risk of bacteremia in acute febrile illness. *Am J Med* 2005;118:1417.
- 6) Churpek MM, Snyder A, Han X, et al. Quick Sepsis-related Organ Failure Assessment, Systemic Inflammatory Response Syndrome, and Early Warning Scores for detecting clinical deterioration in infected patients outside the intensive care unit. *Am J Respir Crit Care Med* 2017;195:906-11.
- 7) 日本臨床腫瘍学会. 発熱性好中球減少症 (FN) 診療ガイドライン 改訂第3版. 2024.
- 8) O'Grady NP, Alexander E, Alhazzani W, et al. Society of Critical Care Medicine and the Infectious Diseases Society of America guidelines for evaluating new fever in adult patients in the ICU. *Crit Care Med* 2023;51:1570-86.
- 9) Nicolle LE, Gupta K, Bradley SF, et al. Clinical practice guideline for the management of asymptomatic bacteriuria: 2019 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2019;68:e83-110.
- 10) Wanahita A, Goldsmith EA, Marino BJ, Musher DM. *Clostridium difficile* infection in patients with unexplained leukocytosis. *Am J Med* 2003;115:543-6.
- 11) Mackowiak PA, LeMaistre CF. Drug fever: a critical appraisal of conventional concepts. An analysis of 51 episodes in two Dallas hospitals and 97 episodes reported in the English literature. *Ann Intern Med* 1987;106:728-33.
- 12) Patel RA, Gallagher JC. Drug fever. *Pharmacotherapy* 2010;30:57-69.
- 13) Silveira PC, Ip IK, Goldhaber SZ, et al. Performance of Wells score for deep vein thrombosis in the inpatient setting. *JAMA Intern Med* 2015;175:1112-7.
- 14) Mavros MN, Velmahos GC, Falagas ME. Atelectasis as a cause of postoperative fever: where is the clinical evidence? *Chest* 2011;140:418-24.
- 15) 日本敗血症診療ガイドライン2024特別委員会. 日本敗血症診療ガイドライン2024. 日集中医誌 2024;31:S1165-313.
- 16) Hranjec T, Rosenberger LH, Swenson B, et al. Aggressive versus conservative initiation of antimicrobial treatment in critically ill surgical patients with suspected intensive-care-unit-acquired infection: a quasi-experimental, before and after observational cohort study. *Lancet Infect Dis* 2012;12:774-80.
- 17) Cantais A, Schnell D, Vincent F, et al. Acetaminophen-induced changes in systemic blood pressure in critically ill patients: results of a multicenter cohort study. *Crit Care Med* 2016;44:2192-8.

- 18) Stuck AE, Minder CE, Frey FJ. Risk of infectious complications in patients taking glucocorticosteroids. *Rev Infect Dis* 1989;11:954-63.
- 19) Karageorgopoulos DE, Qu JM, Korbila IP, Zhu YG, Vasileiou VA, Falagas ME. Accuracy of β -D-glucan for the diagnosis of *Pneumocystis jirovecii* pneumonia: a meta-analysis. *Clin Microbiol Infect* 2013;19:39-49.
- 20) Keane J, Gershon S, Wise RP, et al. Tuberculosis associated with infliximab, a tumor necrosis factor alpha-neutralizing agent. *N Engl J Med* 2001;345:1098-104.
- 21) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認。アクテムラ点滴静注用、ソリス点点滴静注、リツキサン点滴静注）。
- 22) Mikulska M, Lanini S, Gudiol C, et al. ESCMID Study Group for Infections in Compromised Hosts (ESGICH) Consensus Document on the safety of targeted and biological therapies: an infectious diseases perspective (Agents targeting lymphoid cells surface antigens [I]: CD19, CD20 and CD52). *Clin Microbiol Infect* 2018;24 Suppl 2:S71-82.
- 23) Coburn B, Morris AM, Tomlinson G, Detsky AS. Does this adult patient with suspected bacteremia require blood cultures? *JAMA* 2012;308:502-11.
- 24) Musher DM, Fainstein V, Young EJ, Pruett TL. Fever patterns: their lack of clinical significance. *Arch Intern Med* 1979;139:1225-8.

4-2 血圧上昇

- 1) 日本高血圧学会. 高血圧管理・治療ガイドライン2025. 2025.（英語版：Hypertens Res 2026;49:9-235）
- 2) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 3) Rastogi R, Sheehan MM, Hu B, et al. Treatment and outcomes of inpatient hypertension among adults with noncardiac admissions. *JAMA Intern Med* 2021;181:345-52.
- 4) Anderson TS, Herzig SJ, Jing B, et al. Clinical outcomes of intensive inpatient blood pressure management in hospitalized older adults. *JAMA Intern Med* 2023;183:715-23.
- 5) Mohandas R, Chamarthi G, Bozorgmehri S, et al. Pro re nata antihypertensive medications and adverse outcomes in hospitalized patients: a propensity-matched cohort study. *Hypertension* 2021;78:516-24.
- 6) Shimbo D, Barrett Bowling C, Levitan EB, et al. Short-term risk of serious fall injuries in older adults initiating and intensifying treatment with antihypertensive medication. *Circ Cardiovasc Qual Outcomes* 2016;9:222-9.
- 7) Park SK, Lee DY, Kim WJ, et al. Comparing the clinical efficacy of resting and antihypertensive medication in patients of hypertensive urgency: a randomized, control trial. *J Hypertens* 2017;35:1474-80.
- 8) 日本脳卒中学会 脳卒中ガイドライン委員会編. 脳卒中治療ガイドライン2021〔改訂2025〕. 協和企画；2025.
- 9) 日本循環器学会, 日本心臓血管外科学会, 日本胸部外科学会, 日本血管外科学会. 2020年改訂版 大動脈瘤・大動脈解離診療ガイドライン. 2020.
- 10) van den Born BH, Lip GYH, Brguljan-Hitij J, et al. ESC Council on hypertension position document on the management of hypertensive emergencies. *Eur Heart J Cardiovasc Pharmacother* 2019;5:37-46.
- 11) Anderson TS, Wray CM, Jing B, et al. Intensification of older adults' outpatient blood pressure treatment at hospital discharge: national retrospective cohort study. *BMJ* 2018;362:k3503.
- 12) Anderson TS, Jing B, Auerbach A, et al. Clinical outcomes after intensifying antihypertensive medication regimens among older adults at hospital discharge. *JAMA Intern Med* 2019;179:1528-36.

4-3 せん妄・不穏

- 1) Girard TD, Exline MC, Carson SS, et al. Haloperidol and ziprasidone for treatment of delirium in critical illness. *N Engl J Med* 2018;379:2506-16.
- 2) Tieges Z, MacLulich AMJ, Anand A, et al. Diagnostic accuracy of the 4AT for delirium detection in older adults: systematic review and meta-analysis. *Age Ageing* 2021;50:733-43.
- 3) Vijayakrishnan R, Ramasubramanian A, Dhand S. Utility of head CT scan for acute inpatient delirium. *Hosp Top* 2015;93:9-12.
- 4) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認。セレネース注、セロクエル錠）。
- 5) Burry L, Mehta S, Perreault MM, et al. Antipsychotics for treatment of delirium in hospitalised non-ICU patients. *Cochrane Database Syst Rev* 2018;(6):CD005594.

4-4 不眠

- 1) Glass J, Lanctôt KL, Herrmann N, Sproule BA, Busto UE. Sedative hypnotics in older people with insomnia: meta-analysis of risks and benefits. *BMJ* 2005;331:1169.
- 2) Kolla BP, Lovely JK, Mansukhani MP, Morgenthaler TI. Zolpidem is independently associated with increased risk of inpatient falls. *J Hosp Med* 2013;8:1-6.
- 3) Bartick MC, Thai X, Schmidt T, Altaye A, Solet JM. Decrease in as-needed sedative use by limiting nighttime sleep disruptions from hospital staff. *J Hosp Med* 2010;5:E20-4.
- 4) Hu RF, Jiang XY, Chen J, et al. Non-pharmacological interventions for sleep promotion in the intensive care unit. *Cochrane Database Syst Rev* 2015;(10):CD008808.
- 5) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 6) Gillis CM, Poyant JO, Degrado JR, Ye L, Anger KE, Owens RL. Inpatient pharmacological sleep aid utilization is common at a tertiary medical center. *J Hosp Med* 2014;9:652-7.
- 7) Sogawa R, Emoto A, Monji A, et al. Association of orexin receptor antagonists with falls during hospitalization. *J Clin Pharm Ther* 2022;47:809-13.
- 8) Nishino T, Kubota Y, Miyagi Y, et al. Association between hypnotic medication use and in-hospital falls among older adults: a multicenter landmark analysis. *PLoS One* 2026;21:e0351299.
- 9) Paolone A, Riar S, Vejayavarnan S, Papaioannou A, Ioannidis G, Lee J. Use of dual orexin receptor antagonists for insomnia and the risk of falls and fractures: a systematic review and meta-analysis. *J Am Geriatr Soc* 2026. doi:10.1111/jgs.70745（オンライン先行公開）

4-5 嘔気・嘔吐、下痢、便秘

- 1) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 2) Zhang C, Zhang R, Zhang S, Xu M, Zhang S. Baclofen for stroke patients with persistent hiccups: a randomized, double-blind, placebo-controlled trial. *Trials* 2014;15:295.
- 3) Polage CR, Solnick JV, Cohen SH. Nosocomial diarrhea: evaluation and treatment of causes other than *Clostridium difficile*. *Clin Infect Dis* 2012;55:982-9.
- 4) 日本化学療法学会・日本感染症学会CDI診療ガイドライン作成委員会編. *Clostridioides difficile*感染症診療ガイドライン2022. 感染症誌 2023;97(Suppl):S1-96.
- 5) Ponc R, Saunders MD, Kimmey MB. Neostigmine for the treatment of acute colonic pseudo-obstruction. *N Engl J Med* 1999;341:137-41.
- 6) Morishita D, Tomita T, Mori S, et al. Senna versus magnesium oxide for the treatment of chronic constipation: a randomized, placebo-controlled trial. *Am J Gastroenterol* 2021;116:152-61.

- 7) 日本消化管学会編. 便通異常症診療ガイドライン2023—慢性便秘症. 南江堂; 2023. (英語版: Ihara E, et al. Digestion 2025;106:62-89)
- 8) Müller-Lissner SA, Kamm MA, Scarpignato C, Wald A. Myths and misconceptions about chronic constipation. Am J Gastroenterol 2005;100:232-42.

4-6 尿量低下・尿閉

- 1) Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney Int Suppl 2012;2:1-138.
- 2) Chrouser K, Fowler KE, Mann JD, et al. Urinary retention evaluation and catheterization algorithm for adult inpatients. JAMA Netw Open 2024;7:e2422281.
- 3) Nyman MA, Schwenk NM, Silverstein MD. Management of urinary retention: rapid versus gradual decompression and risk of complications. Mayo Clin Proc 1997;72:951-6.
- 4) Halbgewachs C, Domes T. Postobstructive diuresis: pay close attention to urinary retention. Can Fam Physician 2015;61:137-42.
- 5) AKI（急性腎障害）診療ガイドライン作成委員会. AKI（急性腎障害）診療ガイドライン2016. 日腎会誌 2017;59:419-533.
- 6) Hashimoto H, Yamada H, Murata M, Watanabe N. Diuretics for preventing and treating acute kidney injury. Cochrane Database Syst Rev 2025;(1):CD014937.
- 7) Chawla LS, Davison DL, Brasha-Mitchell E, et al. Development and standardization of a furosemide stress test to predict the severity of acute kidney injury. Crit Care 2013;17:R207.

4-7 低血糖・高血糖

- 1) 日本糖尿病学会編. 糖尿病診療ガイドライン2024. 南江堂; 2024.
- 2) Joint British Diabetes Societies for Inpatient Care Group. The hospital management of hypoglycaemia in adults with diabetes mellitus. Revised January 2023.
- 3) 各薬剤の電子添文（KEGG MEDICUSで確認。パクスミー点鼻粉末剤、グルカゴンGノボ注射用、アマリール錠は2026年10月に再確認）.
- 4) 厚生労働省医薬食品局安全対策課長，審査管理課医療機器審査管理室長通知. 血糖測定器等に係る添付文書の改訂について（薬食安発1117第1号・薬食機発1117第1号）. 2011年11月17日.（医薬品医療機器総合機構. PMDA医療安全情報 No.28 血糖測定器の取扱い上の注意について. 2011）
- 5) Cryer PE, Axelrod L, Grossman AB, et al. Evaluation and management of adult hypoglycemic disorders: an Endocrine Society clinical practice guideline. J Clin Endocrinol Metab 2009;94:709-28.
- 6) Arora S, Henderson SO, Long T, Menchine M. Diagnostic accuracy of point-of-care testing for diabetic ketoacidosis at emergency-department triage: β -hydroxybutyrate versus the urine dipstick. Diabetes Care 2011;34:852-4.
- 7) 日本糖尿病学会 SGLT2阻害薬の適正使用に関する委員会. 糖尿病治療におけるSGLT2阻害薬の適正使用に関するRecommendation. 2022年7月改訂.
- 8) Moore C, Woollard M. Dextrose 10% or 50% in the treatment of hypoglycaemia out of hospital? A randomised controlled trial. Emerg Med J 2005;22:512-5.

4-8 転倒・転落の後

- 1) 日本医療安全調査機構 医療事故調査・支援センター. 医療事故の再発防止に向けた提言 第9号 入院中に発生した転倒・転落による頭部外傷に係る死亡事例の分析. 2019.
- 2) National Institute for Health and Care Excellence. Head injury: assessment and early management (NG232). 2023.

- 3) Puzio TJ, Murphy PB, Kregel HR, et al. Delayed intracranial hemorrhage after blunt head trauma while on direct oral anticoagulant: systematic review and meta-analysis. *J Am Coll Surg* 2021;232:1007-16.
- 4) 日本整形外科学会, 日本骨折治療学会監修. 大腿骨頸部/転子部骨折診療ガイドライン2021 改訂第3版. 南江堂; 2021.
- 5) Langdon J, Way A, Heaton S, Bernard J, Molloy S. Vertebral compression fractures: new clinical signs to aid diagnosis. *Ann R Coll Surg Engl* 2010;92:163-6.
- 6) Man-Son-Hing M, Nichol G, Lau A, Laupacis A. Choosing antithrombotic therapy for elderly patients with atrial fibrillation who are at risk for falls. *Arch Intern Med* 1999;159:677-85.
- 7) Tokioka F, Okamoto H, Shiotsu A, Hashimoto T. Frequency and clinical features of radiographic head injury caused by inpatient falls: a single-centre retrospective cohort study. *BMJ Open* 2023;13:e066426.
- 8) Stephen S, Wong EWW, Idris AM, Lim AKH. Intracranial haemorrhage detected by cerebral computed tomography after falls in hospital acute medical wards. *BMC Health Serv Res* 2019;19:792.
- 9) de Wit K, Mercuri M, Clayton N, et al. Derivation of the Falls Decision Rule to exclude intracranial bleeding without head CT in older adults who have fallen. *CMAJ* 2023;195:E1614-21.

4-9 ライン・チューブのトラブル

- 1) 日本医療安全調査機構 医療事故調査・支援センター. 医療事故の再発防止に向けた提言 第4号 気管切開術後早期の気管切開チューブ逸脱・迷入に係る死亡事例の分析. 2018.
- 2) McGrath BA, Bates L, Atkinson D, Moore JA; National Tracheostomy Safety Project. Multidisciplinary guidelines for the management of tracheostomy and laryngectomy airway emergencies. *Anaesthesia* 2012;67:1025-41.
- 3) 日本医療安全調査機構 医療事故調査・支援センター. 医療事故の再発防止に向けた提言 第17号 中心静脈カテーテル挿入・抜去に係る死亡事例の分析—第2報（改訂版）—. 2023.
- 4) 医薬品医療機器総合機構. PMDA医療安全情報 No.71 中心静脈に留置するカテーテル使用時の空気塞栓事例について. 2025.
- 5) Parienti JJ, Mongardon N, Mégarbane B, et al. Intravascular complications of central venous catheterization by insertion site. *N Engl J Med* 2015;373:1220-9.
- 6) O'Grady NP, Alexander M, Burns LA, et al. Guidelines for the prevention of intravascular catheter-related infections. *Clin Infect Dis* 2011;52:e162-93.
- 7) 医薬品医療機器総合機構. PMDA医療安全情報 No.43 胃瘻チューブ取扱い時のリスク. 2014.
- 8) 日本医療安全調査機構 医療事故調査・支援センター. 医療事故の再発防止に向けた提言 第6号 栄養剤投与目的に行われた胃管挿入に係る死亡事例の分析. 2018.
- 9) Gould CV, Umscheid CA, Agarwal RK, et al. Guideline for prevention of catheter-associated urinary tract infections 2009. *Infect Control Hosp Epidemiol* 2010;31:319-26.
- 10) 日本医療機能評価機構. 医療安全情報 No.80 膀胱留置カテーテルによる尿道損傷. 2013. 同 No.142（第2報）. 2018.
- 11) Villanueva C, Hemstreet GP 3rd. Difficult male urethral catheterization: a review of different approaches. *Int Braz J Urol* 2008;34:401-11.
- 12) 日本がん看護学会, 日本臨床腫瘍学会, 日本臨床腫瘍薬学会編. がん薬物療法に伴う血管外漏出に関する合同ガイドライン2023年版. 金原出版; 2022.
- 13) 各薬剤の電子添文（KEGG MEDICUSで確認。サビーン点滴静注用、ノルアドリナリン注、カルチコール注射液、アレビアチン注、注射用エフオーワイ、メイロン静注は2026年10月に確認）。

- 14) ACR Committee on Drugs and Contrast Media. ACR Manual on Contrast Media. American College of Radiology; 2024.

4-10 薬疹とアナフィラキシー

- 1) 日本アレルギー学会Anaphylaxis対策委員会編. アナフィラキシーガイドライン2022. 日本アレルギー学会; 2022.
- 2) 日本医療安全調査機構 医療事故調査・支援センター. 医療事故の再発防止に向けた提言 第3号 注射剤によるアナフィラキシーに係る死亡事例の分析. 2018.
- 3) Simons FE, Gu X, Simons KJ. Epinephrine absorption in adults: intramuscular versus subcutaneous injection. J Allergy Clin Immunol 2001;108:871-3.
- 4) 各薬剤の電子添文（KEGG MEDICUSで確認。ボスミン注、アドレナリン注0.1%シリンジ、ボララミン注、グルカゴンGノボ注射用、塩酸バンコマイシン点滴静注用は2026年10月に確認）。
- 5) Shaker MS, Wallace DV, Golden DBK, et al. Anaphylaxis—a 2020 practice parameter update, systematic review, and Grading of Recommendations, Assessment, Development and Evaluation (GRADE) analysis. J Allergy Clin Immunol 2020;145:1082-123.
- 6) 日本皮膚科学会. 重症多形滲出性紅斑 スティーヴンス・ジョンソン症候群・中毒性表皮壊死症診療ガイドライン補遺2025. 日皮会誌 2025;135:701-14.
- 7) 日本皮膚科学会. 薬剤性過敏症症候群診療ガイドライン2023. 日皮会誌 2024;134:559-80.
- 8) 厚生労働省. 重篤副作用疾患別対応マニュアル（急性汎発性発疹性膿疱症、血管性浮腫）。

第5部 入院疾患の管理

5-1 市中肺炎

- 1) 日本呼吸器学会成人肺炎診療ガイドライン2024作成委員会. 成人肺炎診療ガイドライン2024. 日本呼吸器学会; 2024.
- 2) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認）。
- 3) Metlay JP, Waterer GW, Long AC, et al. Diagnosis and treatment of adults with community-acquired pneumonia. An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America. Am J Respir Crit Care Med 2019;200:e45-67.
- 4) Jones BE, Ramirez JA, Oren E, et al. Diagnosis and management of community-acquired pneumonia: an official American Thoracic Society clinical practice guideline. Am J Respir Crit Care Med 2026;212:24-44.
- 5) Dequin PF, Meziani F, Quenot JP, et al. Hydrocortisone in severe community-acquired pneumonia. N Engl J Med 2023;388:1931-41.
- 6) Uranga A, España PP, Bilbao A, et al. Duration of antibiotic treatment in community-acquired pneumonia: a multicenter randomized clinical trial. JAMA Intern Med 2016;176:1257-65.
- 7) Dinh A, Ropers J, Duran C, et al. Discontinuing β -lactam treatment after 3 days for patients with community-acquired pneumonia in non-critical care wards (PTC): a double-blind, randomised, placebo-controlled, non-inferiority trial. Lancet 2021;397:1195-203.
- 8) 日本呼吸器学会, 日本感染症学会, 日本ワクチン学会 合同委員会. 65歳以上の成人に対する肺炎球菌ワクチン接種に関する考え方（第8版）. 2026.
- 9) 日本感染症学会, 日本呼吸器学会, 日本ワクチン学会. 2026年度の新型コロナワクチン定期接種に関する見解. 2026.
- 10) 日本感染症学会, 日本呼吸器学会, 日本ワクチン学会. 成人のRSウイルスワクチンに関する見解. 2025.

- 11) National Institute for Health and Care Excellence. Pneumonia in adults: diagnosis and management (CG191). 2014. (2025年にNG250へ改訂)
- 12) 厚生労働省. 重篤副作用疾患別対応マニュアル 間質性肺炎(肺臓炎、胞隔炎、肺線維症). 平成18年11月(令和元年9月改定).
- 13) Ohnishi H, Yokoyama A, Kondo K, et al. Comparative study of KL-6, surfactant protein-A, surfactant protein-D, and monocyte chemoattractant protein-1 as serum markers for interstitial lung diseases. *Am J Respir Crit Care Med* 2002;165:378-81.
- 14) Tasaka S, Hasegawa N, Kobayashi S, et al. Serum indicators for the diagnosis of pneumocystis pneumonia. *Chest* 2007;131:1173-80.
- 15) Collard HR, Ryerson CJ, Corte TJ, et al. Acute exacerbation of idiopathic pulmonary fibrosis. An international working group report. *Am J Respir Crit Care Med* 2016;194:265-75.
- 16) Postma DF, van Werkhoven CH, van Elden LJ, et al. Antibiotic treatment strategies for community-acquired pneumonia in adults. *N Engl J Med* 2015;372:1312-23.
- 17) Garin N, Genné D, Carballo S, et al. β -Lactam monotherapy vs β -lactam-macrolide combination treatment in moderately severe community-acquired pneumonia: a randomized noninferiority trial. *JAMA Intern Med* 2014;174:1894-901.
- 18) Sligl WI, Asadi L, Eurich DT, Tjosvold L, Marrie TJ, Majumdar SR. Macrolides and mortality in critically ill patients with community-acquired pneumonia: a systematic review and meta-analysis. *Crit Care Med* 2014;42:420-32.

5-2 誤嚥性肺炎・院内肺炎

- 1) 日本呼吸器学会. 成人肺炎診療ガイドライン2024. 2024.
- 2) Mandell LA, Niederman MS. Aspiration pneumonia. *N Engl J Med* 2019;380:651-63.
- 3) Kalil AC, Metersky ML, Klompas M, et al. Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clin Infect Dis* 2016;63:e61-111.
- 4) Metlay JP, Waterer GW, Long AC, et al. Diagnosis and treatment of adults with community-acquired pneumonia. An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America. *Am J Respir Crit Care Med* 2019;200:e45-67.
- 5) Bai AD, Srivastava S, Digby GC, et al. Anaerobic antibiotic coverage in aspiration pneumonia and the associated benefits and harms: a retrospective cohort study. *Chest* 2024;166:39-48.
- 6) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認) .
- 7) Maeda K, Koga T, Akagi J. Tentative nil per os leads to poor outcomes in older adults with aspiration pneumonia. *Clin Nutr* 2016;35:1147-52.
- 8) Parente DM, Cunha CB, Mylonakis E, Timbrook TT. The clinical utility of methicillin-resistant *Staphylococcus aureus* (MRSA) nasal screening to rule out MRSA pneumonia: a diagnostic meta-analysis with antimicrobial stewardship implications. *Clin Infect Dis* 2018;67:1-7.
- 9) Pappas PG, Kauffman CA, Andes DR, et al. Clinical practice guideline for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2016;62:e1-50.
- 10) Chastre J, Wolff M, Fagon JY, et al. Comparison of 8 vs 15 days of antibiotic therapy for ventilator-associated pneumonia in adults: a randomized trial. *JAMA* 2003;290:2588-98.
- 11) 日本老年医学会. 高齢者ケアの意思決定プロセスに関するガイドライン：人工的水分・栄養補給の導入を中心として. 2012.
- 12) Caldeira D, Alarcão J, Vaz-Carneiro A, Costa J. Risk of pneumonia associated with use of angiotensin converting enzyme inhibitors and angiotensin receptor blockers: systematic review and meta-analysis. *BMJ* 2012;345:e4260.

- 13) Lee JS, Chui PY, Ma HM, et al. Does low dose angiotensin converting enzyme inhibitor prevent pneumonia in older people with neurologic dysphagia: a randomized placebo-controlled trial. *J Am Med Dir Assoc* 2015;16:702-7.

5-3 インフルエンザ・COVID-19

- 1) 厚生労働省新型コロナウイルス感染症対策本部ほか. 新型コロナウイルス感染症の令和6年4月以降の医療提供体制及び公費支援等について（令和6年3月5日事務連絡）. 2024.
- 2) 日本感染症学会, 日本呼吸器学会, 日本化学療法学会, 日本臨床微生物学会, 日本環境感染症学会. 5学会による新型コロナウイルス感染症 診療の指針2025. 2025.
- 3) Chartrand C, Leeflang MM, Minion J, Brewer T, Pai M. Accuracy of rapid influenza diagnostic tests: a meta-analysis. *Ann Intern Med* 2012;156:500-11.
- 4) Dinnes J, Sharma P, Berhane S, et al. Rapid, point-of-care antigen tests for diagnosis of SARS-CoV-2 infection. *Cochrane Database Syst Rev* 2022;7:CD013705.
- 5) Uyeki TM, Bernstein HH, Bradley JS, et al. Clinical practice guidelines by the Infectious Diseases Society of America: 2018 update on diagnosis, treatment, chemoprophylaxis, and institutional outbreak management of seasonal influenza. *Clin Infect Dis* 2019;68:e1-47.
- 6) 日本感染症学会インフルエンザ委員会. 日本感染症学会提言「抗インフルエンザ薬の使用について」. 2019年10月24日.
- 7) Muthuri SG, Venkatesan S, Myles PR, et al. Effectiveness of neuraminidase inhibitors in reducing mortality in patients admitted to hospital with influenza A H1N1pdm09 virus infection: a meta-analysis of individual participant data. *Lancet Respir Med* 2014;2:395-404.
- 8) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認。タミフルカプセル、ラビアクタ点滴静注液、ゾフルーザ錠、ベクルリー点滴静注用、パキロピッドバック、ゾコーバ錠、アクテムラ点滴静注用、オルミエント錠）.
- 9) Kumar D, Ison MG, Mira JP, et al. Combining baloxavir marboxil with standard-of-care neuraminidase inhibitor in patients hospitalised with severe influenza (FLAGSTONE): a randomised, parallel-group, double-blind, placebo-controlled, superiority trial. *Lancet Infect Dis* 2022;22:718-30.
- 10) Lansbury L, Rodrigo C, Leonardi-Bee J, Nguyen-Van-Tam J, Lim WS. Corticosteroids as adjunctive therapy in the treatment of influenza. *Cochrane Database Syst Rev* 2019;2:CD010406.
- 11) WHO Solidarity Trial Consortium. Remdesivir and three other drugs for hospitalised patients with COVID-19: final results of the WHO Solidarity randomised trial and updated meta-analyses. *Lancet* 2022;399:1941-53.
- 12) RECOVERY Collaborative Group. Dexamethasone in hospitalized patients with Covid-19. *N Engl J Med* 2021;384:693-704.
- 13) RECOVERY Collaborative Group. Tocilizumab in patients admitted to hospital with COVID-19 (RECOVERY): a randomised, controlled, open-label, platform trial. *Lancet* 2021;397:1637-45.
- 14) RECOVERY Collaborative Group. Baricitinib in patients admitted to hospital with COVID-19 (RECOVERY): a randomised, controlled, open-label, platform trial and updated meta-analysis. *Lancet* 2022;400:359-68.
- 15) Langford BJ, So M, Raybardhan S, et al. Bacterial co-infection and secondary infection in patients with COVID-19: a living rapid review and meta-analysis. *Clin Microbiol Infect* 2020;26:1622-9.
- 16) 日本感染症学会. インフルエンザ病院内感染対策の考え方について（高齢者施設を含めて）（日本感染症学会提言2012）. 2012.
- 17) Hayden FG, Shinkai M, Clark TW, et al. Ensitrelvir for Covid-19 postexposure prophylaxis in household contacts. *N Engl J Med* 2026;394:1905-15.

5-4 尿路感染症

- 1) 日本感染症学会, 日本化学療法学会. JAID/JSC感染症治療ガイドライン2015—尿路感染症・男性性器感染症—. 日化療会誌 2016;64:1-30.
- 2) 厚生労働省. 抗微生物薬適正使用の手引き 第四版 薬剤耐性菌感染症の抗菌薬適正使用編. 2026.
- 3) Trautner BW, Cortes-Penfield NW, Gupta K, et al. Clinical practice guidelines by the Infectious Diseases Society of America: 2025 guidelines on management and treatment of complicated urinary tract infections. Clin Infect Dis 2026;82(Suppl 3):i24-91.
- 4) Schuler F, Froböse N, Schaumburg F. Prevalence and risk factors for bacteremia in patients with *Staphylococcus aureus* bacteriuria: a retrospective cohort study. Int J Infect Dis 2020;98:467-9.
- 5) Pappas PG, Kauffman CA, Andes DR, et al. Clinical practice guideline for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. Clin Infect Dis 2016;62:e1-50.
- 6) Nicolle LE, Gupta K, Bradley SF, et al. Clinical practice guideline for the management of asymptomatic bacteriuria: 2019 update by the Infectious Diseases Society of America. Clin Infect Dis 2019;68:e83-110.
- 7) 日本泌尿器科学会, 日本尿路結石症学会, 日本泌尿器内視鏡・ロボティクス学会. 尿路結石症診療ガイドライン 第3版. 2023.
- 8) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認) .
- 9) Harris PNA, Tambyah PA, Lye DC, et al. Effect of piperacillin-tazobactam vs meropenem on 30-day mortality for patients with *E coli* or *Klebsiella pneumoniae* bloodstream infection and ceftriaxone resistance: a randomized clinical trial. JAMA 2018;320:984-94.
- 10) Sandberg T, Skoog G, Hermansson AB, et al. Ciprofloxacin for 7 days versus 14 days in women with acute pyelonephritis: a randomised, open-label and double-blind, placebo-controlled, non-inferiority trial. Lancet 2012;380:484-90.
- 11) Yahav D, Franceschini E, Koppel F, et al. Seven versus 14 days of antibiotic therapy for uncomplicated gram-negative bacteremia: a noninferiority randomized controlled trial. Clin Infect Dis 2019;69:1091-8.
- 12) Drekonja DM, Trautner B, Amundson C, et al. Effect of 7 vs 14 days of antibiotic therapy on resolution of symptoms among afebrile men with urinary tract infection: a randomized clinical trial. JAMA 2021;326:324-31.
- 13) Lafaurie M, Chevret S, Fontaine JP, et al. Antimicrobial for 7 or 14 days for febrile urinary tract infection in men: a multicenter noninferiority double-blind, placebo-controlled, randomized clinical trial. Clin Infect Dis 2023;76:2154-62.
- 14) Hooton TM, Bradley SF, Cardenas DD, et al. Diagnosis, prevention, and treatment of catheter-associated urinary tract infection in adults: 2009 international clinical practice guidelines from the Infectious Diseases Society of America. Clin Infect Dis 2010;50:625-63.
- 15) Petty LA, Vaughn VM, Flanders SA, et al. Risk factors and outcomes associated with treatment of asymptomatic bacteriuria in hospitalized patients. JAMA Intern Med 2019;179:1519-27.
- 16) Dasgupta M, Brymer C, Elsayed S. Treatment of asymptomatic UTI in older delirious medical in-patients: a prospective cohort study. Arch Gerontol Geriatr 2017;72:127-34.
- 17) Cai T, Mazzoli S, Mondaini N, et al. The role of asymptomatic bacteriuria in young women with recurrent urinary tract infections: to treat or not to treat? Clin Infect Dis 2012;55:771-7.

5-5 蜂窩織炎・壊死性軟部組織感染症

- 1) Weng QY, Raff AB, Cohen JM, et al. Costs and consequences associated with misdiagnosed lower extremity cellulitis. JAMA Dermatol 2017;153:141-6.

- 2) Nawijn F, Smeeing DPJ, Houwert RM, et al. Time is of the essence when treating necrotizing soft tissue infections: a systematic review and meta-analysis. *World J Emerg Surg* 2020;15:4.
- 3) 日本形成外科学会, 日本創傷外科学会, 日本頭蓋顎顔面外科学会. 形成外科診療ガイドライン3 2021年版 創傷疾患 (第II編 感染創診療ガイドライン). 金原出版; 2021.
- 4) Stevens DL, Bisno AL, Chambers HF, et al. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2014;59:e10-52.
- 5) Gottlieb M, Avila J, Chottiner M, Peksa GD. Point-of-care ultrasonography for the diagnosis of skin and soft tissue abscesses: a systematic review and meta-analysis. *Ann Emerg Med* 2020;76:67-77.
- 6) Moran GJ, Krishnadasan A, Mower WR, et al. Effect of cephalexin plus trimethoprim-sulfamethoxazole vs cephalexin alone on clinical cure of uncomplicated cellulitis: a randomized clinical trial. *JAMA* 2017;317:2088-96.
- 7) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認) .
- 8) Fernando SM, Tran A, Cheng W, et al. Necrotizing soft tissue infection: diagnostic accuracy of physical examination, imaging, and LRINEC score: a systematic review and meta-analysis. *Ann Surg* 2019;269:58-65.
- 9) Babiker A, Li X, Lai YL, et al. Effectiveness of adjunctive clindamycin in β -lactam antibiotic-treated patients with invasive β -haemolytic streptococcal infections in US hospitals: a retrospective multicentre cohort study. *Lancet Infect Dis* 2021;21:697-710.
- 10) 厚生労働省. 感染症法に基づく医師の届出のお願い: 劇症型溶血性レンサ球菌感染症 (届出基準) .
- 11) 国立感染症研究所. 国内における劇症型溶血性レンサ球菌感染症の増加について. 2024.
- 12) Centers for Disease Control and Prevention. Toxic shock syndrome (other than streptococcal) (TSS): 2011 case definition. National Notifiable Diseases Surveillance System.
- 13) 日本感染症学会. 感染症クイック・リファレンス: ピブリオ・バルニフィカス感染症. 2025.
- 14) Hepburn MJ, Dooley DP, Skidmore PJ, et al. Comparison of short-course (5 days) and standard (10 days) treatment for uncomplicated cellulitis. *Arch Intern Med* 2004;164:1669-74.
- 15) Cranendonk DR, Opmeer BC, van Agtmael MA, et al. Antibiotic treatment for 6 days versus 12 days in patients with severe cellulitis: a multicentre randomized, double-blind, placebo-controlled, non-inferiority trial. *Clin Microbiol Infect* 2020;26:606-12.
- 16) Senneville É, Albalawi Z, van Asten SA, et al. IWGDF/IDSA guidelines on the diagnosis and treatment of diabetes-related foot infections (IWGDF/IDSA 2023). *Clin Infect Dis* 2023:ciad527.
- 17) Webb E, Neeman T, Bowden FJ, et al. Compression therapy to prevent recurrent cellulitis of the leg. *N Engl J Med* 2020;383:630-9.
- 18) Thomas KS, Crook AM, Nunn AJ, et al. Penicillin to prevent recurrent leg cellulitis. *N Engl J Med* 2013;368:1695-703.
- 19) Perl B, Gottehrer NP, Raveh D, Schlesinger Y, Rudensky B, Yinnon AM. Cost-effectiveness of blood cultures for adult patients with cellulitis. *Clin Infect Dis* 1999;29:1483-8.
- 20) Gunderson CG, Martinello RA. A systematic review of bacteremias in cellulitis and erysipelas. *J Infect* 2012;64:148-55.
- 21) van Daalen FV, Kallen MC, van den Bosch CMA, Hulscher MEJL, Geerlings SE, Prins JM. Clinical condition and comorbidity as determinants for blood culture positivity in patients with skin and soft-tissue infections. *Eur J Clin Microbiol Infect Dis* 2017;36:1853-8.
- 22) Parks T, Wilson C, Curtis N, Norrby-Teglund A, Sriskandan S. Polyspecific intravenous immunoglobulin in clindamycin-treated patients with streptococcal toxic shock syndrome: a systematic review and meta-analysis. *Clin Infect Dis* 2018;67:1434-6.

- 23) Madsen MB, Hjortrup PB, Hansen MB, et al. Immunoglobulin G for patients with necrotising soft tissue infection (INSTINCT): a randomised, blinded, placebo-controlled trial. *Intensive Care Med* 2017;43:1585-93.

5-6 胆管炎・胆嚢炎

- 1) 急性胆管炎・胆嚢炎診療ガイドライン改訂出版委員会編. 急性胆管炎・胆嚢炎診療ガイドライン2018 (第3版). 医学図書出版; 2018.
- 2) Trowbridge RL, Rutkowski NK, Shojania KG. Does this patient have acute cholecystitis? *JAMA* 2003;289:80-6.
- 3) Kiewiet JJ, Leeuwenburgh MM, Bipat S, et al. A systematic review and meta-analysis of diagnostic performance of imaging in acute cholecystitis. *Radiology* 2012;264:708-20.
- 4) Adedeji OA, McAdam WA. Murphy's sign, acute cholecystitis and elderly people. *J R Coll Surg Edinb* 1996;41:88-9.
- 5) Gurusamy KS, Giljaca V, Takwoingi Y, et al. Ultrasound versus liver function tests for diagnosis of common bile duct stones. *Cochrane Database Syst Rev* 2015;(2):CD011548.
- 6) Giljaca V, Gurusamy KS, Takwoingi Y, et al. Endoscopic ultrasound versus magnetic resonance cholangiopancreatography for common bile duct stones. *Cochrane Database Syst Rev* 2015;(2):CD011549.
- 7) Kiriama S, Takada T, Hwang TL, et al. Clinical application and verification of the TG13 diagnostic and severity grading criteria for acute cholangitis: an international multicenter observational study. *J Hepatobiliary Pancreat Sci* 2017;24:329-37.
- 8) Hakuta R, Hamada T, Nakai Y, et al. No association of timing of endoscopic biliary drainage with clinical outcomes in patients with non-severe acute cholangitis. *Dig Dis Sci* 2018;63:1937-45.
- 9) Gutt CN, Encke J, Königer J, et al. Acute cholecystitis: early versus delayed cholecystectomy, a multicenter randomized trial (ACDC study, NCT00447304). *Ann Surg* 2013;258:385-93.
- 10) Loozen CS, van Santvoort HC, van Duijvendijk P, et al. Laparoscopic cholecystectomy versus percutaneous catheter drainage for acute cholecystitis in high risk patients (CHOCOLATE): multicentre randomised clinical trial. *BMJ* 2018;363:k3965.
- 11) Riall TS, Zhang D, Townsend CM Jr, Kuo YF, Goodwin JS. Failure to perform cholecystectomy for acute cholecystitis in elderly patients is associated with increased morbidity, mortality, and cost. *J Am Coll Surg* 2010;210:668-77.
- 12) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認）.
- 13) Srinu D, Shah J, Jena A, et al. Conventional vs short duration of antibiotics in patients with moderate or severe cholangitis: noninferiority randomized trial. *Am J Gastroenterol* 2024;119:176-82.
- 14) Rahimian J, Wilson T, Oram V, Holzman RS. Pyogenic liver abscess: recent trends in etiology and mortality. *Clin Infect Dis* 2004;39:1654-9.
- 15) Zerem E, Hadzic A. Sonographically guided percutaneous catheter drainage versus needle aspiration in the management of pyogenic liver abscess. *AJR Am J Roentgenol* 2007;189:W138-42.
- 16) Molton JS, Chan M, Kalimuddin S, et al. Oral vs intravenous antibiotics for patients with *Klebsiella pneumoniae* liver abscess: a randomized, controlled noninferiority study. *Clin Infect Dis* 2020;71:952-9.
- 17) Hussain I, Ishrat S, Ho DCW, et al. Endogenous endophthalmitis in *Klebsiella pneumoniae* pyogenic liver abscess: systematic review and meta-analysis. *Int J Infect Dis* 2020;101:259-68.

- 18) Mohan BP, Meyyur Aravamudan V, Khan SR, et al. Prevalence of colorectal cancer in cryptogenic pyogenic liver abscess patients. Do they need screening colonoscopy? A systematic review and meta-analysis. Dig Liver Dis 2019;51:1641-5.
- 19) 厚生労働省. 感染症法に基づく医師の届出のお願い：アメーバ赤痢（届出基準）.
- 20) Boerma D, Rauws EA, Keulemans YC, et al. Wait-and-see policy or laparoscopic cholecystectomy after endoscopic sphincterotomy for bile-duct stones: a randomised trial. Lancet 2002;360:761-5.
- 21) 日本消化器病学会編. 胆石症診療ガイドライン2021（改訂第3版）. 南江堂；2021.

5-7 敗血症

- 1) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. 日集中医誌 2024;31:S1165-313.
- 2) Singer M, Deutschman CS, Seymour CW, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA 2016;315:801-10.
- 3) Seymour CW, Liu VX, Iwashyna TJ, et al. Assessment of clinical criteria for sepsis: for the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA 2016;315:762-74.
- 4) Prescott HC, Antonelli M, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2026. Crit Care Med 2026;54:725-812.
- 5) Henning DJ, Carey JR, Oedorf K, et al. The absence of fever is associated with higher mortality and decreased antibiotic and IV fluid administration in emergency department patients with suspected septic shock. Crit Care Med 2017;45:e575-82.
- 6) Seymour CW, Gesten F, Prescott HC, et al. Time to treatment and mortality during mandated emergency care for sepsis. N Engl J Med 2017;376:2235-44.
- 7) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認）.
- 8) 日本化学療法学会, 日本TDM学会 抗菌薬TDMガイドライン作成委員会. 抗菌薬TDM臨床実践ガイドライン2022. 日化療会誌 2022;70:1-72.
- 9) Abdul-Aziz MH, Hammond NE, Brett SJ, et al. Prolonged vs intermittent infusions of β -lactam antibiotics in adults with sepsis or septic shock: a systematic review and meta-analysis. JAMA 2024;332:638-48.
- 10) Venkatesh B, Finfer S, Cohen J, et al. Adjunctive glucocorticoid therapy in patients with septic shock. N Engl J Med 2018;378:797-808.
- 11) Annane D, Renault A, Brun-Buisson C, et al. Hydrocortisone plus fludrocortisone for adults with septic shock. N Engl J Med 2018;378:809-18.
- 12) Holst LB, Haase N, Wetterslev J, et al. Lower versus higher hemoglobin threshold for transfusion in septic shock. N Engl J Med 2014;371:1381-91.
- 13) Daneman N, Rishu A, Pinto R, et al. Antibiotic treatment for 7 versus 14 days in patients with bloodstream infections. N Engl J Med 2025;392:1065-78.
- 14) Prescott HC, Angus DC. Enhancing recovery from sepsis: a review. JAMA 2018;319:62-75.

5-8 CDI (C. difficile感染症)

- 1) 日本化学療法学会, 日本感染症学会 CDI診療ガイドライン作成委員会編. Clostridioides difficile感染症診療ガイドライン2022. 感染症誌 2023;97(Suppl):S1-96.
- 2) 日本環境感染学会 CDI対策ガイドライン策定委員会編. Clostridioides difficile感染対策ガイド 改訂版. 環境感染誌 2026;41(Suppl III).
- 3) McDonald LC, Gerding DN, Johnson S, et al. Clinical practice guidelines for Clostridium difficile infection in adults and children: 2017 update by the Infectious Diseases Society of America

- (IDSA) and Society for Healthcare Epidemiology of America (SHEA). Clin Infect Dis 2018;66:e1-48.
- 4) Wanahita A, Goldsmith EA, Marino BJ, Musher DM. *Clostridium difficile* infection in patients with unexplained leukocytosis. Am J Med 2003;115:543-6.
 - 5) Hensgens MP, Goorhuis A, Dekkers OM, Kuijper EJ. Time interval of increased risk for *Clostridium difficile* infection after exposure to antibiotics. J Antimicrob Chemother 2012;67:742-8.
 - 6) Johnson S, Laverne V, Skinner AM, et al. Clinical practice guideline by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA): 2021 focused update guidelines on management of *Clostridioides difficile* infection in adults. Clin Infect Dis 2021;73:e1029-44.
 - 7) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認）。
 - 8) Louie TJ, Miller MA, Mullane KM, et al. Fidaxomicin versus vancomycin for *Clostridium difficile* infection. N Engl J Med 2011;364:422-31.
 - 9) Wilcox MH, Gerding DN, Poxton IR, et al. Bezlotoxumab for prevention of recurrent *Clostridium difficile* infection. N Engl J Med 2017;376:305-17.
 - 10) Tariq R, Singh S, Gupta A, et al. Association of gastric acid suppression with recurrent *Clostridium difficile* infection: a systematic review and meta-analysis. JAMA Intern Med 2017;177:784-91.
 - 11) Van Hise NW, Bryant AM, Hennessey EK, Crannage AJ, Khoury JA, Manian FA. Efficacy of oral vancomycin in preventing recurrent *Clostridium difficile* infection in patients treated with systemic antimicrobial agents. Clin Infect Dis 2016;63:651-3.
 - 12) Johnson SW, Brown SV, Priest DH. Effectiveness of oral vancomycin for prevention of healthcare facility-onset *Clostridioides difficile* infection in targeted patients during systemic antibiotic exposure. Clin Infect Dis 2020;71:1133-9.

5-9 細菌性髄膜炎

- 1) van de Beek D, de Gans J, Spanjaard L, et al. Clinical features and prognostic factors in adults with bacterial meningitis. N Engl J Med 2004;351:1849-59.
- 2) 日本神経学会，日本神経治療学会，日本神経感染症学会監修．細菌性髄膜炎診療ガイドライン2014．南江堂；2014．
- 3) Nakao JH, Jafri FN, Shah K, Newman DH. Jolt accentuation of headache and other clinical signs: poor predictors of meningitis in adults. Am J Emerg Med 2014;32:24-8.
- 4) Tunkel AR, Hartman BJ, Kaplan SL, et al. Practice guidelines for the management of bacterial meningitis. Clin Infect Dis 2004;39:1267-84.
- 5) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認）。
- 6) de Gans J, van de Beek D. Dexamethasone in adults with bacterial meningitis. N Engl J Med 2002;347:1549-56.
- 7) Charlier C, Perrodeau É, Leclercq A, et al. Clinical features and prognostic factors of listeriosis: the MONALISA national prospective cohort study. Lancet Infect Dis 2017;17:510-9.
- 8) National Institute for Health and Care Excellence. Meningitis (bacterial) and meningococcal disease: recognition, diagnosis and management. NICE guideline NG240. 2024.
- 9) 日本神経感染症学会，日本神経学会，日本神経治療学会監修．単純ヘルペス脳炎診療ガイドライン2017．南江堂；2017．
- 10) 国立感染症研究所．侵襲性髄膜炎菌感染症発生時対応ガイドライン〔第二版〕．2025．
- 11) Mount HR, Boyle SD. Aseptic and bacterial meningitis: evaluation, treatment, and prevention. Am Fam Physician 2017;96:314-22.

- 12) Doran KS, Fulde M, Gratz N, et al. Host-pathogen interactions in bacterial meningitis. *Acta Neuropathol* 2016;131:185-209.
- 13) Brouwer MC, McIntyre P, Prasad K, van de Beek D. Corticosteroids for acute bacterial meningitis. *Cochrane Database Syst Rev* 2015;(9):CD004405.

5-10 黄色ブドウ球菌菌血症とCRBSI

- 1) Daneman N, Rishu A, Pinto R, et al. Antibiotic treatment for 7 versus 14 days in patients with bloodstream infections. *N Engl J Med* 2025;392:1065-78.
- 2) 日本循環器学会. 2026年改訂版 感染性心内膜炎診療ガイドライン. 2026.
- 3) 日本化学療法学会, 日本感染症学会 MRSA感染症の診療ガイドライン作成委員会. MRSA感染症の診療ガイドライン2024. *日化療会誌* 2024;72:252-321.
- 4) Liu C, Chambers HF, Kern WV, et al. 2026 Consensus statements by the Infectious Diseases Society of America (IDSA) and European Society of Clinical Microbiology and Infectious Diseases (ESCMID) on *Staphylococcus aureus* bacteremia: risk stratification, diagnostic evaluation, and management of adults and children. *Clin Infect Dis* 2026. Online ahead of print.
- 5) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認) .
- 6) 社会保険診療報酬支払基金. 審査情報提供事例 (薬剤) 251 セファゾリンナトリウム水和物 (感染症16) . 2016.
- 7) *Staphylococcus aureus* Network Adaptive Platform (SNAP) Trial Group. Cefazolin for methicillin-susceptible *Staphylococcus aureus* bacteremia. *N Engl J Med* 2026;394:2329-39.
- 8) Tong SYC, Lye DC, Yahav D, et al. Effect of vancomycin or daptomycin with vs without an antistaphylococcal β -lactam on mortality, bacteremia, relapse, or treatment failure in patients with MRSA bacteremia: a randomized clinical trial. *JAMA* 2020;323:527-37.
- 9) Thwaites GE, Scarborough M, Szubert A, et al. Adjunctive rifampicin for *Staphylococcus aureus* bacteraemia (ARREST): a multicentre, randomised, double-blind, placebo-controlled trial. *Lancet* 2018;391:668-78.
- 10) Kaasch AJ, López-Cortés LE, Rodríguez-Baño J, et al. Efficacy and safety of an early oral switch in low-risk *Staphylococcus aureus* bloodstream infection (SABATO): an international, open-label, parallel-group, randomised, controlled, non-inferiority trial. *Lancet Infect Dis* 2024;24:523-34.
- 11) Mermel LA, Allon M, Bouza E, et al. Clinical practice guidelines for the diagnosis and management of intravascular catheter-related infection: 2009 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2009;49:1-45.
- 12) 日本医真菌学会. 侵襲性カンジダ症に対するマネジメントのための臨床実践ガイドライン. 2021.
- 13) Berbari EF, Kanj SS, Kowalski TJ, et al. 2015 Infectious Diseases Society of America (IDSA) clinical practice guidelines for the diagnosis and treatment of native vertebral osteomyelitis in adults. *Clin Infect Dis* 2015;61:e26-46.
- 14) Bernard L, Dinh A, Ghout I, et al. Antibiotic treatment for 6 weeks versus 12 weeks in patients with pyogenic vertebral osteomyelitis: an open-label, non-inferiority, randomised, controlled trial. *Lancet* 2015;385:875-82.
- 15) Park KH, Cho OH, Lee JH, et al. Optimal duration of antibiotic therapy in patients with hematogenous vertebral osteomyelitis at low risk and high risk of recurrence. *Clin Infect Dis* 2016;62:1262-9.
- 16) Li HK, Rombach I, Zambellas R, et al. Oral versus intravenous antibiotics for bone and joint infection. *N Engl J Med* 2019;380:425-36.
- 17) McDanel JS, Perencevich EN, Diekema DJ, et al. Comparative effectiveness of beta-lactams versus vancomycin for treatment of methicillin-susceptible *Staphylococcus aureus* bloodstream infections among 122 hospitals. *Clin Infect Dis* 2015;61:361-7.

- 18) Stryjewski ME, Szczech LA, Benjamin DK Jr, et al. Use of vancomycin or first-generation cephalosporins for the treatment of hemodialysis-dependent patients with methicillin-susceptible *Staphylococcus aureus* bacteremia. Clin Infect Dis 2007;44:190-6.

5-11 感染性心内膜炎

- 1) 日本循環器学会. 2026年改訂版 感染性心内膜炎診療ガイドライン. 2026.
- 2) Fowler VG Jr, Durack DT, Selton-Suty C, et al. The 2023 Duke-International Society for Cardiovascular Infectious Diseases criteria for infective endocarditis: updating the modified Duke criteria. Clin Infect Dis 2023;77:518-26.
- 3) Berge A, Krantz A, Östlund H, Naclér P, Rasmussen M. The DENOVA score efficiently identifies patients with monomicrobial *Enterococcus faecalis* bacteremia where echocardiography is not necessary. Infection 2019;47:45-50.
- 4) Sunnerhagen T, Törnell A, Vikbrant M, Nilson B, Rasmussen M. HANDOC: a handy score to determine the need for echocardiography in non- β -hemolytic streptococcal bacteremia. Clin Infect Dis 2018;66:693-8.
- 5) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）.
- 6) Kang DH, Kim YJ, Kim SH, et al. Early surgery versus conventional treatment for infective endocarditis. N Engl J Med 2012;366:2466-73.
- 7) Iversen K, Ihlemann N, Gill SU, et al. Partial oral versus intravenous antibiotic treatment of endocarditis. N Engl J Med 2019;380:415-24.

5-12 発熱性好中球減少症

- 1) 日本臨床腫瘍学会. 発熱性好中球減少症（FN）診療ガイドライン 改訂第3版. 2024.
- 2) Taplitz RA, Kennedy EB, Bow EJ, et al. Outpatient management of fever and neutropenia in adults treated for malignancy: American Society of Clinical Oncology and Infectious Diseases Society of America clinical practice guideline update. J Clin Oncol 2018;36:1443-53.
- 3) Klastersky J, Paesmans M, Rubenstein EB, et al. The Multinational Association for Supportive Care in Cancer risk index: a multinational scoring system for identifying low-risk febrile neutropenic cancer patients. J Clin Oncol 2000;18:3038-51.
- 4) Carmona-Bayonas A, Jiménez-Fonseca P, Virizuela Echaburu J, et al. Prediction of serious complications in patients with seemingly stable febrile neutropenia: validation of the Clinical Index of Stable Febrile Neutropenia in a prospective cohort of patients from the FINITE study. J Clin Oncol 2015;33:465-71.
- 5) Freifeld A, Marchigiani D, Walsh T, et al. A double-blind comparison of empirical oral and intravenous antibiotic therapy for low-risk febrile patients with neutropenia during cancer chemotherapy. N Engl J Med 1999;341:305-11.
- 6) Kern WV, Cometta A, De Bock R, et al. Oral versus intravenous empirical antimicrobial therapy for fever in patients with granulocytopenia who are receiving cancer chemotherapy. N Engl J Med 1999;341:312-8.
- 7) 各薬剤の電子添文（KEGG MEDICUSで2026年9～10月に確認）.
- 8) Paul M, Soares-Weiser K, Leibovici L. Beta lactam monotherapy versus beta lactam-aminoglycoside combination therapy for fever with neutropenia: systematic review and meta-analysis. BMJ 2003;326:1111.
- 9) Freifeld AG, Bow EJ, Sepkowitz KA, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the Infectious Diseases Society of America. Clin Infect Dis 2011;52:e56-93.

- 10) Walsh TJ, Teppler H, Donowitz GR, et al. Caspofungin versus liposomal amphotericin B for empirical antifungal therapy in patients with persistent fever and neutropenia. *N Engl J Med* 2004;351:1391-402.
- 11) Aguilar-Guisado M, Espigado I, Martín-Peña A, et al. Optimisation of empirical antimicrobial therapy in patients with haematological malignancies and febrile neutropenia (How Long study): an open-label, randomised, controlled phase 4 trial. *Lancet Haematol* 2017;4:e573-83.
- 12) Bucaneve G, Micozzi A, Menichetti F, et al. Levofloxacin to prevent bacterial infection in patients with cancer and neutropenia. *N Engl J Med* 2005;353:977-87.
- 13) Gardner A, Mattiuzzi G, Faderl S, et al. Randomized comparison of cooked and noncooked diets in patients undergoing remission induction therapy for acute myeloid leukemia. *J Clin Oncol* 2008;26:5684-8.
- 14) Stella F, Marasco V, Levati GV, et al. Nonrestrictive diet does not increase infections during post-HSCT neutropenia: data from a multicenter randomized trial. *Blood Adv* 2023;7:5996-6004.
- 15) Sonbol MB, Jain T, Firwana B, et al. Neutropenic diets to prevent cancer infections: updated systematic review and meta-analysis. *BMJ Support Palliat Care* 2019;9:425-33.

5-13 心不全

- 1) 日本循環器学会, 日本心不全学会. 2025年改訂版 心不全診療ガイドライン. 2025.
- 2) Mebazaa A, Gheorghiade M, Piña IL, et al. Practical recommendations for prehospital and early in-hospital management of patients presenting with acute heart failure syndromes. *Crit Care Med* 2008;36(1 Suppl):S129-39.
- 3) 日本循環器学会, 日本心臓リハビリテーション学会. 2021年改訂版 心血管疾患におけるリハビリテーションに関するガイドライン. 2021.
- 4) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 5) Felker GM, Lee KL, Bull DA, et al. Diuretic strategies in patients with acute decompensated heart failure. *N Engl J Med* 2011;364:797-805.
- 6) Mullens W, Dauw J, Martens P, et al. Acetazolamide in acute decompensated heart failure with volume overload. *N Engl J Med* 2022;387:1185-95.
- 7) Trullàs JC, Morales-Rull JL, Casado J, et al. Combining loop with thiazide diuretics for decompensated heart failure: the CLOROTIC trial. *Eur Heart J* 2023;44:411-21.
- 8) Devereux G, Cotton S, Nath M, et al. Bisoprolol in patients with chronic obstructive pulmonary disease at high risk of exacerbation: the BICS randomized clinical trial. *JAMA* 2024;332:462-70.
- 9) Velazquez EJ, Morrow DA, DeVore AD, et al. Angiotensin-neprilysin inhibition in acute decompensated heart failure. *N Engl J Med* 2019;380:539-48.
- 10) Voors AA, Angermann CE, Teerlink JR, et al. The SGLT2 inhibitor empagliflozin in patients hospitalized for acute heart failure: a multinational randomized trial. *Nat Med* 2022;28:568-74.
- 11) Mebazaa A, Davison B, Chioncel O, et al. Safety, tolerability and efficacy of up-titration of guideline-directed medical therapies for acute heart failure (STRONG-HF): a multinational, open-label, randomised, trial. *Lancet* 2022;400:1938-52.
- 12) Ezekowitz JA, Colin-Ramirez E, Ross H, et al. Reduction of dietary sodium to less than 100 mmol in heart failure (SODIUM-HF): an international, open-label, randomised, controlled trial. *Lancet* 2022;399:1391-400.
- 13) Ellison DH, Felker GM. Diuretic treatment in heart failure. *N Engl J Med* 2017;377:1964-75.

5-14 急性冠症候群

- 1) 日本循環器学会, ほか. 急性冠症候群ガイドライン (2018年改訂版) . 2019.

- 2) Templin C, Ghadri JR, Diekmann J, et al. Clinical features and outcomes of takotsubo (stress) cardiomyopathy. *N Engl J Med* 2015;373:929-38.
- 3) Ghadri JR, Wittstein IS, Prasad A, et al. International expert consensus document on takotsubo syndrome (part II): diagnostic workup, outcome, and management. *Eur Heart J* 2018;39:2047-62.
- 4) 日本循環器学会, ほか. 2020年JCSガイドライン フォーカスアップデート版 冠動脈疾患患者における抗血栓療法. 2020.
- 5) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認) .
- 6) Byrne RA, Rossello X, Coughlan JJ, et al. 2023 ESC Guidelines for the management of acute coronary syndromes. *Eur Heart J* 2023;44:3720-826.
- 7) 日本心臓リハビリテーション学会. 急性冠症候群の心臓リハビリテーション標準プログラム (2023年版) . 2023.
- 8) Fukumoto Y, Tsutsui H, Tsuchihashi M, Masumoto A, Takeshita A. The incidence and risk factors of cholesterol embolization syndrome, a complication of cardiac catheterization: a prospective study. *J Am Coll Cardiol* 2003;42:211-6.
- 9) Kristensen AMD, Rossello X, Atar D, et al. Beta-blockers after myocardial infarction with normal ejection fraction. *N Engl J Med* 2026;394:540-50.
- 10) Watanabe H, Morimoto T, Natsuaki M, et al. Comparison of clopidogrel monotherapy after 1 to 2 months of dual antiplatelet therapy with 12 months of dual antiplatelet therapy in patients with acute coronary syndrome: the STOPDAPT-2 ACS randomized clinical trial. *JAMA Cardiol* 2022;7:407-17.
- 11) Kunadian V, Mossop H, Shields C, et al. Invasive treatment strategy for older patients with myocardial infarction. *N Engl J Med* 2024;391:1673-84.

5-15 心房細動

- 1) 日本循環器学会, 日本不整脈心電学会. 2024年JCS/JHRSガイドライン フォーカスアップデート版 不整脈治療. 2024.
- 2) 日本循環器学会, 日本不整脈心電学会. 2020年改訂版 不整脈薬物治療ガイドライン. 2020.
- 3) 各薬剤の電子添文 (KEGG MEDICUSで2026年9月に確認) .
- 4) Van Gelder IC, Groenveld HF, Crijns HJ, et al. Lenient versus strict rate control in patients with atrial fibrillation. *N Engl J Med* 2010;362:1363-73.
- 5) Kirchhof P, Camm AJ, Goette A, et al. Early rhythm-control therapy in patients with atrial fibrillation. *N Engl J Med* 2020;383:1305-16.
- 6) Pluymaekers NAHA, Dudink EAMP, Luermans JGLM, et al. Early or delayed cardioversion in recent-onset atrial fibrillation. *N Engl J Med* 2019;380:1499-508.
- 7) Cardiac Arrhythmia Suppression Trial (CAST) Investigators. Preliminary report: effect of encainide and flecainide on mortality in a randomized trial of arrhythmia suppression after myocardial infarction. *N Engl J Med* 1989;321:406-12.
- 8) Okumura K, Tomita H, Nakai M, et al. A novel risk stratification system for ischemic stroke in Japanese patients with non-valvular atrial fibrillation. *Circ J* 2021;85:1254-62.
- 9) Yasuda S, Kaikita K, Akao M, et al. Antithrombotic therapy for atrial fibrillation with stable coronary disease. *N Engl J Med* 2019;381:1103-13.
- 10) Joglar JA, Chung MK, Armbruster AL, et al. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation. *Circulation* 2024;149:e1-156.
- 11) 日本循環器学会, 日本血管外科学会. 2022年改訂版 末梢動脈疾患ガイドライン. 2022.
- 12) Voskoboinik A, Kalman JM, De Silva A, et al. Alcohol abstinence in drinkers with atrial fibrillation. *N Engl J Med* 2020;382:20-8.

5-16 深部静脈血栓症・肺塞栓症

- 1) 日本循環器学会, 日本肺高血圧・肺循環学会. 肺血栓塞栓症・深部静脈血栓症および肺高血圧症に関するガイドライン (2025年改訂版). 2025.
- 2) 日本医療安全調査機構. 医療事故の再発防止に向けた提言 第2号 急性肺血栓塞栓症に係る死亡事例の分析. 2017.
- 3) Jiménez D, Aujesky D, Moores L, et al. Simplification of the pulmonary embolism severity index for prognostication in patients with acute symptomatic pulmonary embolism. *Arch Intern Med* 2010;170:1383-9.
- 4) 各薬剤の電子添文 (KEGG MEDICUSで2026年9～10月に確認) .
- 5) Resuscitation Council UK. 2025 Resuscitation Guidelines: Adult advanced life support guidelines. 2025.
- 6) Meyer G, Vicaut E, Danays T, et al. Fibrinolysis for patients with intermediate-risk pulmonary embolism. *N Engl J Med* 2014;370:1402-11.
- 7) EINSTEIN-PE Investigators. Oral rivaroxaban for the treatment of symptomatic pulmonary embolism. *N Engl J Med* 2012;366:1287-97.
- 8) Agnelli G, Buller HR, Cohen A, et al. Oral apixaban for the treatment of acute venous thromboembolism. *N Engl J Med* 2013;369:799-808.
- 9) Hokusai-VTE Investigators. Edoxaban versus warfarin for the treatment of symptomatic venous thromboembolism. *N Engl J Med* 2013;369:1406-15.
- 10) Raschke RA, Reilly BM, Guidry JR, et al. The weight-based heparin dosing nomogram compared with a "standard care" nomogram: a randomized controlled trial. *Ann Intern Med* 1993;119:874-81.
- 11) Garcia DA, Baglin TP, Weitz JI, et al. Parenteral anticoagulants: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest* 2012;141(2 Suppl):e24S-43S.
- 12) Raskob GE, van Es N, Verhamme P, et al. Edoxaban for the treatment of cancer-associated venous thromboembolism. *N Engl J Med* 2018;378:615-24.
- 13) Agnelli G, Becattini C, Meyer G, et al. Apixaban for the treatment of venous thromboembolism associated with cancer. *N Engl J Med* 2020;382:1599-607.
- 14) Righini M, Galanaud JP, Guenneguez H, et al. Anticoagulant therapy for symptomatic calf deep vein thrombosis (CACTUS): a randomised, double-blind, placebo-controlled trial. *Lancet Haematol* 2016;3:e556-62.
- 15) Kirkilesis G, Kakkos SK, Bicknell C, et al. Treatment of distal deep vein thrombosis. *Cochrane Database Syst Rev* 2020;4:CD013422.
- 16) Ageno W, Bertù L, Bucherini E, et al. Rivaroxaban treatment for six weeks versus three months in patients with symptomatic isolated distal deep vein thrombosis: randomised controlled trial. *BMJ* 2022;379:e072623.

5-17 COPD増悪

- 1) 日本呼吸器学会COPDガイドライン第6版作成委員会編. COPD (慢性閉塞性肺疾患) 診断と治療のためのガイドライン2022 [第6版] . メディカルレビュー社 ; 2022.
- 2) Couturaud F, Bertoletti L, Pastre J, et al. Prevalence of pulmonary embolism among patients with COPD hospitalized with acutely worsening respiratory symptoms. *JAMA* 2021;325:59-68.
- 3) Jiménez D, Agustí A, Tabernero E, et al. Effect of a pulmonary embolism diagnostic strategy on clinical outcomes in patients hospitalized for COPD exacerbation: a randomized clinical trial. *JAMA* 2021;326:1277-85.

- 4) Austin MA, Wills KE, Blizzard L, et al. Effect of high flow oxygen on mortality in chronic obstructive pulmonary disease patients in prehospital setting: randomised controlled trial. *BMJ* 2010;341:c5462.
- 5) Global Initiative for Chronic Obstructive Lung Disease. Global strategy for prevention, diagnosis and management of COPD: 2026 report. 2025.
- 6) 各薬剤の電子添文（KEGG MEDICUSで2026年9月に確認）。
- 7) Stefan MS, Rothberg MB, Priya A, Pekow PS, Au DH, Lindenauer PK. Association between β -blocker therapy and outcomes in patients hospitalised with acute exacerbations of chronic obstructive lung disease with underlying ischaemic heart disease, heart failure or hypertension. *Thorax* 2012;67:977-84.
- 8) Leuppi JD, Schuetz P, Bingisser R, et al. Short-term vs conventional glucocorticoid therapy in acute exacerbations of chronic obstructive pulmonary disease: the REDUCE randomized clinical trial. *JAMA* 2013;309:2223-31.
- 9) Sivapalan P, Lapperre TS, Janner J, et al. Eosinophil-guided corticosteroid therapy in patients admitted to hospital with COPD exacerbation (CORTICO-COP): a multicentre, randomised, controlled, open-label, non-inferiority trial. *Lancet Respir Med* 2019;7:699-709.
- 10) Vollenweider DJ, Frei A, Steurer-Stey CA, et al. Antibiotics for exacerbations of chronic obstructive pulmonary disease. *Cochrane Database Syst Rev* 2018;10:CD010257.
- 11) Osadnik CR, Tee VS, Carson-Chahhoud KV, et al. Non-invasive ventilation for the management of acute hypercapnic respiratory failure due to exacerbation of chronic obstructive pulmonary disease. *Cochrane Database Syst Rev* 2017;7:CD004104.
- 12) Plant PK, Owen JL, Elliott MW. Early use of non-invasive ventilation for acute exacerbations of chronic obstructive pulmonary disease on general respiratory wards: a multicentre randomised controlled trial. *Lancet* 2000;355:1931-5.
- 13) Puhan MA, Gimeno-Santos E, Cates CJ, et al. Pulmonary rehabilitation following exacerbations of chronic obstructive pulmonary disease. *Cochrane Database Syst Rev* 2016;12:CD005305.
- 14) Greening NJ, Williams JE, Hussain SF, et al. An early rehabilitation intervention to enhance recovery during hospital admission for an exacerbation of chronic respiratory disease: randomised controlled trial. *BMJ* 2014;349:g4315.
- 15) Aubier M, Murciano D, Milic-Emili J, et al. Effects of the administration of O₂ on ventilation and blood gases in patients with chronic obstructive pulmonary disease during acute respiratory failure. *Am Rev Respir Dis* 1980;122:747-54.
- 16) Robinson TD, Freiberg DB, Regnis JA, Young IH. The role of hypoventilation and ventilation-perfusion redistribution in oxygen-induced hypercapnia during acute exacerbations of chronic obstructive pulmonary disease. *Am J Respir Crit Care Med* 2000;161:1524-9.
- 17) Abdo WF, Heunks LMA. Oxygen-induced hypercapnia in COPD: myths and facts. *Crit Care* 2012;16:323.

5-18 喘息発作

- 1) Niimi A, Fukunaga K, Taniguchi M, et al. Executive summary: Japanese guidelines for adult asthma (JGL) 2021. *Allergol Int* 2023;72:207-26.
- 2) Global Initiative for Asthma. Global strategy for asthma management and prevention, 2026 update. 2026.
- 3) Kirkland SW, Vandenberghe C, Voaklander B, et al. Combined inhaled beta-agonist and anticholinergic agents for emergency management in adults with asthma. *Cochrane Database Syst Rev* 2017;1:CD001284.
- 4) Cates CJ, Welsh EJ, Rowe BH. Holding chambers (spacers) versus nebulisers for beta-agonist treatment of acute asthma. *Cochrane Database Syst Rev* 2013;(9):CD000052.

- 5) Manser R, Reid D, Abramson M. Corticosteroids for acute severe asthma in hospitalised patients. *Cochrane Database Syst Rev* 2001;(1):CD001740.
- 6) 厚生労働省. 重篤副作用疾患別対応マニュアル 非ステロイド性抗炎症薬による喘息発作（アスピリン喘息、解熱鎮痛薬喘息、アスピリン不耐喘息、NSAIDs過敏喘息）. 2006（2022年改定）.
- 7) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）.
- 8) Nair P, Milan SJ, Rowe BH. Addition of intravenous aminophylline to inhaled beta2-agonists in adults with acute asthma. *Cochrane Database Syst Rev* 2012;12:CD002742.
- 9) Kew KM, Kirtchuk L, Michell CI. Intravenous magnesium sulfate for treating adults with acute asthma in the emergency department. *Cochrane Database Syst Rev* 2014;(5):CD010909.
- 10) Goodacre S, Cohen J, Bradburn M, et al. Intravenous or nebulised magnesium sulphate versus standard therapy for severe acute asthma (3Mg trial): a double-blind, randomised controlled trial. *Lancet Respir Med* 2013;1:293-300.

5-19 胸水

- 1) 日本呼吸器外科学会. 膿胸治療ガイドライン. 2023.
- 2) 日本肺癌学会. 肺癌診療ガイドライン2025年版. 2025.
- 3) 厚生労働省. 結核医療の基準（平成21年厚生労働省告示第16号，令和7年3月27日改正）.
- 4) Roberts ME, Rahman NM, Maskell NA, et al. British Thoracic Society Guideline for pleural disease. *Thorax* 2023;78(Suppl 3):s1-s42.
- 5) Asciak R, Bedawi EO, Bhatnagar R, et al. British Thoracic Society Clinical Statement on pleural procedures. *Thorax* 2023;78(Suppl 3):s43-s68.
- 6) Ault MJ, Rosen BT, Scher J, et al. Thoracentesis outcomes: a 12-year experience. *Thorax* 2015;70:127-32.
- 7) Light RW, Macgregor MI, Luchsinger PC, Ball WC Jr. Pleural effusions: the diagnostic separation of transudates and exudates. *Ann Intern Med* 1972;77:507-13.
- 8) Bielsa S, Porcel JM, Castellote J, et al. Solving the Light's criteria misclassification rate of cardiac and hepatic transudates. *Respirology* 2012;17:721-6.
- 9) Roth BJ, O'Meara TF, Cragun WH. The serum-effusion albumin gradient in the evaluation of pleural effusions. *Chest* 1990;98:546-9.
- 10) Aggarwal AN, Agarwal R, Sehgal IS, Dhooria S. Adenosine deaminase for diagnosis of tuberculous pleural effusion: a systematic review and meta-analysis. *PLoS One* 2019;14:e0213728.
- 11) Arnold DT, De Fonseca D, Perry S, et al. Investigating unilateral pleural effusions: the role of cytology. *Eur Respir J* 2018;52:1801254.
- 12) Maskell NA, Davies CW, Nunn AJ, et al. U.K. controlled trial of intrapleural streptokinase for pleural infection. *N Engl J Med* 2005;352:865-74.
- 13) Rahman NM, Maskell NA, West A, et al. Intrapleural use of tissue plasminogen activator and DNase in pleural infection. *N Engl J Med* 2011;365:518-26.
- 14) ユニタルク胸膜腔内注入用懸濁剤4g 電子添文（KEGG MEDICUSで2026年10月に確認）.
- 15) Davies HE, Mishra EK, Kahan BC, et al. Effect of an indwelling pleural catheter vs chest tube and talc pleurodesis for relieving dyspnea in patients with malignant pleural effusion: the TIME2 randomized controlled trial. *JAMA* 2012;307:2383-9.
- 16) Thomas R, Fysh ETH, Smith NA, et al. Effect of an indwelling pleural catheter vs talc pleurodesis on hospitalization days in patients with malignant pleural effusion: the AMPLE randomized clinical trial. *JAMA* 2017;318:1903-12.
- 17) Glargaard S, Thomsen JH, Tuxen C, et al. A randomized controlled trial of thoracentesis in acute heart failure. *Circulation* 2025;151:1150-61.

5-20 気胸

- 1) American College of Surgeons Committee on Trauma. Advanced Trauma Life Support: student course manual. 10th ed. Chicago: American College of Surgeons; 2018.
- 2) Laan DV, Vu TD, Thiels CA, et al. Chest wall thickness and decompression failure: a systematic review and meta-analysis comparing anatomic locations in needle thoracostomy. *Injury* 2016;47:797-804.
- 3) Alrajhi K, Woo MY, Vaillancourt C. Test characteristics of ultrasonography for the detection of pneumothorax: a systematic review and meta-analysis. *Chest* 2012;141:703-8.
- 4) 日本気胸・嚢胞性肺疾患学会編. 気胸・嚢胞性肺疾患 規約・用語・ガイドライン 2009年版. 金原出版; 2009.
- 5) Roberts ME, Rahman NM, Maskell NA, et al. British Thoracic Society Guideline for pleural disease. *Thorax* 2023;78(Suppl 3):s1-42.
- 6) Brown SGA, Ball EL, Perrin K, et al. Conservative versus interventional treatment for spontaneous pneumothorax. *N Engl J Med* 2020;382:405-15.
- 7) Carson-Chahhoud KV, Wakai A, van Agteren JE, et al. Simple aspiration versus intercostal tube drainage for primary spontaneous pneumothorax in adults. *Cochrane Database Syst Rev* 2017;9:CD004479.
- 8) Marx T, Joly LM, Parmentier AL, et al. Simple aspiration versus drainage for complete pneumothorax: a randomized noninferiority trial. *Am J Respir Crit Care Med* 2023;207:1475-85.
- 9) Hallifax RJ, McKeown E, Sivakumar P, et al. Ambulatory management of primary spontaneous pneumothorax: an open-label, randomised controlled trial. *Lancet* 2020;396:39-49.
- 10) Asciak R, Bedawi EO, Bhatnagar R, et al. British Thoracic Society Clinical Statement on pleural procedures. *Thorax* 2023;78(Suppl 3):s43-68.
- 11) キシロカイン注ポリアンブ1%の電子添文（KEGG MEDICUSで2026年10月に確認）.
- 12) 日本気胸・嚢胞性肺疾患学会 学術委員会. 低圧持続吸引器の圧単位に係る声明. 2019.
- 13) Parienti JJ, Mongardon N, Mégarbane B, et al. Intravascular complications of central venous catheterization by insertion site. *N Engl J Med* 2015;373:1220-9.
- 14) Gordon CE, Feller-Kopman D, Balk EM, et al. Pneumothorax following thoracentesis: a systematic review and meta-analysis. *Arch Intern Med* 2010;170:332-9.
- 15) Joseph J, Sahn SA. Thoracic endometriosis syndrome: new observations from an analysis of 110 cases. *Am J Med* 1996;100:164-70.
- 16) Walker SP, Bibby AC, Halford P, et al. Recurrence rates in primary spontaneous pneumothorax: a systematic review and meta-analysis. *Eur Respir J* 2018;52:1800864.
- 17) Northfield TC. Oxygen therapy for spontaneous pneumothorax. *Br Med J* 1971;4:86-8.
- 18) Park CB, Moon MH, Jeon HW, et al. Does oxygen therapy increase the resolution rate of primary spontaneous pneumothorax? *J Thorac Dis* 2017;9:5239-43.

5-21 CKD患者の入院管理

- 1) 日本腎臓学会編. エビデンスに基づくCKD診療ガイドライン2023. 東京医学社; 2023.
- 2) 日本腎臓学会編. CKD診療ガイド2024. 東京医学社; 2024.
- 3) Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int* 2024;105(4S):S117-314.
- 4) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）.

- 5) Heerspink HJL, Stefánsson BV, Correa-Rotter R, et al. Dapagliflozin in patients with chronic kidney disease. *N Engl J Med* 2020;383:1436-46.
- 6) EMPA-KIDNEY Collaborative Group. Empagliflozin in patients with chronic kidney disease. *N Engl J Med* 2023;388:117-27.
- 7) Bhandari S, Mehta S, Khwaja A, et al. Renin-angiotensin system inhibition in advanced chronic kidney disease. *N Engl J Med* 2022;387:2021-32.
- 8) 日本腎臓学会編. 慢性腎臓病に対する食事療法基準2014年版. 日腎会誌 2014;56:553-99.
- 9) UK Kidney Association. Clinical practice guideline: management of hyperkalaemia in adults. Updated July 2026.
- 10) de Brito-Ashurst I, Varagunam M, Rafferty MJ, Yaqoob MM. Bicarbonate supplementation slows progression of CKD and improves nutritional status. *J Am Soc Nephrol* 2009;20:2075-84.
- 11) 日本透析医学会. 慢性腎臓病に伴う骨・ミネラル代謝異常の診療ガイドライン（2025年改訂版）. 透析会誌 2026;59:127-224.
- 12) Pfeffer MA, Burdmann EA, Chen CY, et al. A trial of darbepoetin alfa in type 2 diabetes and chronic kidney disease. *N Engl J Med* 2009;361:2019-32.
- 13) Singh AK, Szczec L, Tang KL, et al. Correction of anemia with epoetin alfa in chronic kidney disease. *N Engl J Med* 2006;355:2085-98.
- 14) 日本腎臓学会. HIF-PH阻害薬適正使用に関するrecommendation（2020年9月29日版）. 日腎会誌 2020;62:711-6.

5-22 透析患者の入院管理

- 1) 日本透析医学会. 維持血液透析ガイドライン：血液透析処方. 透析会誌 2013;46:587-632.
- 2) Foley RN, Gilbertson DT, Murray T, Collins AJ. Long interdialytic interval and mortality among patients receiving hemodialysis. *N Engl J Med* 2011;365:1099-107.
- 3) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）.
- 4) 日本透析医学会. 透析の開始と継続に関する意思決定プロセスについての提言. 透析会誌 2020;53:173-217.
- 5) 日本透析医学会. 慢性腎臓病に伴う骨・ミネラル代謝異常の診療ガイドライン（2025年改訂版）. 透析会誌 2026;59:127-224.
- 6) 日本透析医学会. 2015年版 慢性腎臓病患者における腎性貧血治療のガイドライン. 透析会誌 2016;49:89-158.
- 7) Lok CE, Huber TS, Lee T, et al. KDOQI clinical practice guideline for vascular access: 2019 update. *Am J Kidney Dis* 2020;75(4 Suppl 2):S1-164.
- 8) 日本透析医学会. 2011年版 慢性血液透析用バスキュラーアクセスの作製および修復に関するガイドライン. 透析会誌 2011;44:855-937.
- 9) Allon M, Takeshian A, Shanklin N. Effect of insulin-plus-glucose infusion with or without epinephrine on fasting hyperkalemia. *Kidney Int* 1993;43:212-7.
- 10) 日本透析医学会. 慢性透析患者の食事療法基準. 透析会誌 2014;47:287-91.
- 11) UK Kidney Association. Clinical practice guideline: management of hyperkalaemia in adults. Updated July 2026.
- 12) 日本透析医学会. 血液透析患者における心血管合併症の評価と治療に関するガイドライン. 透析会誌 2011;44:337-425.
- 13) Apple FS, Murakami MM, Pearce LA, Herzog CA. Predictive value of cardiac troponin I and T for subsequent death in end-stage renal disease. *Circulation* 2002;106:2941-5.
- 14) 日本循環器学会, 日本不整脈心電学会. 2024年JCS/JHRSガイドライン フォーカスアップデート版 不整脈治療. 2024.

- 15) 日本透析医学会 腹膜透析ガイドライン改訂ワーキンググループ編. 腹膜透析ガイドライン2019. 医学図書出版; 2019.
- 16) Li PK, Chow KM, Cho Y, et al. ISPD peritonitis guideline recommendations: 2022 update on prevention and treatment. *Perit Dial Int* 2022;42:110-53.

5-23 ネフローゼ症候群

- 1) 厚生労働科学研究費補助金難治性疾患政策研究事業 難治性腎障害に関する調査研究班編. エビデンスに基づくネフローゼ症候群診療ガイドライン2020. 東京医学社; 2020.
- 2) Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group. KDIGO 2021 clinical practice guideline for the management of glomerular diseases. *Kidney Int* 2021;100(4S):S1-276.
- 3) Fiorentino M, Bolognani D, Tesar V, et al. Renal biopsy in patients with diabetes: a pooled meta-analysis of 48 studies. *Nephrol Dial Transplant* 2017;32:97-110.
- 4) Liang S, Zhang XG, Cai GY, et al. Identifying parameters to distinguish non-diabetic renal diseases from diabetic nephropathy in patients with type 2 diabetes mellitus: a meta-analysis. *PLoS One* 2013;8:e64184.
- 5) 日本腎臓学会 腎生検ガイドブック改訂委員会編. 腎生検ガイドブック2020. 東京医学社; 2020.
- 6) 日本輸血・細胞治療学会ガイドライン委員会. 輸血療法実践ガイド. 日本輸血細胞治療学会誌 2026;72:521-70.
- 7) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 8) Fervenza FC, Appel GB, Barbour SJ, et al. Rituximab or cyclosporine in the treatment of membranous nephropathy. *N Engl J Med* 2019;381:36-46.
- 9) Fliser D, Zurbrüggen I, Mutschler E, et al. Coadministration of albumin and furosemide in patients with the nephrotic syndrome. *Kidney Int* 1999;55:629-34.
- 10) Kitsios GD, Mascari P, Ettunsi R, Gray AW. Co-administration of furosemide with albumin for overcoming diuretic resistance in patients with hypoalbuminemia: a meta-analysis. *J Crit Care* 2014;29:253-9.

5-24 上部消化管出血

- 1) 藤城光弘, 井口幹崇, 小野敏嗣, ほか. 非静脈瘤性上部消化管出血における内視鏡診療ガイドライン (第2版) . 日本消化器内視鏡学会雑誌 2024;66:1515-38.
- 2) Srygley FD, Gerardo CJ, Tran T, et al. Does this patient have a severe upper gastrointestinal bleed? *JAMA* 2012;307:1072-9.
- 3) 日本輸血・細胞治療学会ガイドライン委員会. 輸血療法実践ガイド. 日本輸血細胞治療学会誌 2026;72:521-70.
- 4) Villanueva C, Colomo A, Bosch A, et al. Transfusion strategies for acute upper gastrointestinal bleeding. *N Engl J Med* 2013;368:11-21.
- 5) de Franchis R, Bosch J, Garcia-Tsao G, et al. Baveno VII - Renewing consensus in portal hypertension. *J Hepatol* 2022;76:959-74.
- 6) Laine L, Barkun AN, Saltzman JR, et al. ACG clinical guideline: upper gastrointestinal and ulcer bleeding. *Am J Gastroenterol* 2021;116:899-917.
- 7) Stanley AJ, Laine L, Dalton HR, et al. Comparison of risk scoring systems for patients presenting with upper gastrointestinal bleeding: international multicentre prospective study. *BMJ* 2017;356:i6432.
- 8) Lau JYW, Yu Y, Tang RSY, et al. Timing of endoscopy for acute upper gastrointestinal bleeding. *N Engl J Med* 2020;382:1299-308.

- 9) Sugimoto M, Kamada T, Ito M, et al. Evidence-based clinical practice guidelines for peptic ulcer disease 2026. J Gastroenterol 2026;61:1361-81.
- 10) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 11) HALT-IT Trial Collaborators. Effects of a high-dose 24-h infusion of tranexamic acid on death and thromboembolic events in patients with acute gastrointestinal bleeding (HALT-IT): an international randomised, double-blind, placebo-controlled trial. Lancet 2020;395:1927-36.
- 12) 日本消化器病学会, 日本肝臓学会. 肝硬変診療ガイドライン2020（改訂第3版）. 2020.
- 13) Gralnek IM, Stanley AJ, Morris AJ, et al. Endoscopic diagnosis and management of nonvariceal upper gastrointestinal hemorrhage (NVUGIH): European Society of Gastrointestinal Endoscopy (ESGE) Guideline - Update 2021. Endoscopy 2021;53:300-32.
- 14) Gisbert JP, Abaira V. Accuracy of *Helicobacter pylori* diagnostic tests in patients with bleeding peptic ulcer: a systematic review and meta-analysis. Am J Gastroenterol 2006;101:848-63.
- 15) Sachar H, Vaidya K, Laine L. Intermittent vs continuous proton pump inhibitor therapy for high-risk bleeding ulcers: a systematic review and meta-analysis. JAMA Intern Med 2014;174:1755-62.
- 16) Green FW Jr, Kaplan MM, Curtis LE, Levine PH. Effect of acid and pepsin on blood coagulation and platelet aggregation: a possible contributor to prolonged gastroduodenal mucosal hemorrhage. Gastroenterology 1978;74:38-43.

5-25 下部消化管出血

- 1) Nagata N, Kobayashi K, Yamauchi A, et al. Identifying bleeding etiologies by endoscopy affected outcomes in 10,342 cases with hematochezia: CODE BLUE-J study. Am J Gastroenterol 2021;116:2222-34.
- 2) Oakland K, Chadwick G, East JE, et al. Diagnosis and management of acute lower gastrointestinal bleeding: guidelines from the British Society of Gastroenterology. Gut 2019;68:776-89.
- 3) Srygley FD, Gerardo CJ, Tran T, et al. Does this patient have a severe upper gastrointestinal bleed? JAMA 2012;307:1072-9.
- 4) Longstreth GF, Yao JF. Epidemiology, clinical features, high-risk factors, and outcome of acute large bowel ischemia. Clin Gastroenterol Hepatol 2009;7:1075-80.
- 5) Sengupta N, Feuerstein JD, Jairath V, et al. Management of patients with acute lower gastrointestinal bleeding: an updated ACG guideline. Am J Gastroenterol 2023;118:208-31.
- 6) 日本消化管学会. 大腸憩室症（憩室出血・憩室炎）ガイドライン. 2017（英語版：Nagata N, Ishii N, Manabe N, et al. Guidelines for colonic diverticular bleeding and colonic diverticulitis: Japan Gastroenterological Association. Digestion 2019;99 Suppl 1:1-26）。
- 7) Niikura R, Nagata N, Yamada A, et al. Efficacy and safety of early vs elective colonoscopy for acute lower gastrointestinal bleeding. Gastroenterology 2020;158:168-75.
- 8) Tsay C, Shung D, Stemmer Frumento K, et al. Early colonoscopy does not improve outcomes of patients with lower gastrointestinal bleeding: systematic review of randomized trials. Clin Gastroenterol Hepatol 2020;18:1696-703.
- 9) ニフレック配合内用剤 電子添文（KEGG MEDICUSで2026年10月に確認）。
- 10) Kobayashi K, Nagata N, Furumoto Y, et al. Effectiveness and adverse events of endoscopic clipping versus band ligation for colonic diverticular hemorrhage: a large-scale multicenter cohort study. Endoscopy 2022;54:735-44.
- 11) 日本輸血・細胞治療学会ガイドライン委員会. 輸血療法実践ガイド. 日本輸血細胞治療学会誌 2026;72:521-70.

- 12) HALT-IT Trial Collaborators. Effects of a high-dose 24-h infusion of tranexamic acid on death and thromboembolic events in patients with acute gastrointestinal bleeding (HALT-IT): an international randomised, double-blind, placebo-controlled trial. *Lancet* 2020;395:1927-36.
- 13) Chan FK, Leung Ki EL, Wong GL, et al. Risks of bleeding recurrence and cardiovascular events with continued aspirin use after lower gastrointestinal hemorrhage. *Gastroenterology* 2016;151:271-7.
- 14) Omori J, Kaise M, Nagata N, et al. Characteristics, outcomes, and risk factors of surgery for acute lower gastrointestinal bleeding: nationwide cohort study of 10,342 hematochezia cases. *J Gastroenterol* 2024;59:24-33.

5-26 腸閉塞

- 1) 急性腹症診療ガイドライン2025改訂出版委員会編. 急性腹症診療ガイドライン2025 第2版. 医学書院; 2025.
- 2) Ten Broek RPG, Krielen P, Di Saverio S, et al. Bologna guidelines for diagnosis and management of adhesive small bowel obstruction (ASBO): 2017 update of the evidence-based guidelines from the world society of emergency surgery ASBO working group. *World J Emerg Surg* 2018;13:24.
- 3) Millet I, Taourel P, Ruyer A, et al. Value of CT findings to predict surgical ischemia in small bowel obstruction: a systematic review and meta-analysis. *Eur Radiol* 2015;25:1823-35.
- 4) Millet I, Boutot D, Faget C, et al. Assessment of strangulation in adhesive small bowel obstruction on the basis of combined CT findings: implications for clinical care. *Radiology* 2017;285:798-808.
- 5) Katano T, Shimura T, Nishie H, et al. The first management using intubation of a nasogastric tube with Gastrografin enterography or long tube for non-strangulated acute small bowel obstruction: a multicenter, randomized controlled trial. *J Gastroenterol* 2020;55:858-67.
- 6) Ceresoli M, Coccolini F, Catena F, et al. Water-soluble contrast agent in adhesive small bowel obstruction: a systematic review and meta-analysis of diagnostic and therapeutic value. *Am J Surg* 2016;211:1114-25.
- 7) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）.
- 8) Keenan JE, Turley RS, McCoy CC, et al. Trials of nonoperative management exceeding 3 days are associated with increased morbidity in patients undergoing surgery for uncomplicated adhesive small bowel obstruction. *J Trauma Acute Care Surg* 2014;76:1367-72.
- 9) Pisano M, Zorcolo L, Merli C, et al. 2017 WSES guidelines on colon and rectal cancer emergencies: obstruction and perforation. *World J Emerg Surg* 2018;13:36.
- 10) Tian BWCA, Vigutto G, Tan E, et al. WSES consensus guidelines on sigmoid volvulus management. *World J Emerg Surg* 2023;18:34.
- 11) Al-Mashat A, Fareed A, Senanayake T, Smith SR, Gani J. Do nasogastric or nasoenteric tubes improve outcomes from adhesional small bowel obstruction: a systematic review and meta-analysis. *BMC Surg* 2025;25:545.
- 12) Shinohara K, Asaba Y, Ishida T, Maeta T, Suzuki M, Mizukami Y. Nonoperative management without nasogastric tube decompression for adhesive small bowel obstruction. *Am J Surg* 2022;223:1179-82.

5-27 急性膵炎

- 1) 急性膵炎診療ガイドライン2021改訂出版委員会編. 急性膵炎診療ガイドライン2021 第5版. 金原出版; 2021.

- 2) 武田和憲, 大槻 眞, 北川元二, 他. 急性膵炎の診断基準・重症度判定基準最終改訂案. 厚生労働科学研究費補助金難治性疾患克服研究事業 難治性膵疾患に関する調査研究 平成17年度総括・分担研究報告書. 2006;27-34.
- 3) Rompianesi G, Hann A, Komolafe O, et al. Serum amylase and lipase and urinary trypsinogen and amylase for diagnosis of acute pancreatitis. *Cochrane Database Syst Rev* 2017;4:CD012010.
- 4) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 5) Tenner S, Dubner H, Steinberg W. Predicting gallstone pancreatitis with laboratory parameters: a meta-analysis. *Am J Gastroenterol* 1994;89:1863-6.
- 6) de-Madaria E, Buxbaum JL, Maisonneuve P, et al. Aggressive or moderate fluid resuscitation in acute pancreatitis. *N Engl J Med* 2022;387:989-1000.
- 7) Bolado F, Buxbaum JL, Vaillo-Rocamora A, et al. Early weight-based aggressive vs. non-aggressive goal-directed fluid resuscitation in the early phase of acute pancreatitis: an open-label multicenter randomized controlled trial (the WATERFALL trial), design, and rationale. *Front Med (Lausanne)* 2020;7:440.
- 8) Bakker OJ, van Brunschot S, van Santvoort HC, et al. Early versus on-demand nasoenteric tube feeding in acute pancreatitis. *N Engl J Med* 2014;371:1983-93.
- 9) Schepers NJ, Hallensleben NDL, Besselink MG, et al. Urgent endoscopic retrograde cholangiopancreatography with sphincterotomy versus conservative treatment in predicted severe acute gallstone pancreatitis (APEC): a multicentre randomised controlled trial. *Lancet* 2020;396:167-76.
- 10) Akshintala VS, Kanthasamy K, Bhullar FA, et al. Incidence, severity, and mortality of post-ERCP pancreatitis: an updated systematic review and meta-analysis of 145 randomized controlled trials. *Gastrointest Endosc* 2023;98:1-6.e12.
- 11) Dumonceau JM, Kapral C, Aabakken L, et al. ERCP-related adverse events: European Society of Gastrointestinal Endoscopy (ESGE) Guideline. *Endoscopy* 2020;52:127-49.
- 12) Banks PA, Bollen TL, Dervenis C, et al. Classification of acute pancreatitis--2012: revision of the Atlanta classification and definitions by international consensus. *Gut* 2013;62:102-11.
- 13) van Santvoort HC, Besselink MG, Bakker OJ, et al. A step-up approach or open necrosectomy for necrotizing pancreatitis. *N Engl J Med* 2010;362:1491-502.
- 14) Boxhoorn L, van Dijk SM, van Grinsven J, et al. Immediate versus postponed intervention for infected necrotizing pancreatitis. *N Engl J Med* 2021;385:1372-81.
- 15) da Costa DW, Bouwense SA, Schepers NJ, et al. Same-admission versus interval cholecystectomy for mild gallstone pancreatitis (PONCHO): a multicentre randomised controlled trial. *Lancet* 2015;386:1261-8.
- 16) Nordback I, Pelli H, Lappalainen-Lehto R, et al. The recurrence of acute alcohol-associated pancreatitis can be reduced: a randomized controlled trial. *Gastroenterology* 2009;136:848-55.
- 17) 日本動脈硬化学会. 動脈硬化性疾患予防ガイドライン2022年版. 2022.
- 18) Das SL, Singh PP, Phillips AR, et al. Newly diagnosed diabetes mellitus after acute pancreatitis: a systematic review and meta-analysis. *Gut* 2014;63:818-31.
- 19) D'Ambra V, Ricci C, Ingaldi C, et al. Non-aggressive lactated Ringer's resuscitation improves outcomes and has a favorable safety profile in acute pancreatitis: a comprehensive systematic review and component network meta-analysis of randomized controlled trials using GRADE and CiNeMA approaches. *Pancreatology* 2026 (オンライン先行) . doi:10.1016/j.pan.2026.08.007.
- 20) Chen H, Lu X, Xu B, Meng C, Xie D. Lactated Ringer solution is superior to normal saline solution in managing acute pancreatitis: an updated meta-analysis of randomized controlled trials. *J Clin Gastroenterol* 2022;56:e114-20.

- 21) de-Madaria E, Herrera-Marante I, González-Camacho V, et al. Fluid resuscitation with lactated Ringer's solution vs normal saline in acute pancreatitis: a triple-blind, randomized, controlled trial. *United European Gastroenterol J* 2018;6:63-72.

5-28 肝硬変

- 1) 日本消化器病学会, 日本肝臓学会. 肝硬変診療ガイドライン2020 (改訂第3版). 2020.
- 2) Biggins SW, Angeli P, Garcia-Tsao G, et al. Diagnosis, evaluation, and management of ascites, spontaneous bacterial peritonitis and hepatorenal syndrome: 2021 practice guidance by the American Association for the Study of Liver Diseases. *Hepatology* 2021;74:1014-48.
- 3) Kim WR, Biggins SW, Kremers WK, et al. Hyponatremia and mortality among patients on the liver-transplant waiting list. *N Engl J Med* 2008;359:1018-26.
- 4) European Association for the Study of the Liver. EASL Clinical Practice Guidelines for the management of patients with decompensated cirrhosis. *J Hepatol* 2018;69:406-60.
- 5) Runyon BA, Montano AA, Akriviadis EA, Antillon MR, Irving MA, McHutchison JG. The serum-ascites albumin gradient is superior to the exudate-transudate concept in the differential diagnosis of ascites. *Ann Intern Med* 1992;117:215-20.
- 6) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 7) Sort P, Navasa M, Arroyo V, et al. Effect of intravenous albumin on renal impairment and mortality in patients with cirrhosis and spontaneous bacterial peritonitis. *N Engl J Med* 1999;341:403-9.
- 8) Akriviadis EA, Runyon BA. Utility of an algorithm in differentiating spontaneous from secondary bacterial peritonitis. *Gastroenterology* 1990;98:127-33.
- 9) China L, Freemantle N, Forrest E, et al. A randomized trial of albumin infusions in hospitalized patients with cirrhosis. *N Engl J Med* 2021;384:808-17.
- 10) Caraceni P, Riggio O, Angeli P, et al. Long-term albumin administration in decompensated cirrhosis (ANSWER): an open-label randomised trial. *Lancet* 2018;391:2417-29.
- 11) 日本輸血・細胞治療学会ガイドライン委員会. 輸血療法実践ガイド. 日本輸血細胞治療学会誌 2026;72:521-70.
- 12) Bass NM, Mullen KD, Sanyal A, et al. Rifaximin treatment in hepatic encephalopathy. *N Engl J Med* 2010;362:1071-81.
- 13) Tan JL, Lokan T, Chinnaratha MA, Veysey M. Risk of bleeding after abdominal paracentesis in patients with chronic liver disease and coagulopathy: a systematic review and meta-analysis. *JGH Open* 2024;8:e70013.
- 14) Rowley MW, Agarwal S, Seetharam AB, Hirsch KS. Real-time ultrasound-guided paracentesis by radiologists: near zero risk of hemorrhage without correction of coagulopathy. *J Vasc Interv Radiol* 2019;30:259-64.
- 15) Tripodi A, Chantarangkul V, Primignani M, et al. Thrombin generation in plasma from patients with cirrhosis supplemented with normal plasma: considerations on the efficacy of treatment with fresh-frozen plasma. *Intern Emerg Med* 2012;7:139-44.

5-29 DKA・HHS

- 1) 日本糖尿病学会. 糖尿病診療ガイドライン2024. 第20章 糖尿病における急性代謝失調・シックデイ (感染症を含む). 2024.
- 2) Umpierrez GE, Davis GM, ElSayed NA, et al. Hyperglycemic crises in adults with diabetes: a consensus report. *Diabetes Care* 2024;47:1257-75.
- 3) 日本糖尿病学会 SGLT2阻害薬の適正使用に関する委員会. 糖尿病治療におけるSGLT2阻害薬の適正使用に関するRecommendation. 2022年7月改訂.

- 4) Imagawa A, Hanafusa T, Awata T, et al. Report of the Committee of the Japan Diabetes Society on the Research of Fulminant and Acute-onset Type 1 Diabetes Mellitus: new diagnostic criteria of fulminant type 1 diabetes mellitus (2012). *J Diabetes Investig* 2012;3:536-9.
- 5) Kitabchi AE, Umpierrez GE, Miles JM, Fisher JN. Hyperglycemic crises in adult patients with diabetes. *Diabetes Care* 2009;32:1335-43.
- 6) Self WH, Evans CS, Jenkins CA, et al. Clinical effects of balanced crystalloids vs saline in adults with diabetic ketoacidosis: a subgroup analysis of cluster randomized clinical trials. *JAMA Netw Open* 2020;3:e2024596.
- 7) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 8) Joint British Diabetes Societies for Inpatient Care Group. The management of diabetic ketoacidosis in adults. Revised March 2023.
- 9) Morris LR, Murphy MB, Kitabchi AE. Bicarbonate therapy in severe diabetic ketoacidosis. *Ann Intern Med* 1986;105:836-40.
- 10) Mustafa OG, Haq M, Dashora U, et al. Management of hyperosmolar hyperglycaemic state (HHS) in adults: an updated guideline from the Joint British Diabetes Societies (JBDS) for Inpatient Care Group. *Diabet Med* 2023;40:e15005.
- 11) Kuppermann N, Ghetty S, Schunk JE, et al. Clinical trial of fluid infusion rates for pediatric diabetic ketoacidosis. *N Engl J Med* 2018;378:2275-87.

5-30 副腎不全

- 1) 日本内分泌学会, 日本小児内分泌学会, 日本ステロイドホルモン学会, 厚生労働科学研究費補助金政策研究事業「副腎ホルモン産生異常に関する調査研究」班. 副腎クラーゼを含む副腎皮質機能低下症の診断と治療に関する指針. *日内分泌会誌* 2015;91(Suppl. Sep.):1-78.
- 2) Beuschlein F, Else T, Bancos I, et al. European Society of Endocrinology and Endocrine Society joint clinical guideline: diagnosis and therapy of glucocorticoid-induced adrenal insufficiency. *J Clin Endocrinol Metab* 2024;109:1657-83.
- 3) Hahner S, Loeffler M, Bleicken B, et al. Epidemiology of adrenal crisis in chronic adrenal insufficiency: the need for new prevention strategies. *Eur J Endocrinol* 2010;162:597-602.
- 4) Hahner S, Spinnler C, Fassnacht M, et al. High incidence of adrenal crisis in educated patients with chronic adrenal insufficiency: a prospective study. *J Clin Endocrinol Metab* 2015;100:407-16.
- 5) Bornstein SR, Allolio B, Arlt W, et al. Diagnosis and treatment of primary adrenal insufficiency: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab* 2016;101:364-89.
- 6) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 7) 日本内分泌学会. 免疫チェックポイント阻害薬による内分泌障害の診療ガイドライン. *日内分泌会誌* 2018;94(Suppl. Nov.):1-11.
- 8) Annane D, Pastores SM, Rochweg B, et al. Guidelines for the diagnosis and management of critical illness-related corticosteroid insufficiency (CIRCI) in critically ill patients (Part I): Society of Critical Care Medicine (SCCM) and European Society of Intensive Care Medicine (ESICM) 2017. *Crit Care Med* 2017;45:2078-88.
- 9) Sprung CL, Annane D, Keh D, et al. Hydrocortisone therapy for patients with septic shock. *N Engl J Med* 2008;358:111-24.
- 10) 日本版敗血症診療ガイドライン2024特別委員会. 日本版敗血症診療ガイドライン2024. *日集中医誌* 2024;31:S1165-313.
- 11) Prete A, Taylor AE, Bancos I, et al. Prevention of adrenal crisis: cortisol responses to major stress compared to stress dose hydrocortisone delivery. *J Clin Endocrinol Metab* 2020;105:2262-74.

5-31 脳卒中急性期の内科管理

- 1) 日本脳卒中学会 脳卒中ガイドライン委員会（改訂2025）． 脳卒中治療ガイドライン2021〔改訂2025〕（改訂項目の公開版）． 2025.
- 2) Middleton S, McElduff P, Ward J, et al. Implementation of evidence-based treatment protocols to manage fever, hyperglycaemia, and swallowing dysfunction in acute stroke (QASC): a cluster randomised controlled trial. *Lancet* 2011;378:1699-706.
- 3) Johnston KC, Bruno A, Pauls Q, et al. Intensive vs standard treatment of hyperglycemia and functional outcome in patients with acute ischemic stroke: the SHINE randomized clinical trial. *JAMA* 2019;322:326-35.
- 4) Wang Y, Wang Y, Zhao X, et al. Clopidogrel with aspirin in acute minor stroke or transient ischemic attack. *N Engl J Med* 2013;369:11-9.
- 5) Gao Y, Chen W, Pan Y, et al. Dual antiplatelet treatment up to 72 hours after ischemic stroke. *N Engl J Med* 2023;389:2413-24.
- 6) Johnston SC, Easton JD, Farrant M, et al. Clopidogrel and aspirin in acute ischemic stroke and high-risk TIA. *N Engl J Med* 2018;379:215-25.
- 7) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）．
- 8) Fischer U, Koga M, Strbian D, et al. Early versus later anticoagulation for stroke with atrial fibrillation. *N Engl J Med* 2023;388:2411-21.
- 9) Werring DJ, Dehbi HM, Ahmed N, et al. Optimal timing of anticoagulation after acute ischaemic stroke with atrial fibrillation (OPTIMAS): a multicentre, blinded-endpoint, phase 4, randomised controlled trial. *Lancet* 2024;404:1731-41.
- 10) Fischer U, Trelle S, Branca M, et al. Early versus Late initiation of direct oral Anticoagulants in post-ischaemic stroke patients with atrial fibrillation (ELAN): protocol for an international, multicentre, randomised-controlled, two-arm, open, assessor-blinded trial. *Eur Stroke J* 2022;7:487-95.
- 11) Amarenco P, Bogousslavsky J, Callahan A 3rd, et al. High-dose atorvastatin after stroke or transient ischemic attack. *N Engl J Med* 2006;355:549-59.
- 12) RESTART Collaboration. Effects of antiplatelet therapy after stroke due to intracerebral haemorrhage (RESTART): a randomised, open-label trial. *Lancet* 2019;393:2613-23.
- 13) FOCUS Trial Collaboration. Effects of fluoxetine on functional outcomes after acute stroke (FOCUS): a pragmatic, double-blind, randomised, controlled trial. *Lancet* 2019;393:265-74.
- 14) Vahedi K, Hofmeijer J, Juettler E, et al. Early decompressive surgery in malignant infarction of the middle cerebral artery: a pooled analysis of three randomised controlled trials. *Lancet Neurol* 2007;6:215-22.
- 15) 日本高血圧学会． 高血圧管理・治療ガイドライン2025． 2025． （英語版：Hypertens Res 2026;49:9-235）
- 16) 警察庁交通局運転免許課長． 一定の病気に係る免許の可否等の運用基準（丁運発第68号）． 2022.
- 17) He J, Zhang Y, Xu T, et al. Effects of immediate blood pressure reduction on death and major disability in patients with acute ischemic stroke: the CATIS randomized clinical trial. *JAMA* 2014;311:479-89.
- 18) ENOS Trial Investigators. Efficacy of nitric oxide, with or without continuing antihypertensive treatment, for management of high blood pressure in acute stroke (ENOS): a partial-factorial randomised controlled trial. *Lancet* 2015;385:617-28.
- 19) Sandset EC, Bath PM, Boysen G, et al. The angiotensin-receptor blocker candesartan for treatment of acute stroke (SCAST): a randomised, placebo-controlled, double-blind trial. *Lancet* 2011;377:741-50.

5-32 オンコロジック・エマージェンシー

- 1) 日本臨床腫瘍学会. 腫瘍崩壊症候群 (TLS) 診療ガイドランス 第2版. 2021.
- 2) Cairo MS, Coiffier B, Reiter A, Younes A. Recommendations for the evaluation of risk and prophylaxis of tumour lysis syndrome (TLS) in adults and children with malignant diseases: an expert TLS panel consensus. *Br J Haematol* 2010;149:578-86.
- 3) Cairo MS, Bishop M. Tumour lysis syndrome: new therapeutic strategies and classification. *Br J Haematol* 2004;127:3-11.
- 4) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 5) Spina M, Nagy Z, Ribera JM, et al. FLORENCE: a randomized, double-blind, phase III pivotal study of febuxostat versus allopurinol for the prevention of tumor lysis syndrome (TLS) in patients with hematologic malignancies at intermediate to high TLS risk. *Ann Oncol* 2015;26:2155-61.
- 6) Cortes J, Moore JO, Maziarz RT, et al. Control of plasma uric acid in adults at risk for tumor lysis syndrome: efficacy and safety of rasburicase alone and rasburicase followed by allopurinol compared with allopurinol alone—results of a multicenter phase III study. *J Clin Oncol* 2010;28:4207-13.
- 7) National Institute for Health and Care Excellence. Spinal metastases and metastatic spinal cord compression (NG234). 2023.
- 8) 日本臨床腫瘍学会. 骨転移診療ガイドライン 改訂第2版. 2022.
- 9) van der Sande JJ, Kröger R, Boogerd W. Multiple spinal epidural metastases; an unexpectedly frequent finding. *J Neurol Neurosurg Psychiatry* 1990;53:1001-3.
- 10) Husband DJ. Malignant spinal cord compression: prospective study of delays in referral and treatment. *BMJ* 1998;317:18-21.
- 11) Patchell RA, Tibbs PA, Regine WF, et al. Direct decompressive surgical resection in the treatment of spinal cord compression caused by metastatic cancer: a randomised trial. *Lancet* 2005;366:643-8.
- 12) Rowell NP, Gleeson FV. Steroids, radiotherapy, chemotherapy and stents for superior vena caval obstruction in carcinoma of the bronchus: a systematic review. *Clin Oncol (R Coll Radiol)* 2002;14:338-51.
- 13) 日本肺癌学会. 肺癌診療ガイドライン—悪性胸膜中皮腫・胸腺腫瘍含む—2024年版. 2024.
- 14) Burtis WJ, Brady TG, Orloff JJ, et al. Immunochemical characterization of circulating parathyroid hormone-related protein in patients with humoral hypercalcemia of cancer. *N Engl J Med* 1990;322:1106-12.
- 15) Schneider BJ, Naidoo J, Santomasso BD, et al. Management of immune-related adverse events in patients treated with immune checkpoint inhibitor therapy: ASCO guideline update. *J Clin Oncol* 2021;39:4073-126.

5-33 急性単関節炎

- 1) Margaretten ME, Kohlwes J, Moore D, Bent S. Does this adult patient have septic arthritis? *JAMA* 2007;297:1478-88.
- 2) Ravn C, Neyt J, Benito N, et al. Guideline for management of septic arthritis in native joints (SANJO). *J Bone Jt Infect* 2023;8:29-37.
- 3) 日本痛風・核酸代謝学会 ガイドライン改訂委員会. 高尿酸血症・痛風の治療ガイドライン 第3版. 2018.

- 4) Zhang W, Doherty M, Bardin T, et al. European League Against Rheumatism recommendations for calcium pyrophosphate deposition. Part I: terminology and diagnosis. *Ann Rheum Dis* 2011;70:563-70.
- 5) Papanicolas LE, Hakendorf P, Gordon DL. Concomitant septic arthritis in crystal monoarthritis. *J Rheumatol* 2012;39:157-60.
- 6) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）。
- 7) 社会保険診療報酬支払基金. 審査情報提供事例（薬剤）251 セファゾリンナトリウム水和物（感染症16）. 2016.
- 8) Workowski KA, Bachmann LH, Chan PA, et al. Sexually transmitted infections treatment guidelines, 2021. *MMWR Recomm Rep* 2021;70(4):1-187.
- 9) Li HK, Rombach I, Zambellas R, et al. Oral versus intravenous antibiotics for bone and joint infection. *N Engl J Med* 2019;380:425-36.
- 10) Gjika E, Beaulieu JY, Vakalopoulos K, et al. Two weeks versus four weeks of antibiotic therapy after surgical drainage for native joint bacterial arthritis: a prospective, randomised, non-inferiority trial. *Ann Rheum Dis* 2019;78:1114-21.
- 11) Richette P, Doherty M, Pascual E, et al. 2016 updated EULAR evidence-based recommendations for the management of gout. *Ann Rheum Dis* 2017;76:29-42.
- 12) Terkeltaub RA, Furst DE, Bennett K, et al. High versus low dosing of oral colchicine for early acute gout flare: twenty-four-hour outcome of the first multicenter, randomized, double-blind, placebo-controlled, parallel-group, dose-comparison colchicine study. *Arthritis Rheum* 2010;62:1060-8.
- 13) Janssens HJ, Janssen M, van de Lisdonk EH, et al. Use of oral prednisolone or naproxen for the treatment of gout arthritis: a double-blind, randomised equivalence trial. *Lancet* 2008;371:1854-60.
- 14) Rainer TH, Cheng CH, Janssens HJ, et al. Oral prednisolone in the treatment of acute gout: a pragmatic, multicenter, double-blind, randomized trial. *Ann Intern Med* 2016;164:464-71.
- 15) FitzGerald JD, Dalbeth N, Mikuls T, et al. 2020 American College of Rheumatology guideline for the management of gout. *Arthritis Care Res (Hoboken)* 2020;72:744-60.
- 16) Zhang W, Doherty M, Pascual E, et al. EULAR recommendations for calcium pyrophosphate deposition. Part II: management. *Ann Rheum Dis* 2011;70:571-5.
- 17) Taylor TH, Mecchella JN, Larson RJ, et al. Initiation of allopurinol at first medical contact for acute attacks of gout: a randomized clinical trial. *Am J Med* 2012;125:1126-34.e7.
- 18) Hill EM, Sky K, Sit M, et al. Does starting allopurinol prolong acute treated gout? A randomized clinical trial. *J Clin Rheumatol* 2015;21:120-5.

5-34 終末期の症状緩和

- 1) 日本循環器学会, 日本心不全学会. 2021年改訂版 循環器疾患における緩和ケアについての提言. 2021.
- 2) Hui D, dos Santos R, Chisholm G, et al. Clinical signs of impending death in cancer patients. *Oncologist* 2014;19:681-7.
- 3) Hui D, Dos Santos R, Chisholm G, et al. Bedside clinical signs associated with impending death in patients with advanced cancer: preliminary findings of a prospective, longitudinal cohort study. *Cancer* 2015;121:960-7.
- 4) 日本緩和医療学会 緩和医療ガイドライン委員会編. 終末期がん患者の輸液療法に関するガイドライン 2013年版. 金原出版; 2013.
- 5) 日本緩和医療学会 ガイドライン統括委員会編. がん患者の治療抵抗性の苦痛と鎮静に関する基本的な考え方の手引き 2023年版. 金原出版; 2023.

- 6) 日本緩和医療学会 ガイドライン統括委員会編. 進行性疾患患者の呼吸困難の緩和に関する診療ガイドライン 2023年版. 金原出版; 2023.
- 7) Abernethy AP, McDonald CF, Frith PA, et al. Effect of palliative oxygen versus room air in relief of breathlessness in patients with refractory dyspnoea: a double-blind, randomised controlled trial. *Lancet* 2010;376:784-93.
- 8) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 9) Agar MR, Lawlor PG, Quinn S, et al. Efficacy of oral risperidone, haloperidol, or placebo for symptoms of delirium among patients in palliative care: a randomized clinical trial. *JAMA Intern Med* 2017;177:34-42.
- 10) Hui D, Frisbee-Hume S, Wilson A, et al. Effect of lorazepam with haloperidol vs haloperidol alone on agitated delirium in patients with advanced cancer receiving palliative care: a randomized clinical trial. *JAMA* 2017;318:1047-56.
- 11) van Esch HJ, van Zuylen L, Geijteman ECT, et al. Effect of prophylactic subcutaneous scopolamine butylbromide on death rattle in patients at the end of life: the SILENCE randomized clinical trial. *JAMA* 2021;326:1268-76.
- 12) Bruera E, Hui D, Dalal S, et al. Parenteral hydration in patients with advanced cancer: a multicenter, double-blind, placebo-controlled randomized trial. *J Clin Oncol* 2013;31:111-8.
- 13) 日本老年医学会. 高齢者ケアの意思決定プロセスに関するガイドライン—人工的水分・栄養補給の導入を中心として. 2012.
- 14) Maeda I, Morita T, Yamaguchi T, et al. Effect of continuous deep sedation on survival in patients with advanced cancer (J-Proval): a propensity score-weighted analysis of a prospective cohort study. *Lancet Oncol* 2016;17:115-22.
- 15) Yokomichi N, Yamaguchi T, Maeda I, et al. Effect of continuous deep sedation on survival in the last days of life of cancer patients: a multicenter prospective cohort study. *Palliat Med* 2022;36:189-99.
- 16) Ekström MP, Bornefalk-Hermansson A, Abernethy AP, Currow DC. Safety of benzodiazepines and opioids in very severe respiratory disease: national prospective study. *BMJ* 2014;348:g445.

5-35 化学療法中の患者を受け持つとき

- 1) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認) .
- 2) 日本肝臓学会 肝炎診療ガイドライン作成委員会編. B型肝炎治療ガイドライン (第5版) . 2026.
- 3) 日本癌治療学会編. 制吐薬適正使用ガイドライン 2023年10月改訂 第3版. 金原出版; 2023.
- 4) Iihara H, Abe M, Wada M, et al. 2023 Japan Society of Clinical Oncology clinical practice guidelines update for antiemesis. *Int J Clin Oncol* 2024;29:873-88.
- 5) Hashimoto H, Abe M, Tokuyama O, et al. Olanzapine 5 mg plus standard antiemetic therapy for the prevention of chemotherapy-induced nausea and vomiting (J-FORCE): a multicentre, randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Oncol* 2020;21:242-9.
- 6) 日本臨床腫瘍研究グループ. 有害事象共通用語規準 v5.0 日本語訳JCOG版 (CTCAE v5.0-JCOG) .
- 7) Elad S, Cheng KKF, Lalla RV, et al. MASCC/ISOO clinical practice guidelines for the management of mucositis secondary to cancer therapy. *Cancer* 2020;126:4423-31.
- 8) Brahmer JR, Abu-Sbeih H, Ascierto PA, et al. Society for Immunotherapy of Cancer (SITC) clinical practice guideline on immune checkpoint inhibitor-related adverse events. *J Immunother Cancer* 2021;9:e002435.
- 9) Schneider BJ, Naidoo J, Santomaso BD, et al. Management of immune-related adverse events in patients treated with immune checkpoint inhibitor therapy: ASCO guideline update. *J Clin Oncol* 2021;39:4073-126.
- 10) 日本臨床腫瘍学会編. がん免疫療法ガイドライン 第3版. 金原出版; 2023.

- 11) 日本がん看護学会, 日本臨床腫瘍学会, 日本臨床腫瘍薬学会編. がん薬物療法に伴う血管外漏出に関する合同ガイドライン2023年版. 金原出版; 2022.
- 12) 日本がん看護学会, 日本臨床腫瘍学会, 日本臨床腫瘍薬学会編. がん薬物療法における職業性曝露対策ガイドライン 2019年版. 金原出版; 2019.

5-36 炎症性腸疾患と大腸憩室炎

- 1) Manabe N, Haruma K, Nakajima A, et al. Characteristics of colonic diverticulitis and factors associated with complications: a Japanese multicenter, retrospective, cross-sectional study. *Dis Colon Rectum* 2015;58:1174-81.
- 2) 厚生労働科学研究費補助金 難治性疾患政策研究事業「難治性炎症性腸管障害に関する調査研究」(久松班). 潰瘍性大腸炎・クローン病 診断基準・治療指針 令和7年度改訂版(令和8年3月31日). 2026.
- 3) Lamb CA, Kennedy NA, Raine T, et al. British Society of Gastroenterology consensus guidelines on the management of inflammatory bowel disease in adults. *Gut* 2019;68(Suppl 3):s1-s106.
- 4) Harbord M, Eliakim R, Bettenworth D, et al. Third European evidence-based consensus on diagnosis and management of ulcerative colitis. Part 2: current management. *J Crohns Colitis* 2017;11:769-84.
- 5) 電子添文: ロベミンカプセル、ブスコパン錠、水溶性プレドニン、レミケード点滴静注用、セフメタゾン静注用、ゾシン静注用、ヘパリンカルシウム皮下注. KEGG MEDICUS (2026年10月5日閲覧).
- 6) Grainge MJ, West J, Card TR. Venous thromboembolism during active disease and remission in inflammatory bowel disease: a cohort study. *Lancet* 2010;375:657-63.
- 7) Nakase H, Uchino M, Shinzaki S, et al. Evidence-based clinical practice guidelines for inflammatory bowel disease 2020. *J Gastroenterol* 2021;56:489-526 (日本消化器病学会 炎症性腸疾患 (IBD) 診療ガイドライン2020 改訂第2版の英語版).
- 8) Turner D, Walsh CM, Steinhart AH, et al. Response to corticosteroids in severe ulcerative colitis: a systematic review of the literature and a meta-regression. *Clin Gastroenterol Hepatol* 2007;5:103-10.
- 9) Travis SP, Farrant JM, Ricketts C, et al. Predicting outcome in severe ulcerative colitis. *Gut* 1996;38:905-10.
- 10) 日本消化管学会. 大腸憩室症(憩室出血・憩室炎)ガイドライン. 日本消化管学会雑誌 2017;1(Suppl).
- 11) Peery AF, Shaukat A, Strate LL. AGA clinical practice update on medical management of colonic diverticulitis: expert review. *Gastroenterology* 2021;160:906-11.e1.
- 12) Sawyer RG, Claridge JA, Nathens AB, et al. Trial of short-course antimicrobial therapy for intraabdominal infection. *N Engl J Med* 2015;372:1996-2005.
- 13) Lambrichts DPV, Bolkenstein HE, van der Does DCHE, et al. Multicentre study of non-surgical management of diverticulitis with abscess formation. *Br J Surg* 2019;106:458-66.
- 14) Meyer J, Orzi LA, Combescure C, et al. Risk of colorectal cancer in patients with acute diverticulitis: a systematic review and meta-analysis of observational studies. *Clin Gastroenterol Hepatol* 2019;17:1448-56.e17.
- 15) Chabok A, Pählman L, Hjern F, et al. Randomized clinical trial of antibiotics in acute uncomplicated diverticulitis. *Br J Surg* 2012;99:532-9.
- 16) Daniels L, Ünlü Ç, de Korte N, et al. Randomized clinical trial of observational versus antibiotic treatment for a first episode of CT-proven uncomplicated acute diverticulitis. *Br J Surg* 2017;104:52-61.
- 17) van Dijk ST, Daniels L, Ünlü Ç, et al. Long-term effects of omitting antibiotics in uncomplicated acute diverticulitis. *Am J Gastroenterol* 2018;113:1045-52.

第6部 法令・書類・医療安全

6-1 死亡確認と死亡診断書

- 1) 厚生労働省医政局, 厚生労働省政策統括官(統計・情報システム管理、労使関係担当). 死亡診断書(死体検案書)記入マニュアル 令和8年度版. 2026.
- 2) Academy of Medical Royal Colleges. A code of practice for the diagnosis and confirmation of death: 2025 update. London: Academy of Medical Royal Colleges; 2025.
- 3) 厚生労働省医政局医事課長. 医師による異状死体の届出の徹底について(医政医発0208第3号). 2019.
- 4) Dhanani S, Hornby L, van Beinum A, et al. Resumption of cardiac activity after withdrawal of life-sustaining measures. N Engl J Med 2021;384:345-52.
- 5) 厚生労働省. 死亡診断書(死体検案書)について: よくある質問. 2026年10月1日閲覧.
- 6) 厚生労働省医政局医事課長. 医師法第20条ただし書の適切な運用について(医政医発0831第1号). 2012.
- 7) 医師法(昭和23年法律第201号)第19条、第20条、第21条、第33条の3.
- 8) 厚生労働省医政局医事課. 「医師による異状死体の届出の徹底について」に関する質疑応答集(Q&A)について(事務連絡). 2019.
- 9) 日本法医学会. 「異状死」ガイドライン. 日法医誌 1994;48:357-8.
- 10) 日本医師会. 研修ワークブック 院内調査のすすめ方 2023年度研修資料. 2023.
- 11) 医療法(昭和23年法律第205号)第6条の10.
- 12) Shojania KG, Burton EC, McDonald KM, Goldman L. Changes in rates of autopsy-detected diagnostic errors over time: a systematic review. JAMA 2003;289:2849-56.
- 13) 死体解剖保存法(昭和24年法律第204号)第7条.
- 14) 日本医療安全調査機構. ご遺族(ご家族)の皆様へ 病理解剖について(説明用紙)とその利用について. 2020.
- 15) 戸籍法(昭和22年法律第224号)第86条.
- 16) 墓地、埋葬等に関する法律(昭和23年法律第48号)第3条.

6-2 感染症法に基づく届出

- 1) 感染症の予防及び感染症の患者に対する医療に関する法律(平成10年法律第114号) 第8条、第12条、第14条、第17～20条、第26条、第26条の2、第30条、第53条の11、第53条の14、第53条の15、第77条.
- 2) 厚生労働省. 感染症法に基づく医師の届出のお願い(届出の対象となる感染症の種類、感染症法における感染症の分類). 2026年10月閲覧.
- 3) 感染症の予防及び感染症の患者に対する医療に関する法律施行規則(平成10年厚生省令第99号) 第4条、第4条の3.
- 4) 埼玉県狭山保健所. 感染症発生届等について(医療機関の方へ). 2026年10月3日閲覧.
- 5) 厚生労働省. 医師及び指定届出機関の管理者が都道府県知事に届け出る基準(平成18年3月8日健感発第0308001号の別紙、令和8年4月6日施行). 2026.
- 6) 厚生労働省. 感染症法に基づく医師の届出ハンドブック(令和6年度厚生労働行政推進調査事業費補助金による). 2026年10月閲覧.
- 7) 厚生労働省健康局結核感染症課長. 感染症の予防及び感染症の患者に対する医療に関する法律における結核患者の入退院及び就業制限の取扱いについて(平成19年9月7日健感発第0907001号、平成26年1月29日一部改正). 2014.

- 8) 阿彦忠之, 加藤誠也, ほか. 感染症法に基づく結核の接触者健康診断の手引き 改訂第6版. 結核予防会; 2022.
- 9) 食品衛生法 (昭和22年法律第233号). 第63条.
- 10) 食品衛生法施行規則 (昭和23年厚生省令第23号). 第72条.
- 11) 厚生労働省医政局地域医療計画課. インフルエンザ及びノロウイルス感染症の院内感染に関する保健所への報告及び相談について (平成27年3月9日事務連絡). 2015.
- 12) 厚生労働省健康・生活衛生局感染症対策部感染症対策課長. 感染症の予防及び感染症の患者に対する医療に関する法律第12条第1項及び第14条第2項に基づく届出の基準等について (一部改正) (令和8年2月5日感感発0205第2号). 2026.

6-3 針刺し・血液曝露

- 1) HIV感染症および血友病におけるチーム医療の構築と医療水準の向上を目指した研究班. 抗HIV治療ガイドライン (2026年3月版). 2026.
- 2) Schillie S, Murphy TV, Sawyer M, et al. CDC guidance for evaluating health-care personnel for hepatitis B virus protection and for administering postexposure management. MMWR Recomm Rep 2013;62(RR-10):1-19.
- 3) Moorman AC, de Perio MA, Goldschmidt R, et al. Testing and clinical management of health care personnel potentially exposed to hepatitis C virus — CDC guidance, United States, 2020. MMWR Recomm Rep 2020;69(6):1-8.
- 4) Centers for Disease Control and Prevention. Updated U.S. Public Health Service guidelines for the management of occupational exposures to HBV, HCV, and HIV and recommendations for postexposure prophylaxis. MMWR Recomm Rep 2001;50(RR-11):1-52.
- 5) 厚生労働省労働基準局長. C型肝炎、エイズ及びMRSA感染症に係る労災保険における取扱いについて (平成5年10月29日基発第619号、平成22年9月9日基発0909第1号で一部改正). 2010.
- 6) 厚生労働省健康局疾病対策課長. 「HIV検査の実施について」の改廃について (平成16年10月29日健疾感発第1029004号). 2004.
- 7) 日本環境感染学会ワクチン委員会. 医療関係者のためのワクチンガイドライン 第5版. 環境感染誌 2026;41(Suppl II):S1-54.
- 8) 各薬剤の電子添文 (KEGG MEDICUSで2026年10月に確認).
- 9) Kofman AD, Struble KA, Heneine W, et al. 2025 US Public Health Service guidelines for the management of occupational exposures to human immunodeficiency virus and recommendations for post-exposure prophylaxis in healthcare settings. Infect Control Hosp Epidemiol 2025;46:863-73.
- 10) 國島広之, 吉川徹, 網中真由美, ほか. 職業感染制御委員会 医療機関におけるC型肝炎ウイルス曝露後検査の進め方. 環境感染誌 2022;37:31-2.
- 11) 地方公務員災害補償基金理事長. 抗HBs人免疫グロブリン製剤及びB型肝炎ワクチンに関する療養補償の取扱いについて (昭和62年10月12日地企企第27号、平成30年4月1日第3次改正). 2018.
- 12) 宮城労働局. 労災診療費算定基準Q&A. 2026年10月閲覧.
- 13) 厚生労働省. 労働者災害補償保険法の施行に関する事務に使用する文書の様式を定める件 (昭和35年労働省告示第10号、令和2年厚生労働省告示第397号で一部改正). 2020.
- 14) 厚生労働省. 事業主の皆さまへ 速やかな「労災保険給付」のため請求書の作成にご協力をお願いします. 2026年10月閲覧.
- 15) Siegel JD, Rhinehart E, Jackson M, Chiarello L; Healthcare Infection Control Practices Advisory Committee. 2007 Guideline for isolation precautions: preventing transmission of infectious agents in healthcare settings. Am J Infect Control 2007;35(10 Suppl 2):S65-164.
- 16) Occupational Safety and Health Administration. Bloodborne pathogens. 29 CFR 1910.1030 (2026年10月閲覧).

6-4 医療用麻薬の取り扱い

- 1) 麻薬及び向精神薬取締法（昭和28年法律第14号）．第2条、第3～5条、第7条、第13条、第17条、第27～29条、第33～35条、第39条、第41条、第50条の21～23、第58条の2、第66条の2．
- 2) 厚生労働省医薬食品局監視指導・麻薬対策課．病院・診療所における麻薬管理マニュアル．2011（2015年の法改正による修正を含む）．
- 3) 厚生労働省医薬局監視指導・麻薬対策課．がんの痛みの治療における医療用麻薬の自己管理マニュアル～医療従事者の役割～．2024．
- 4) 麻薬及び向精神薬取締法施行規則（昭和28年厚生省令第14号）．第9条の3、第10条の2、第40条、第41条．麻薬及び向精神薬取締法施行令（昭和28年政令第57号）．第3条、第4条．
- 5) 厚生労働省医薬・生活衛生局監視指導・麻薬対策課．医療用麻薬適正使用ガイダンス～がん疼痛及び慢性疼痛治療における医療用麻薬の使用と管理のガイダンス～．2017．
- 6) 保険医療機関及び保険医療養担当規則（昭和32年厚生省令第15号）．第20条．
- 7) 療担規則及び薬担規則並びに療担基準に基づき厚生労働大臣が定める揭示事項等（平成18年厚生労働省告示第107号）．第10、第10の2（令和8年厚生労働省告示第15号による改正まで）．
- 8) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認）．
- 9) 厚生労働省医薬・生活衛生局監視指導・麻薬対策課．病院・診療所・飼育動物診療施設・薬局における覚醒剤原料取扱いの手引き（覚醒剤取締法上の取扱い）．2020．

付録

付録C 抗菌薬の腎機能別用量表

- 1) 各薬剤の電子添文（KEGG MEDICUSで2026年10月に確認．ピクシリン、ユナシン-S、ゾシン、セファメジンα、セフメタゾン、ロセフィン、セフトアジジム、セフェピム塩酸塩、メロペン、塩酸バンコマイシン、ゲンタシン、キュビシン、ザイボックス、クラビット、ジスロマック、アネメトロ、フラジール、ダラシンS、サワシリン、オーグメンチン、ケフレックス、バクタ、ビブラマイシン、ゾピラックス、バルトレックス、ジフルカン、ファンガード、タミフル）．
- 2) 日本呼吸器学会成人肺炎診療ガイドライン2024作成委員会編．成人肺炎診療ガイドライン2024．日本呼吸器学会；2024（参考資料：代表的な抗菌薬名と用法・用量）．
- 3) 日本腎臓学会編．CKD診療ガイド2012．東京医学社；2012（付表：腎機能低下時の薬剤投与量）．
- 4) 日本化学療法学会，日本TDM学会 抗菌薬TDMガイドライン作成委員会．抗菌薬TDM臨床実践ガイドライン2022．日化療誌 2022;70:1-72．

内科大全 病棟編

入院時の指示・薬・検査・当直・入院疾患の管理

2026年10月08日01:03 下書き（入稿用ではない）

本書に記載した薬剤の用量・用法、診断・治療の基準は、発行時点の電子添文とガイドラインに基づく。実際の使用にあたっては、最新の電子添文とガイドラインを確認すること。

本書の内容の無断での複製・転載を禁じる。